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Towards model-informed precision dosing of piperacillin: multicenter systematic external evaluation of pharmacokinetic models in critically ill adults with a focus on Bayesian forecasting

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Supplementary Data (ESM_Main)

Evaluation dataset

Clinical data of intensive care unit (ICU) patients treated with piperacillin (PIP) were available from four previously published studies including 11 different German centers (study 1: 207 patients, 9 centers, prospective randomized controlled trial TARGET, DRKS00011159; study 2: 282 patients, General Hospital Heidenheim, retrospective, reference number 351/14 (Ethical Committee University of Ulm); study 3: 12 patients, University Hospital Regensburg, prospective-observational, reference number 18-1020-101 (Ethical Committee University of Regensburg); study 4: 60 patients, University Hospital LMU Munich, prospective-observational, NCT01793012).

Intravenous PIP/tazobactam (TZP) was administered as continuous infusion (CI) in studies 1-3 and as intermittent infusions (II, 30 minutes) in study 4. After an optional loading dose at therapy onset (at the discretion of the responsible physician), the total daily PIP dose (8-16 g) was based on renal function (studies 1-4) and based on individual therapeutic drug monitoring (TDM) results (studies 1-2). Blood samples were obtained at least once daily at steady state in studies 1-3 (CI). In study 4, the first sample was taken right before the next dose, followed by multiple measurements at predefined time points during and after PIP administration (e.g., second sample: 15 minutes after start of infusion). All centers measured total PIP plasma concentrations using validated mass spectrometry or chromatography methods. Further information on dosing, sample collection, and analysis is presented in the respective original publications.

Only patients with at least two TDM samples were included in the study. Missing values of patient characteristics ('covariates') were (i) linearly interpolated between available datapoints of a patient, (ii) replaced by the reference value of the respective model codes if entirely missing in a patient (e.g. WT=70kg, given $V_d = V_{d\_typical} \cdot (WT/70)^1$), or (iii) replaced by the median of the patient population underlying the model, if continuous or categorical covariates were missing for an entire study. The aggregated dataset included a total of 561 ICU patients with 3654 PIP concentrations and is summarized in the main manuscript **(Table 1)**.

Literature research and population pharmacokinetic model selection

A systematic literature review of piperacillin (PIP) population pharmacokinetic (PopPK) models was conducted in PubMed up to 27 June 2022. The following search terms were used:

(1) ((piperacillin) OR (piperacillin[nm])) AND ((((pharmacokinetics[MeSH Terms]) OR (pharmacokinetic*[Title/Abstract])) OR (population*[Title/Abstract])) AND (model*[Title/Abstract]))
(2) ((piperacillin) OR (piperacillin[nm])) AND (population pharmacokinetic*[Title/Abstract] OR PKPD)
(3) piperacillin AND (population pharmacokinetic OR NONMEM OR NLME OR nonlinear mixed effect OR non linear mixed effect OR model) AND (ICU OR intensive care unit OR critically ill) (= modified search term [1])

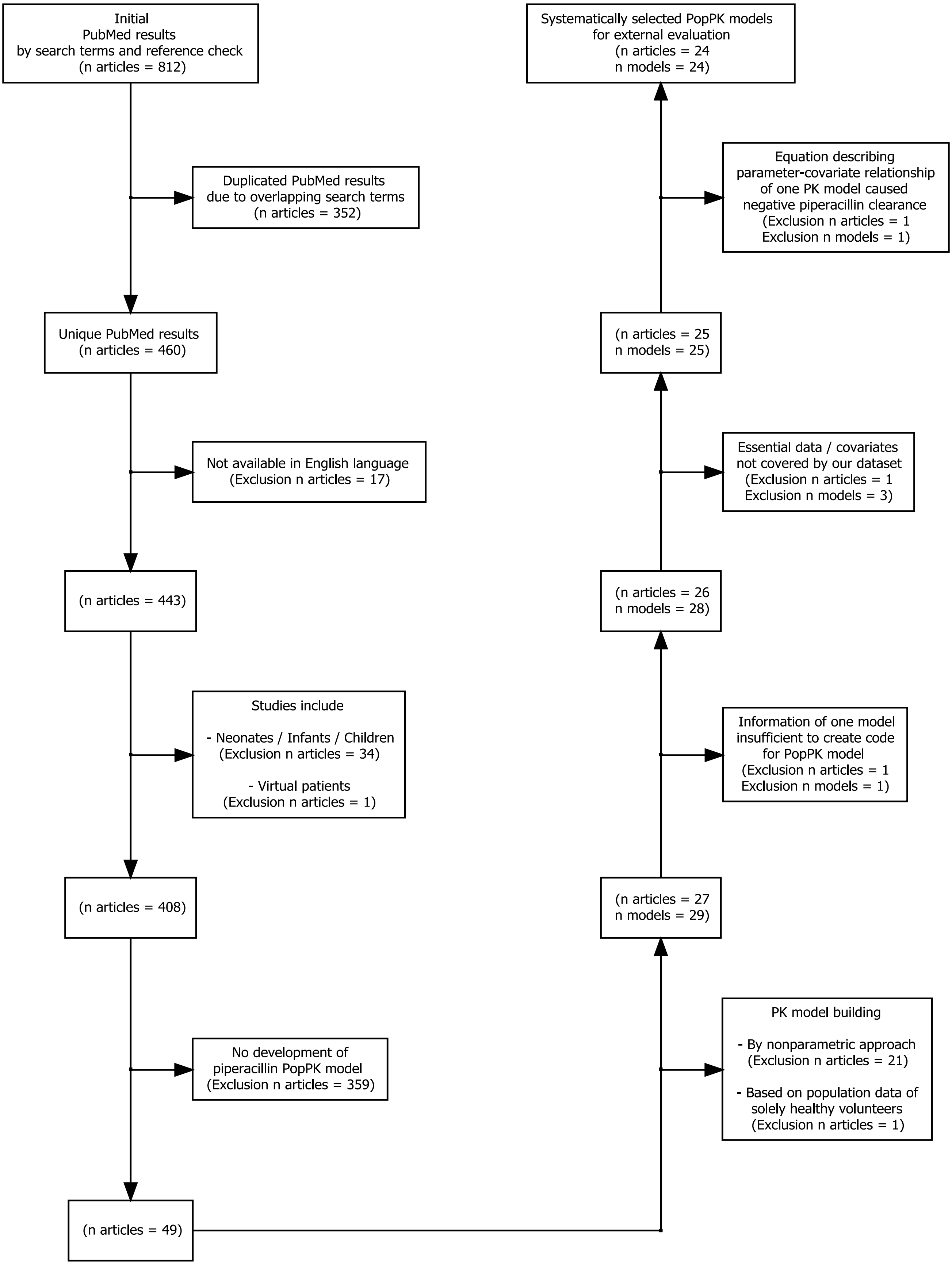
+ 44 relevant articles added by reference check.

Models were included according to the following selection criteria:

(1) Publication written in English
(2) Models based on clinical data from ill, non-virtual, adult patients
(3) Development of a parametric PopPK model.

PK models were excluded if the model description was inadequate, or if the external dataset did not cover essential information for evaluation. The corresponding author of one study with insufficiently described model properties (missing exponential error) was contacted via email but did not reply [2]. Two models were excluded because therapeutic drug monitoring data of amikacin was necessary for model evaluation [3]. One model was excluded because our evaluation dataset did not include information on specific membranes used in renal replacement therapy patients [4]. One model was excluded because the pharmacokinetic formula caused negative piperacillin clearance based on our external evaluation data [5].

Fig. S1 Flowchart of the literature research and population pharmacokinetic (PopPK) model selection



Software used for model reconstruction and processing

Detailed coding of the 24 identified PopPK models for piperacillin was performed by two persons (four-eye-principle, quality check) using the nonlinear mixed-effects modeling (NONMEM® 7.5, Icon Development Solutions, USA) software. The models were reconstructed using the specified pharmacometric properties (formulas and parameters), as described in the original publications, without any modifications. Model processing using the evaluation dataset was performed without any additional fitting of the model to the data (NONMEM option: MAXEVAL = 0) supported by PsN 5.3.0 (Perl-speaks-NONMEM, Uppsala University, Sweden) in conjunction with Pirana 21.11.1 (Pirana Modelling Workbench, Certara, USA). Model results were evaluated and illustrated employing R 4.1.3 (R Foundation for Statistical Computing, Austria) in RStudio® 4.2.2, statistical software (RStudio, Inc., USA).

Overview of the pharmacometric studies and population pharmacokinetic models for piperacillin

A clinical database with a median of 21 patients (range 4-78) was used for model development and two thirds (n=16) considered critically ill/ICU patients only. Model properties and pharmacokinetic parameters were highly variable. A 2-compartment structural model was commonly chosen (n=17). The most frequently integrated covariates were renal function (creatinine clearance (CLCR) or estimated glomerular filtration rate (eGFR), n=17) and measures of body size (e.g. total body weight, n=13). Interindividual variability was included in all models, on different parameters.

Table S1 Summary of the pharmacometric studies, focusing on the characteristics and parameters of the population pharmacokinetic models for piperacillin

Reference	**Study population**	**No. of patients (♂/♀) {RRT}**	**Piperacillin/tazobactam administration**		**Piperacillin therapeutic drug monitoring (TDM)**			**Piperacillin population pharmacokinetic model**			
			Method	*Dosage*	*Ana-lytical method*	*Type of measured concentr.*	*No. of plasma samples*	*Structure*	*Covariates included*	*PK equations and parameter values*	*Variabilities* [a]
Andersen et al. 2018 [6]	Less critically ill septic non-ICU patients without RRT	22 (12/10) {0}	Intermittent Infusion (II)	4/0.5 g every 8 hours as a 3 min iv. bolus	Ultra HPLC	Unbound fraction	7 per pat., 154 in total	2 CMT	CL_{CR} (CG) → on CL	TVCL = 2.23+0.0757*CL_{CR} CL = TVCL*EXP(IIV_CL) Q = 3.54 V1 = 12.4 V2 = 3.48	IIV_CL = 0.0845 RV_{prop} = 0.115
Asín-Prieto et al. 2013 [7]	Critically ill ICU patients undergoing continuous RRT with different degree of remaining renal function	16 (12/4) {16}	Intermittent Infusion (II)	4/0.5 g every 4/6 or 8 hours as a 20 min iv. infusion	HPLC-UV	Total	9 per pat., 144 in total	2 CMT	CL_{CR} (CG) → on CL RRT → on CL WT → on V2	TVCL = 2.77+(CL_{CR}/1.95)^4.55 +0.64*RRT CL = TVCL*EXP(IIV_CL) Q = 20.53 V1 = 20.86*EXP(IIV_V1) V2 = 21.41^(WT/75.3)	IIV_CL = 0.177 IIV_V1 = 0.4463 Correlation of IIV_CL and IIV_V1 = 0.4946 RV_{add} = 0.19 (log scale)
Bue et al. 2020 [8]	Critically ill ICU patients treated with continuous RRT	10 (6/4) {10}	Intermittent Infusion (II)	4/0.5 g every 8 hours as a 3 min iv. bolus	Ultra HPLC	Unbound fraction	8 per pat., 80 in total	2 CMT	WT → on CL,Q, V1,V2	TVCL = 3.3*(WT/70)^0.75 CL = TVCL*EXP(IIV_CL) TVQ = 15.4*(WT/70)^0.75 Q = TVQ*EXP(IIV_Q) V1 = 6.77*(WT/70) TVV2 = 10.0*(WT/70) V2 = TVV2*EXP(IIV_V2)	IIV_CL = 0.0118 IIV_Q = 0.2626 IIV_V2 = 0.0654 RV_{prop} = 0.166

Bulitta et al. 2007 [9]	Cystic fibrosis patients	8 (5/3) {NA}	Intermittent Infusion (II)	Single 4/0.5 g as a 5 min iv. bolus (one patient with 3 g bolus)	HPLC	Total	21 per pat., 546 in total	2 CMT	LBM[c] (Cheymol) → on CL,Q, V1,V2	TVCL = 11.3*(LBM/53.0)^0.75 CL = TVCL*EXP(IIV_CL) Q = 12.8*(LBM/53.0)^0.75 TVV1 = 6.49*(LBM/53.0) V1 = TVV1*EXP(IIV_V1) TVV2 = 3.12*(LBM/53.0) V2 = TVV2*EXP (IIV_V2)	IIV_CL = 0.0108 IIV_V1 = 0.0654 IIV_V2 = 0.1106 RV_{prop} = 0.132 RV_{add} = 1.88
Chen et al. 2016 [10]	Hospitalized patients with known or suspected nosocomial infection, no RRT	50 (31/19) {0}	Intermittent Infusion (II) or Extended Infusion (EI)	4/0.5 g by either 30 min (II) or 3-hour (EI) iv. infusion (min 4 to max 14 days)	HPLC	Total	NA per pat., 210 in total	1 CMT	CL_{CR} (CG) → on CL WT → on V	TVCL = 9.14+(4.6^(CL_{CR}/68.7) CL = TVCL*EXP (IIV_CL) TVV = 12.2+(9.49^(WT/61.1) V = TVV*EXP (IIV_V)	IIV_CL = 0.0923 IIV_V = 0.1349 RV_{prop} = 0.0932
Chung et al. 2015 [11]	Obese or nonobese hospitalized patients with known or suspected bacterial infection	27 (17/10) {NA}	Extended Infusion (EI)	4/0.5 g or 6/0.75 g every 8 hours as a 4-hour iv. infusion	HPLC	Total	NA per pat., 198 in total	1 CMT	Measured CL_{CR}[e] → on CL BMI → on CL WT → on V	TVCL = 11.3+(0.0646*(CL_{CR}-105))+(0.0579*(BMI-35)) CL = TVCL*EXP (IIV_CL) TVV = 31.3+(0.132*(WT-120)) V = TVV*EXP (IIV_V)	IIV_CL = 0.0217 IIV_V = 0.094 RV_{prop} = 0.155 RV_{add} = 5.27
Delattre et al. 2012 (Model 1 for PIP) [3]	Critically ill septic ICU patients receiving piperacillin comedicated with Amikacin	22 (NA/NA) {0}	Intermittent Infusion (II)	4 g every 6 hours as a 30 min iv. infusion	HPLC	Total	6 per pat., 132 in total	2 CMT	CL_{CR} (CG, calculated for a nominal 70 kg subject) → on CL WT → on CL,Q, V1,V2	TVCL = (9.97*(WT/70)^0.75)*((CL_{CR}/100)^0.62) CL = TVCL*EXP (IIV_CL) Q = (5.48*(WT/70)^0.75) TVV1 = 24.0*(WT/70) V1 = TVV1*EXP (IIV_V1) V2 = 8.11*(WT/70)	IIV_CL = 0.2352 IIV_V1 = 0.1874 RV_{prop} = 0.307 RV_{add} = 1.90
Fillâtre et al. 2021 [12]	Case-Control-Study with critically ill septic ICU patients. Cases with and Control without ECMO support	42 (34/8) {8}	Extended Infusion (EI)	4/0.5 g as a 4 hour iv. infusion. Dosage interval depending on renal function indicated by CL_{CR} and RRT	HPLC-UV	Total	5.3 per pat., 221 in total	2 CMT	none	CL = 5.01*EXP (IIV_CL) Q = 7.01 V1 = 17.60*EXP (IIV_V1) V2 = 10.40*EXP (IIV_V2)	IIV_CL = 0.36 IIV_Q = 0 IIV_V1 = 0.1089 IIV_V2 = 0.7056 RV_{prop} = 0.04 RV_{add} = 5.15

Hahn et al. 2021 [13]	Critically ill ICU patients with VA-ECMO support. Half of population underwent RRT	26 (19/7) {13}	Extended Infusion (EI)	4/0.5 g as a 40 min iv. bolus. Adaption of dosage and infusion interval depending on renal function indicated by CL_{CR} and RRT	LC-MS/MS	Total	NA per pat., 244 in total	2 CMT	CL_{CR} (CG) → on CL ECMO → on CL CVVHDF → on V1 Note: RRT used instead of CVVHDF (Dialysis mode not available)	TVCL = 9.4*(1-0.092*ECMO) +(0.115*(CL_{CR}-54.7)) CL = TVCL*EXP (IIV_CL) Q = 17.2 TVV1 = 6.56*(1+1.46*CVVHDF) V1 = TVV1*EXP (IIV_V1) V2 = 14.2*EXP (IIV_V2)	IIV_CL = 0.0523 IIV_V1 = 0.291 IIV_V2 = 0.138 RV_{prop} = 0.313
Hamada et al. 2013 [14]	Patients with Community-Acquired-Pneumonia (CAP)	53 (39/14) {NA}	Intermittent Infusion (II)	4/0.5 g every 8 hours as iv. infusion	NA	Total	NA per pat., 157 in total	1 CMT	CL_{CR} (CG) → on CL	TVCL = 8.22+0.0561*(CL_{CR}-71.4) CL = TVCL*EXP (IIV_CL) V = 13.7	IIV_CL = 0.0319 RV_{prop} = 0.115
Ishihara et al. 2020 [15]	Late elderly patients with pneumonia, no RRT	18 (14/4) {0}	Intermittent Infusion (II)	Three times a day 4/0.5 g or 2/0.25 g if CL_{CR} < 50ml/min	HPLC-UV	Total	NA per pat., 100 in total	2 CMT	CL_{CR} (CG) → on CL	TVCL = 4.58+0.061*(CL_{CR}-37.4) CL = TVCL*EXP (IIV_CL) Q = 20.7*EXP (IIV_Q) V1 = 5.39*EXP (IIV_V1) V2 = 6.96	IIV_CL = 0.0705 IIV_Q = 0.311 IIV_V1 = 0.389 RV_{prop} = 0.000927 RV_{add} = 25.1
Jeon et al. 2014 [16]	Burn ICU patients	50 (40/10) {5}	Intermittent Infusion (II)	4/0.5 g every 8 hours as 30 min iv. infusion	HPLC-MS/MS	Total	NA	2 CMT	CL_{CR} (CG) → on CL Day after burn injury (DAI) → on CL SEPSIS → on V1	TVCL = 16.6*(CL_{CR}/132)+(DAI*(-0.0874)) CL = TVCL*EXP (IIV_CL) Q = 0.636*EXP (IIV_Q) TVV1 = 25.3+(SEPSIS*14.8) V1 = TVV1*EXP (IIV_V1) V2 = 16.1 Note: No patient with burn reported in our external dataset. DAI set to 0 for evaluation	IIV_CL = 0.1181 IIV_Q = 0.5963 IIV_V1 = 0.1653 Correlation of IIV_CL and IIV_V1 = 0.434 RV_{prop} = 0.185 RV_{add} = 0.359
Kim et al. 2016 [17]	Korean patients with acute infections	33 (17/16) {NA}	Intermittent Infusion (II)	4/0.5 g or if CL_{CR} < 50 ml/min 2/0.25 g every 8 hours as a 60 min iv. infusion	LC–MS/MS	Total	NA	2 CMT	CL_{CR} (CG) → on CL WT → on V1	TVCL = 2.9+4.03*(CL_{CR}/47) CL = TVCL*EXP (IIV_CL) Q = 2.29 TVV1 = 19.50*(WT/60) V1 = TVV1*EXP (IIV_V1) V2 = 3.76	IIV_CL = 0.279 IIV_V1 = 0.179 RV_{prop} = 0.399

Kim et al. 2022 [18]	Critically ill Korean ICU patients. Half of population with ECMO support	38 (25/13) {8}	Intermittent Infusion (II)	2/0.25 g, 3/0.375 g, 4/0.5 g, every 6 or 8 hours as a 30 min iv. infusion	LC- MS/MS	Total	NA per pat., 226 in total	2 CMT	$eGFR_{CYSC}$ (CKD-EPI) → on CL Note: External dataset eGFR (CL_{CR}) calculated by CG WT → on CL, Q, V1, V2 ECMO → on V1	TVCL = 5.05*((WT/70)^0.75)^ (0.00932*(eGFR-52.77)) CL = TVCL*EXP (IIV_CL) TVVE1 = 7.38*(WT/70) VE1 = TVVE1* EXP (IIV_VE1) TVVE0 = 16.5*(WT/70) VE0 = TVVE0*EXP (IIV_VE0) V1 = (ECMO*VE1+ (1-ECMO)*VE0) Q = 6.28*((WT/70)^0.75) V2 = 6.27*(WT/70)	IIV_CL = 0.1076 IIV_VE1 = 0.1912 IIV_VE0 = 0.3552 RV_{prop} = 0.269
Klastrup et al. 2020 [19]	Critically ill ICU patients without RRT, no ECMO support	78 (53/25) {0}	Initial Bolus followed by Continuous Infusion (CI)	Loading dose of 4/0.5 g over 3 min followed by daily continuous dosage (16,12,8 g) adjusted according to serum creatinine levels	Ultra HPLC	Unbound fraction	NA per pat., 196 in total	1 CMT	CL_{CR} (CG) → on CL	TVCL = 2.25+ (0.119* CL_{CR}) CL = TVCL*EXP (IIV_CL) V = 35.8	IIV_CL = 0.2848 RV_{prop} = 0.226
Li et al. 2005 [20]	Multicenter hospitalized patients with complicated intraabdominal infection, no septic shock, no RRT, no treatment with sympatho- mimetic agents	56 (52 for final piperacillin pop. PK model building) (41/15) {0}	Continuous Infusion (CI) or Intermittent Infusion (II)	12/1.5 g as CI over 24 hours (n=24) or 3/0.375 g as 30 min iv. infusion every 6 hours (n=28), Dose adjusting for both groups by CL_{CR} (CG)	HPLC	Total	NA per pat., 184 in total	1 CMT	CL_{CR} (CG) → on CL WT → on V	TVCL = 5.05+(9.6*(CL_{CR}/89) CL = TVCL*EXP (IIV_CL) TVV = 22.3*(WT/81.8) V = TVV*EXP (IIV_V)	IIV_CL = 0.0739 IIV_V = 0.0616 RV_{prop} = 0.185 RV_{add} = 1.77

Roberts et al. 2015 [21]	Multidrug PK study with critically ill patients undergoing different intensity rates of continuous RRT	7 receiving piperacillin (NA/NA), {7}	Intermittent Infusion (II)	4/0.5 g every 6, 8 or 12 hours as iv. infusion	LC-MS/MS	Total	NA per pat., 29 in total	1 CMT	WT → on V	CL = 3.54*EXP (IIV_CL) TVV = 18.7*(WT/82) V = TVV*EXP (IIV_V)	IIV_CL = 0.1547 IIV_V = 0.0714 RV_{prop} = 0.44
Roberts et al. 2010 [22]	Critically ill septic ICU patients with normal renal function	16 (11/5) {0}	Initial Bolus followed by Continuous Infusion (CI) or Intermittent Infusion (II)	12/1.5 g per day as CI (8) after a loading dose of 4/0.5 g over 20 min as iv. bolus or 4/0.5 g as II (8) over 20 min every 6-8 hours	HPLC-UV	Unbound fraction	27 per pat., 432 in total	2 CMT	WT → on CL	TVCL = 17.1*(WT/70) CL = TVCL*EXP (IIV_CL+IOV_CL) Q = 52*EXP (IIV_Q) V1 = 7.2*EXP (IIV_V1+IOV_V1) V2 = 17.8*EXP (IIV_V2) ALAG=0.07*EXP (IIV_ALAG) Note: Delay of absorption not available for combined external dataset. No use of ALAG for evaluation	IIV_CL = 0.0851 IIV_Q = 0.2247 IIV_V1 = 0.0674 IIV_V2 = 0.4291 IIV_ALAG = 0.1748 IOV_CL = 0.1935 IOV_V1 = 0.0578 RV_{prop} = 0.253 RV_{add} = 3.2
Selig et al. 03_2022 [23]	Synthesis of aggregated piperacillin PK data of 10 previously published studies of critically ill patients receiving continuous RRT and additional four comparable patients of a Military Health System (MHS)	138 (= sum of the different piperacillin study patients) (NA/NA) {138}	Intermittent Infusion (II) or Extended Infusion (EI) or Continuous Infusion (CI)	MHS patients received 2/0.25 – 3/0.375 g every 6 hours as a 30 min iv. infusion	HPLC for MHS patient samples	Total for MHS patient samples	NA	1 CMT	none	CL = 2.7*EXP (IIV_CL) V = 25.83*EXP (IIV_V)	IIV_CL = 0.38 IIV_V = 0.067 RV_{prop} = 0.42
Selig et al. 05_2022 [24]	Critically ill patients with non-burn-Trauma or with Burn	19 (16/3) {NA}	Intermittent Infusion (II)	3/0.375 g - 4/0.5 g every 6 hours as a 30 min iv. infusion	HPLC-UV	Total	4 per pat., 76 in total	2 CMT	CL_{CR} (CG) → on CL	TVCL = 17.56*((CL_{CR}/130)^0.65 CL = TVCL*EXP (IIV_CL) Q = 6.8 V1 = 33.59*EXP (IIV_V1) V2 = 10.5	IIV_CL = 0.17 IIV_V1 = 0.34 RV_{prop} = 0.24

Sukarn- janaset et al. 2019 [25]	Critically ill ICU patients during the early phase of sepsis, no RRT	48 (37/11) {0}	Intermittent Infusion (II)	2/0.25 g or 4/0.5 g every 6 or 8 hours as a 30 min infusion	HPLC	Unbound fraction	5 per pat., 237 in total	2 CMT	CL_{CR} (CG, Jelliffe) MAP → on CL Note: External dataset CL_{CR} calculated by CG. Missing MAP values set to 68 ABW[b] → on V1	TVCL = 5.37+ (0.06*(CL_{CR}-55))+ (0.05*(MAP-68)) CL = TVCL*EXP (IIV_CL) Q = 21.3 TVV1 = 9.35+ (0.26*(ABW-56)) V1 = TVV1*EXP (IIV_V1) V2 = 7.77	IIV_CL = 0.0781 IIV_V1 = 0.2677 RV_{prop} = 0.223
Tamme et al. 2016 [26]	Critically ill ICU patients with septic shock receiving high volume haemodia- filtration	10 (6/4) {10, 7 with complete RRT period during study}	Intermittent Infusion (II)	Single 4/0.5 g dose as 30 min iv. infusion	Ultra HPLC- MS/MS	Total	12 per pat., 101 in total	2 CMT	None	CL = 6.9*EXP (IIV_CL) Q = 10.5*EXP (IIV_Q) V1 = 9.0*EXP (IIV_V1) V2 = 11.2*EXP (IIV_V2)	IIV_CL = 0.0361 IIV_Q = 0.1369 IIV_V1 = 0.0676 IIV_V2 = 0.1156 Combined additive power RV_{const} = 1 RV_{resid} = 0.06 RV_{power} = 1.07
Udy et al. 2015 [27]	Critically ill septic patients with augmented renal clearance	48 (27/11) {0}	Intermittent Infusion (II)	4/0.5 g as a 20 min iv. infusion every 6 hours	HPLC- UV	Unbound fraction	5 per pat., 240 in total	2 CMT	Measured CL_{CR}[e] → on CL	TVCL = 16.3*(CL_{CR}/100) CL = TVCL*EXP (IIV_CL) Q = 37.3 V1 = 19.9*EXP (IIV_V1) V2 = 18.8*EXP (IIV_V2) ALAG = 0.8*EXP (IIV_ALAG) Note: Delay of absorption not available for combined external dataset. No use of ALAG for evaluation	IIV_CL = 0.2728 IIV_V1 = 0.084 IIV_V2 = 0.3764 IIV_ALAG = 0.0000089 RV_{prop} = 0.01 RV_{add} = 0.3

Wallenburg et al. 2022 [28] <u>Note: Joint PopPK model for piperacillin/ tazobactam</u>	Critically ill ICU patients, without RRT, no ECMO support	39 (26/13) {0}	Intermittent Infusion (II)	4/0.5 g as a 30 min iv. infusion every 8 hours	Ultra HPLC-MS/MS	Total	NA per pat., 325 in total	2 CMT <u>Note: Evaluation only available for central PIP Pop-PK model</u>	Measured CL_{CR}[e] → on CL FFM[d] → on CL,Q, V1,V2	$TVCL_{nonrenal}$=4.11* ((FFM/58.2)^0.75) $CL_{nonrenal}$ = $TVCL_{nonrenal}$* EXP (IIV_CL) $TVCL_{renal}$ = 0.0613*CL_{CR} CL_{renal} = $TVCL_{renal}$* EXP (IOV_CLR) CL = $CL_{nonrenal}$ + CL_{renal} Q = 3.27* ((FFM/58.2)^0.75) TVV1 = 29.7*(FFM/58.2) V1 = TVV1*EXP (IIV_V1) TVV2 = 4.26*(FFM/58.2) V2 = TVV2*EXP (IIV_V2)	IIV_CL = 0.0651 IIV_V1 = 0.177 IIV_V2 = 2.5088 IOV_CL_{renal} = 0.0946 RV_{prop} = 0.28

ICU: intensive care unit; pat.: patient; HPLC: high performance liquid chromatography; UV: ultraviolet; MS/MS: tandem mass spectrometry; CMT: compartment; CL_{CR}: creatinine clearance [mL/min]; eGFR: estimated glomerular filtration rate [ml/min]; CG: Cockroft-Gault; CKD-EPI: Chronic Kidney Disease Epidemiology Collaboration; CL: clearance [L/h]; Q: intercompartmental clearance [L/h]; RRT: renal replacement therapy [dichotomous]; WT: total body weight [kg]; ABW: adjusted body weight [kg]; IBW: ideal body weight; LBM: lean body mass [kg]; BMI: body mass index; FFM: fat free mass [kg]; TV: typical value; V1: volume of distribution (central) [L]; V2: volume of distribution (peripheral) [L]; ALAG: delay in absorption [h]; IIV: interindividual variability [ω^2]; IOV: interoccasional variability [ω^2]; RV: residual variability [$_{prop}$ %/100; $_{add}$ mg/L]; CVVHDF: continuous veno-venous haemodiafiltration; ECMO: extracorporeal membrane oxygenation [dichotomous]; NA: not available.

[a] IIV and IOV presented as variance (ω^2); Conversion of coefficient of variation (CV) [%] to variance ω^2 calculated using equation: ω^2 = ln((CV/100)^2 +1)
[b] ABW in external dataset calculated as done by Sukarnjanaset et al.: ABW=IBW+(0.4*Weight-IBW); IBW (Devine formula)
[c] LBM in external dataset calculated using Cheymol equation as done by Bulitta et al.: LBM_{male} = (1.10*WT)-(120*(WT/HT)**2); LBM_{female} = (1.07*WT)-(148*(WT/HT)**2)
[d] FFM in external dataset calculated using Holford equation as done by Wallenburg et al.: FFM_{male} = (9270*WT)/(6680+(216*BMI)); FFM_{female} = (9270*WT)/(8780+(244*BMI))
[e] Due to missing information on CL_{CR} measured by means of urine collection in our combined external dataset, we instead used CL_{CR} calculated using Cockroft-Gault formula

Table S2 Type of piperacillin (PIP) concentrations used in the 24 evaluated population pharmacokinetic models

Model prediction using total PIP concentration (PIP_{total})	Model prediction using free PIP concentration ($PIP_{unbound}$) (for evaluation = 70% of available PIP_{total})
Asin-Prieto_2013	Andersen_2018
Bulitta_2007	Bue_2020
Chen_2016	Klastrup_2020
Chung_2015	Roberts-JA_2010
Delattre_2012	Sukarnjanaset_2019
Fillâtre_2021	Udy_2015
Hahn_2021	
Hamada_2013	
Ishihara_2020	
Jeon_2014	
Kim_2016	
Kim_2022	
Li_2005	
Roberts-DM_2015	
Selig_03-2022	
Selig_05-2022	
Tamme_2016	
Wallenburg_2022	

Table S3 Comparison of the predicted and observed plasma piperacillin concentrations of the 24 evaluated population pharmacokinetic models for piperacillin, based on the total dataset stratified by study.
A priori: Prediction of all concentrations based on dosing history and patient covariates only;
Bayesian 1: Prediction of concentrations considering the first therapeutic drug monitoring (TDM) sample for each patient;
Bayesian 2: Prediction of concentrations considering the first and second TDM sample for each patient;
MPE: Median relative prediction error (accuracy); MAPE: Median absolute relative prediction error (precision); II: intermittent infusion; CI: continuous infusion; NA: not available;
Values presented as median [%] (25% Quantile to 75% Quantile)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Andersen_2018	Total (Multicenter)	24.92 (-17.82 to 55.6)	43.51 (22.6 to 71.34)	18.6 (-10.91 to 47.76)	34.61 (15.26 to 59.55)	22.83 (-8.52 to 50.93)	36.92 (17.37 to 62.01)
Andersen_2018	1 (Target)	22.01 (-14.88 to 53.79)	37.54 (19.42 to 63.96)	11.4 (-8.8 to 37.57)	24 (9.73 to 47.25)	16.64 (-3.58 to 40.9)	24.96 (10.74 to 46.73)
Andersen_2018	2 (Heidenheim)	28.11 (-1.52 to 54.77)	35.23 (17.15 to 57.85)	6.84 (-14.47 to 29.28)	23.51 (10.44 to 41.59)	6.73 (-12.21 to 35.73)	22.33 (9.96 to 40.21)
Andersen_2018	3 (Regensburg)	25.96 (-35.85 to 98.39)	68.93 (38.37 to 98.39)	20.99 (-4.72 to 52.22)	21.17 (12.4 to 52.22)	NA	NA
Andersen_2018	4 (Munich) (II)	25.18 (-34.08 to 56.59)	51.15 (27.67 to 80.66)	27.66 (-10.92 to 55.06)	44.12 (22.53 to 67.44)	27.37 (-11.5 to 55.61)	43.95 (22.59 to 67.49)
Andersen_2018	1-3 (CI)	24.82 (-8.75 to 54.6)	37.26 (18.58 to 61.68)	9.46 (-10.83 to 34.95)	23.79 (10.02 to 45.13)	15.12 (-5.81 to 39.85)	24.5 (10.53 to 45.28)
Asin-Prieto_2013	Total (Multicenter)	-135.55 (-160.31 to -102.73)	135.59 (102.93 to 160.33)	-77.34 (-101.33 to -53.26)	77.68 (54.53 to 101.55)	-55.77 (-79.19 to -31.02)	57.41 (34.71 to 80.33)
Asin-Prieto_2013	1 (Target)	-137.5 (-153.32 to -116.23)	137.5 (116.36 to 153.32)	-79.5 (-99.12 to -54.82)	80.05 (56.75 to 99.42)	-50.23 (-71.72 to -26.88)	51.35 (31.91 to 72.51)
Asin-Prieto_2013	2 (Heidenheim)	-129.27 (-146.49 to -108.92)	129.27 (108.92 to 146.49)	-76.2 (-98.31 to -54.65)	76.2 (55.25 to 98.31)	-55.64 (-73.47 to -31.56)	55.64 (33.37 to 73.47)
Asin-Prieto_2013	3 (Regensburg)	-133.18 (-155.76 to -71.19)	133.18 (75.56 to 155.76)	-75.99 (-95.24 to -42.77)	75.99 (49.6 to 95.24)	NA	NA
Asin-Prieto_2013	4 (Munich) (II)	-138.47 (-175.48 to -90.28)	138.56 (90.61 to 175.55)	-77.11 (-103.77 to -52.84)	77.38 (53.48 to 104.28)	-58.15 (-83.32 to -31.98)	59.92 (36.22 to 84.13)
Asin-Prieto_2013	1-3 (CI)	-134.18 (-150.66 to -113.05)	134.18 (113.12 to 150.66)	-77.94 (-98.73 to -54.67)	78.49 (55.84 to 98.9)	-51.45 (-72.11 to -27.52)	52.43 (32.29 to 72.6)
Bue_2020	Total (Multicenter)	78.25 (39.04 to 119.25)	79.75 (44.94 to 119.77)	72.28 (32.83 to 112.71)	73.87 (39.99 to 113.34)	65.33 (28.67 to 106.88)	67.25 (35.97 to 107.38)
Bue_2020	1 (Target)	92.87 (46.15 to 127.27)	93.71 (52.28 to 127.6)	89.83 (45.85 to 124.93)	90.2 (51.45 to 124.93)	86.66 (50.71 to 121.54)	87.12 (51.61 to 121.54)
Bue_2020	2 (Heidenheim)	83.6 (46.29 to 119.14)	83.92 (47.53 to 119.23)	81.91 (43.91 to 114.33)	83 (45.09 to 114.37)	78.06 (36.84 to 105.66)	78.06 (37.07 to 105.66)
Bue_2020	3 (Regensburg)	93.32 (27.04 to 127.18)	93.32 (47.61 to 127.18)	82.18 (39.6 to 128.11)	82.18 (45.01 to 128.11)	NA	NA
Bue_2020	4 (Munich) (II)	69.41 (33.22 to 108.09)	71.9 (40.91 to 108.91)	62.87 (25.6 to 100.23)	65.58 (34.95 to 101.18)	56.63 (21.28 to 95.45)	60.64 (32.18 to 97.19)
Bue_2020	1-3 (CI)	89.56 (46.11 to 123.47)	90.69 (50.28 to 123.53)	87.73 (44.98 to 121.06)	88.33 (48.86 to 121.29)	85.24 (47.81 to 118.8)	85.31 (49.11 to 118.8)
Bulitta_2007	Total (Multicenter)	-62.97 (-133.56 to -4.38)	67.55 (28.31 to 134.28)	-28.06 (-63.67 to 5.69)	39.79 (18.29 to 68.81)	-17.65 (-57.25 to 11.41)	34.28 (15.04 to 65.79)
Bulitta_2007	1 (Target)	-53.43 (-97.15 to -5.21)	59.14 (30.16 to 98.3)	-10.56 (-39.88 to 18.52)	30.24 (14.77 to 56.25)	3.6 (-20.77 to 29.32)	25.58 (11.44 to 47.68)
Bulitta_2007	2 (Heidenheim)	-64.81 (-98.1 to -20.85)	65.9 (27.24 to 98.1)	-19.15 (-47.6 to 6.4)	32.51 (14.51 to 50.82)	-8.46 (-33.35 to 17.95)	27.69 (13.15 to 43.98)
Bulitta_2007	3 (Regensburg)	-56.14 (-108.67 to -5.6)	73.67 (34.62 to 112.78)	-16.59 (-43.97 to 34.04)	42.52 (27.88 to 53.29)	NA	NA
Bulitta_2007	4 (Munich) (II)	-79.8 (-189.35 to 1.93)	84.36 (27.2 to 189.41)	-41.36 (-78.34 to -5.81)	47.27 (23.01 to 82)	-29.37 (-75.16 to 2.66)	39.12 (17.11 to 80.02)
Bulitta_2007	1-3 (CI)	-55.96 (-97.53 to -10.05)	61.6 (29.26 to 98.5)	-13.98 (-43.08 to 15.42)	30.95 (14.7 to 55.1)	1.02 (-24.55 to 27.48)	26.2 (11.53 to 46.54)
Chen_2016	Total (Multicenter)	-90.23 (-123.3 to -54.99)	90.74 (56.97 to 123.87)	-17.16 (-48.18 to 15.24)	34.05 (16.49 to 60.5)	9.73 (-20.38 to 37.7)	30.82 (14.37 to 54.26)
Chen_2016	1 (Target)	-88.15 (-119.85 to -54.3)	88.74 (56.89 to 120.35)	3.92 (-20.58 to 31.62)	26.01 (13.04 to 50.48)	16.67 (-9.53 to 43.47)	27.91 (14.37 to 51.31)
Chen_2016	2 (Heidenheim)	-85.18 (-107.42 to -54.3)	85.18 (54.91 to 107.42)	-2.09 (-25.88 to 22.24)	24.8 (11.14 to 43.87)	3.06 (-16.17 to 33.49)	23.24 (11.44 to 39.5)
Chen_2016	3 (Regensburg)	-74.4 (-121.7 to -20.85)	104.7 (37.34 to 124.07)	-0.35 (-20.14 to 26.82)	22.58 (17.84 to 31.88)	NA	NA
Chen_2016	4 (Munich) (II)	-94.12 (-135.86 to -56.35)	94.82 (58.41 to 136.23)	-32.79 (-64.5 to -4.96)	42.19 (21.25 to 70.45)	7.68 (-25.51 to 35.58)	32.47 (14.55 to 56.97)
Chen_2016	1-3 (CI)	-86.98 (-115.29 to -54.11)	87.19 (55.83 to 115.78)	1.37 (-22.37 to 28.3)	25.55 (12.21 to 48.22)	14.42 (-11.66 to 41.1)	27.36 (13.39 to 48.95)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Chung_2015	Total (Multicenter)	-33.18 (-68.61 to 4.99)	46.43 (22.91 to 76.77)	-15.63 (-46.59 to 15.22)	33.42 (15.37 to 57.07)	-3.38 (-33.1 to 23.9)	28.01 (12.73 to 51.85)
Chung_2015	1 (Target)	-27.02 (-64.67 to 6.75)	41.84 (18.26 to 71.47)	-2.81 (-29.38 to 24.69)	27.54 (11.86 to 47.49)	8.25 (-17.79 to 32.44)	24.79 (11.57 to 40.77)
Chung_2015	2 (Heidenheim)	-24.32 (-55.52 to 5.59)	34.99 (17.24 to 58.28)	-6.24 (-27.71 to 15.81)	22.09 (10.93 to 39.06)	-2.3 (-21.57 to 19.77)	21.35 (10.06 to 41.11)
Chung_2015	3 (Regensburg)	-14.85 (-87.22 to 35.95)	73.2 (33.22 to 93.51)	-1.63 (-32.13 to 59.17)	40.68 (10.08 to 59.44)	NA	NA
Chung_2015	4 (Munich) (II)	-40.27 (-77.44 to 3.21)	54.61 (29.07 to 87.16)	-27.16 (-56.3 to 7.37)	39.84 (19.68 to 63.45)	-7.87 (-42.29 to 19.65)	30.12 (13.8 to 57.95)
Chung_2015	1-3 (CI)	-25.53 (-60.63 to 6.6)	39.59 (17.77 to 66.77)	-4.34 (-28.87 to 21.15)	25.27 (11.4 to 45.29)	5.33 (-18.86 to 29.59)	24.43 (11.11 to 40.82)
Delattre_2012	Total (Multicenter)	-19.26 (-56.43 to 17.09)	41.4 (18.26 to 70.73)	-13.98 (-37.93 to 12.64)	29.32 (13.49 to 49.96)	-6.16 (-26.83 to 19.96)	24.21 (11.37 to 42.49)
Delattre_2012	1 (Target)	-16.7 (-51.15 to 15.68)	36.48 (16.39 to 62.84)	0.49 (-18.87 to 27.4)	22.11 (10.27 to 43.93)	7.43 (-12.46 to 30.98)	22.63 (10.17 to 41.11)
Delattre_2012	2 (Heidenheim)	-11.34 (-40.54 to 17.47)	30.35 (13.24 to 49.49)	-2.14 (-23.94 to 20)	21.49 (10.02 to 38.83)	-2.23 (-20.8 to 24.22)	22.32 (11.47 to 36.99)
Delattre_2012	3 (Regensburg)	-11.05 (-76.23 to 41.76)	72.49 (32.24 to 87.26)	6.82 (-21.45 to 52.26)	26.67 (11.38 to 52.26)	NA	NA
Delattre_2012	4 (Munich) (II)	-26.97 (-64.83 to 17.3)	50.62 (23.4 to 83.6)	-25.3 (-47.45 to 0.25)	34.24 (17.78 to 54.72)	-11.81 (-31.05 to 11.78)	24.94 (11.8 to 43.83)
Delattre_2012	1-3 (CI)	-13.87 (-46.91 to 17.06)	33.45 (15.18 to 58.77)	-0.85 (-20.52 to 24.64)	21.93 (10.18 to 42.32)	5.83 (-14.93 to 29.65)	22.53 (10.38 to 40.21)
Fillatre_2021	Total (Multicenter)	12.33 (-31.19 to 67.24)	47.33 (22.47 to 83.16)	6.21 (-17.67 to 36.52)	26.91 (11.64 to 51)	13.01 (-11.05 to 41.67)	26.87 (12.17 to 51.2)
Fillatre_2021	1 (Target)	33.66 (-18.2 to 78.62)	55.29 (27.92 to 90.41)	21.04 (-3.85 to 47.1)	30.48 (13.78 to 54.37)	25.25 (0.82 to 56.44)	31.58 (15.44 to 60.44)
Fillatre_2021	2 (Heidenheim)	18.81 (-23.21 to 66.49)	45.9 (21.62 to 75.8)	13.29 (-11.67 to 40.47)	26.43 (12.31 to 48.19)	16.03 (-4.89 to 46.66)	24.88 (9.48 to 48.4)
Fillatre_2021	3 (Regensburg)	36.66 (-49.11 to 73.46)	60.09 (44.22 to 96.3)	8.74 (-6.08 to 40.49)	11.17 (6.64 to 40.49)	NA	NA
Fillatre_2021	4 (Munich) (II)	-1.47 (-37.32 to 55.87)	44.11 (20.26 to 83.19)	-2.49 (-24.3 to 27.51)	25.59 (11.04 to 50.51)	7.31 (-15.53 to 34.72)	25.55 (11.42 to 48.54)
Fillatre_2021	1-3 (CI)	28.25 (-21.11 to 73.12)	51.5 (25.12 to 83.16)	17.35 (-6.55 to 45.24)	28.58 (12.65 to 51.84)	23.86 (-1 to 52.28)	30.46 (13.72 to 56.83)
Hahn_2021	Total (Multicenter)	-48.91 (-86.09 to -11.78)	53.03 (25.3 to 88.07)	-24.6 (-50.69 to 3.79)	33.16 (16.21 to 55.36)	-18.83 (-45.71 to 6.51)	29.33 (13.73 to 50.64)
Hahn_2021	1 (Target)	-51.51 (-84.19 to -20.14)	54.67 (29.18 to 85.73)	-22.65 (-44.66 to 3.56)	31.49 (16.08 to 52.69)	-10.06 (-32.27 to 15.27)	24.83 (11.35 to 43.52)
Hahn_2021	2 (Heidenheim)	-45.51 (-72.03 to -18.29)	47.96 (23.5 to 73.07)	-23.43 (-43.82 to -2.18)	28.63 (12.89 to 47.81)	-17.29 (-35.78 to 6.3)	25.38 (14.03 to 43.18)
Hahn_2021	3 (Regensburg)	-46.02 (-101.08 to 37.56)	79.23 (41.52 to 104.88)	-15.98 (-47.37 to 49.2)	48.71 (23.11 to 67)	NA	NA
Hahn_2021	4 (Munich) (II)	-48.32 (-96.82 to -3.7)	54.78 (23.71 to 99.79)	-27.08 (-54.78 to 5.48)	35.74 (17.31 to 58.48)	-23.62 (-50.42 to 2.99)	31.8 (14.94 to 55.47)
Hahn_2021	1-3 (CI)	-49.18 (-79.14 to -18.8)	51.67 (26.81 to 80.58)	-22.77 (-44.63 to 2.12)	29.97 (15.1 to 50.88)	-11.31 (-33.23 to 13.21)	25.1 (11.7 to 43.61)
Hamada_2013	Total (Multicenter)	-24.02 (-71.81 to 11.56)	41.84 (18.06 to 79.38)	6.25 (-22.34 to 34.87)	29.32 (13.71 to 52.5)	12.45 (-17.74 to 39.8)	30.89 (14.61 to 53.99)
Hamada_2013	1 (Target)	-20.78 (-58.15 to 13.11)	40.03 (17.33 to 69.33)	5.2 (-17.83 to 30.99)	24.1 (11 to 47.02)	14.44 (-8.89 to 36.85)	25.23 (11.84 to 45.43)
Hamada_2013	2 (Heidenheim)	-20.28 (-52.25 to 9.9)	35.55 (16.1 to 57.04)	0.24 (-20.13 to 19.89)	20.12 (10.82 to 40.88)	3.57 (-16.15 to 30.23)	22.48 (9.19 to 40.64)
Hamada_2013	3 (Regensburg)	-8.61 (-78.81 to 40.54)	74.44 (36.21 to 90.11)	11.44 (-17.49 to 47.33)	23.04 (13.71 to 47.33)	NA	NA
Hamada_2013	4 (Munich) (II)	-31.05 (-112.5 to 11.65)	47.18 (19.4 to 115.27)	9.89 (-26.37 to 39.69)	34.74 (16.5 to 59.52)	12.55 (-23.12 to 41.03)	34.74 (16.69 to 60.58)
Hamada_2013	1-3 (CI)	-20.5 (-56.03 to 11.51)	38.22 (16.91 to 64.39)	3.24 (-18.88 to 27.67)	23.33 (10.91 to 44.78)	12.26 (-10.05 to 36.34)	24.69 (11.37 to 44.2)
Ishihara_2020	Total (Multicenter)	4.13 (-42.77 to 37.05)	38.82 (17.72 to 67.33)	8.95 (-24.57 to 37.6)	33.44 (14.74 to 57.41)	6.52 (-23.77 to 32.29)	28.87 (12.66 to 54.91)
Ishihara_2020	1 (Target)	-0.21 (-36.67 to 32.85)	34.76 (15.07 to 61.34)	10.09 (-14.24 to 39.33)	28.58 (12.22 to 51.05)	17.25 (-6.05 to 43.54)	28.17 (12.61 to 50.03)
Ishihara_2020	2 (Heidenheim)	5.39 (-26.52 to 33.3)	29.89 (13.5 to 49.85)	9.81 (-13.04 to 33.74)	25.09 (10.63 to 43.14)	9.81 (-10.51 to 32.43)	23.95 (9.86 to 41.64)
Ishihara_2020	3 (Regensburg)	6.21 (-56.5 to 76.35)	62.68 (45.27 to 95.31)	16.22 (-13.95 to 72.21)	23.22 (17.77 to 72.21)	NA	NA
Ishihara_2020	4 (Munich) (II)	5.93 (-67.72 to 40.65)	46.58 (22.02 to 83.31)	6.92 (-37.99 to 38.67)	38.6 (18.56 to 64.75)	1.52 (-34.43 to 26.99)	29.83 (12.91 to 58.31)
Ishihara_2020	1-3 (CI)	2.57 (-31.71 to 33.4)	32.52 (14.72 to 56.51)	10.02 (-13.82 to 36.64)	27.35 (11.82 to 48.29)	15.48 (-7.19 to 40.31)	27.3 (12.13 to 48.28)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Jeon_2014	Total (Multicenter)	-26.08 (-63.78 to 15.34)	44.97 (20.95 to 75.8)	-7.66 (-36.74 to 22.35)	30.18 (14.33 to 55.17)	7.9 (-15.78 to 35.77)	26.53 (11.32 to 47.83)
Jeon_2014	1 (Target)	-26.56 (-58.32 to 7.84)	39.58 (18.89 to 66.37)	2.56 (-16.13 to 30.5)	22.52 (9.75 to 45.31)	9.73 (-9.71 to 37.26)	23.72 (9.72 to 43.56)
Jeon_2014	2 (Heidenheim)	-13.78 (-43.75 to 16.63)	33.23 (15.7 to 53.51)	4.33 (-19.05 to 30.71)	25.35 (13.02 to 43.63)	4.99 (-13.63 to 28.49)	22.63 (8.37 to 40.39)
Jeon_2014	3 (Regensburg)	-31.06 (-63.13 to 60.9)	61.37 (54.18 to 93.52)	9.26 (-17.61 to 42.96)	22.18 (15.1 to 42.96)	NA	NA
Jeon_2014	4 (Munich) (II)	-33.28 (-73.99 to 18.31)	56.63 (26.88 to 91.34)	-20.49 (-50.4 to 13.55)	35.86 (17.56 to 62.01)	7.88 (-19.33 to 35.91)	27.61 (12.31 to 49.98)
Jeon_2014	1-3 (CI)	-21.18 (-52.98 to 12.89)	36.67 (17.54 to 61.19)	3.55 (-17.57 to 30.62)	23.6 (10.49 to 44.81)	8.06 (-10.24 to 35.58)	23.47 (9.58 to 42.29)
Kim_2016	Total (Multicenter)	-32.03 (-66.52 to 3.13)	44.11 (22.04 to 74.4)	-19.16 (-44.15 to 6.77)	30.26 (14.37 to 50.96)	-9.94 (-33.51 to 14.74)	25.39 (11.89 to 44.51)
Kim_2016	1 (Target)	-29.96 (-65.62 to 3.1)	40.89 (20.04 to 71.38)	-7.49 (-28.23 to 17.68)	24.48 (11.81 to 43.98)	0.32 (-20.41 to 23.77)	21.56 (11.15 to 39.23)
Kim_2016	2 (Heidenheim)	-24.47 (-53.94 to 3.1)	32.88 (17.08 to 57.61)	-11.49 (-30.34 to 10.95)	23.57 (11.49 to 41.1)	-10.13 (-27.68 to 17.59)	23.99 (13.87 to 35.43)
Kim_2016	3 (Regensburg)	-24.72 (-83.19 to 40.45)	76.61 (40.49 to 96.51)	-0.04 (-29.46 to 46.73)	34.11 (8.13 to 47.61)	NA	NA
Kim_2016	4 (Munich) (II)	-37.14 (-73.16 to 2.27)	50.42 (26.24 to 82.69)	-28.73 (-52.24 to -2.61)	36.01 (17.48 to 56.4)	-14.39 (-39.46 to 9.31)	27.19 (12.15 to 47.15)
Kim_2016	1-3 (CI)	-27.43 (-59.8 to 3.45)	38.11 (18.4 to 66.01)	-8.57 (-29.09 to 14.65)	24.32 (11.52 to 43.05)	-1.21 (-22.05 to 22.23)	22.22 (11.38 to 38.78)
Kim_2022	Total (Multicenter)	-9.83 (-45.5 to 27.32)	37 (17.18 to 66.72)	-5.85 (-31.8 to 21.85)	27.28 (12.69 to 47.52)	-0.92 (-24.81 to 22.68)	23.71 (10.93 to 43.36)
Kim_2022	1 (Target)	-8.43 (-49.51 to 26.71)	39.32 (17.18 to 66.81)	3.29 (-21.55 to 30.82)	26.62 (12.25 to 47.28)	11.23 (-13.77 to 34.2)	24.68 (12.07 to 43.81)
Kim_2022	2 (Heidenheim)	-1.05 (-30.05 to 29.27)	29.85 (14.26 to 53.52)	1.92 (-20.86 to 25.05)	23.03 (10.13 to 39.88)	1.39 (-19.09 to 24.99)	22.81 (10.09 to 40)
Kim_2022	3 (Regensburg)	5.63 (-59.47 to 61.98)	59.87 (48.02 to 81.54)	11.26 (-21.96 to 57.2)	32.27 (18.8 to 57.2)	NA	NA
Kim_2022	4 (Munich) (II)	-14.41 (-48.45 to 24.82)	39.44 (18.63 to 75.37)	-12.83 (-38.14 to 15.47)	28.68 (13.82 to 48.83)	-5.37 (-30.02 to 16.38)	23.5 (10.59 to 43.87)
Kim_2022	1-3 (CI)	-5.35 (-41.5 to 28.78)	35.33 (16.03 to 60.75)	2.84 (-21.3 to 28.84)	25.19 (11.41 to 45.37)	9.57 (-14.6 to 33.14)	24.39 (11.54 to 42.38)
Klastrup_2020	Total (Multicenter)	-12.98 (-48.65 to 22.35)	37.51 (17.13 to 66.78)	-7.71 (-34.31 to 19.9)	28.04 (12.59 to 51.88)	6.65 (-20.27 to 36.62)	28.37 (12.46 to 51.23)
Klastrup_2020	1 (Target)	-11.84 (-45.82 to 21.41)	33.98 (15.9 to 61.1)	3.32 (-15.64 to 29.59)	20.87 (9.75 to 43.48)	10.64 (-8.34 to 34.2)	22.05 (9.4 to 41.85)
Klastrup_2020	2 (Heidenheim)	-2.8 (-31.84 to 25.04)	28.14 (13.94 to 48.22)	-0.88 (-22.25 to 21.87)	22.21 (10.7 to 40.76)	0.29 (-18.95 to 28.2)	22.14 (11.68 to 36.99)
Klastrup_2020	3 (Regensburg)	-10.52 (-67.75 to 68.25)	69 (42.23 to 86.41)	2.2 (-19.17 to 27.36)	21.65 (16.23 to 33.59)	NA	NA
Klastrup_2020	4 (Munich) (II)	-18.64 (-59.11 to 20.88)	43.34 (19.79 to 78.87)	-17.9 (-45.35 to 12.5)	32.04 (16.22 to 57.71)	5.47 (-25.11 to 39.6)	31.51 (14.1 to 55.84)
Klastrup_2020	1-3 (CI)	-8.3 (-40 to 23.2)	32.18 (15.29 to 56.02)	1.52 (-18.39 to 27.81)	21.71 (9.97 to 42.69)	8.96 (-11.23 to 32.91)	22.08 (10.33 to 40.45)
Li_2005	Total (Multicenter)	-64.85 (-105.1 to -32.54)	66.66 (36.66 to 105.68)	-19.27 (-45.77 to 6.87)	30.47 (14.18 to 53.06)	-5.97 (-34.08 to 20.85)	26.84 (12.3 to 49.99)
Li_2005	1 (Target)	-62.95 (-95.4 to -30.69)	65.45 (38.01 to 96.12)	-8.67 (-29.37 to 17.29)	25.41 (11.59 to 44.59)	4.33 (-17.2 to 27.29)	22.22 (11.06 to 39.81)
Li_2005	2 (Heidenheim)	-59.18 (-86.57 to -31.98)	59.88 (33.11 to 86.71)	-13.47 (-32.89 to 9.11)	24.49 (11.21 to 41.4)	-6.76 (-25.41 to 20.8)	23.94 (13.51 to 38.15)
Li_2005	3 (Regensburg)	-55.19 (-110.91 to 10.08)	78.93 (36.37 to 113.2)	-0.72 (-28.77 to 26.2)	28.59 (12.78 to 38.35)	NA	NA
Li_2005	4 (Munich) (II)	-70.1 (-133.73 to -33.88)	71.6 (37.03 to 134.75)	-27.84 (-55.48 to -0.08)	36.03 (16.82 to 59.79)	-9.5 (-44.27 to 16.96)	30.09 (12.66 to 55.45)
Li_2005	1-3 (CI)	-60.96 (-90.98 to -30.77)	62.58 (36.47 to 91.41)	-10.23 (-31.03 to 13.93)	25.08 (11.41 to 43.36)	2.55 (-19.21 to 25.92)	22.61 (11.33 to 39.34)
Roberts-DM_2015	Total (Multicenter)	42.26 (-3.09 to 96.32)	56.74 (23.85 to 100.44)	23.49 (-13.27 to 60.28)	39.9 (19.06 to 69.19)	19.52 (-14.56 to 49.92)	34.59 (17.42 to 60.74)
Roberts-DM_2015	1 (Target)	66.18 (16.52 to 105.86)	70.82 (37.63 to 107.41)	48.57 (12.89 to 86.04)	53.95 (29.7 to 90.27)	42.83 (15.23 to 73.13)	47.52 (25.09 to 75.68)
Roberts-DM_2015	2 (Heidenheim)	52.51 (11.41 to 95.42)	57.69 (26.85 to 96.08)	38.57 (5.53 to 73.54)	44.79 (20.39 to 74.76)	28.13 (2.36 to 54.11)	34.91 (15.18 to 60.68)
Roberts-DM_2015	3 (Regensburg)	71.23 (-15.3 to 101.42)	81.26 (32.5 to 111.89)	43.69 (10.02 to 94.61)	57.82 (25.89 to 94.61)	NA	NA
Roberts-DM_2015	4 (Munich) (II)	23.11 (-13.2 to 86.92)	44.12 (18.75 to 95.93)	7.82 (-24.07 to 40.1)	32.1 (15.8 to 58.59)	10.35 (-21.72 to 38.56)	30.44 (15.49 to 53.93)
Roberts-DM_2015	1-3 (CI)	61.15 (13.59 to 101.23)	66.67 (33.04 to 103.32)	45.92 (10.46 to 81.27)	50.63 (26.64 to 83.63)	39.12 (9.57 to 68.97)	45.19 (23 to 71.42)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Roberts-JA_2010	Total (Multicenter)	-66.97 (-125.51 to -15.09)	71.73 (29.56 to 126.18)	-22.34 (-70.31 to 9.9)	37.67 (16.18 to 78.97)	-18.48 (-64.97 to 12.1)	35.61 (15.39 to 79.56)
Roberts-JA_2010	1 (Target)	-63.75 (-111.41 to -17.56)	69.8 (34.09 to 112.15)	1.45 (-23.52 to 29.7)	26.92 (12.91 to 51.31)	12.83 (-11.27 to 39.57)	26.48 (12.22 to 47.93)
Roberts-JA_2010	2 (Heidenheim)	-73.53 (-108.22 to -28.79)	73.65 (33.08 to 108.22)	-48.96 (-80.83 to -15.25)	52.33 (25.44 to 80.83)	-5.36 (-23.46 to 22.1)	22.61 (11.99 to 37.47)
Roberts-JA_2010	3 (Regensburg)	-66.81 (-119.11 to -18.31)	75.33 (37.39 to 123.1)	-4.04 (-24.33 to 9.75)	21.78 (6.26 to 36.67)	NA	NA
Roberts-JA_2010	4 (Munich) (II)	-65.37 (-162.1 to -8.83)	72.78 (25.53 to 163.28)	-32.12 (-92.09 to 2.6)	43.11 (17.18 to 94.97)	-36.73 (-94.06 to -3.22)	44.54 (17.99 to 98.24)
Roberts-JA_2010	1-3 (CI)	-68.05 (-109.21 to -23.54)	71.14 (33.65 to 109.52)	-12.27 (-48.63 to 19.81)	33.15 (15.41 to 61.17)	8.89 (-16.12 to 35.63)	25.4 (12.2 to 46.5)
Selig_03-2022	Total (Multicenter)	64.71 (18.92 to 115.29)	67.72 (31 to 115.48)	12.89 (-20.92 to 51.7)	36.23 (17.09 to 63.48)	15.07 (-14.85 to 43.47)	30.31 (14.89 to 54.14)
Selig_03-2022	1 (Target)	88.5 (43.79 to 123.17)	89.38 (49.43 to 123.24)	43.97 (14.23 to 74.34)	47.48 (24.44 to 76.77)	28.76 (5.4 to 54.27)	34.01 (17.5 to 57.14)
Selig_03-2022	2 (Heidenheim)	75.39 (37.07 to 113.49)	75.97 (40.01 to 113.63)	33.18 (5.32 to 59.7)	38.44 (17.96 to 61.97)	17.64 (-5.9 to 44.24)	26.66 (13.95 to 50.31)
Selig_03-2022	3 (Regensburg)	85.5 (10.81 to 120.07)	93.25 (32.6 to 120.07)	32.39 (10.28 to 88.34)	37.49 (25.44 to 88.34)	NA	NA
Selig_03-2022	4 (Munich) (II)	41.89 (6.76 to 102.9)	50.24 (22.76 to 104.32)	-6.35 (-33.89 to 25.66)	30.93 (14.44 to 56.36)	6.87 (-21.32 to 37.87)	29.28 (14.15 to 53.1)
Selig_03-2022	1-3 (CI)	83.65 (39.94 to 119.78)	84.7 (44.74 to 119.91)	39.65 (10.35 to 69.32)	43.75 (21.82 to 70.59)	26.74 (3 to 52.73)	32.36 (16.47 to 55.09)
Selig_05-2022	Total (Multicenter)	-53.36 (-85 to -19.8)	60.01 (33.17 to 89)	-22.03 (-49.61 to 6.2)	34.54 (16.76 to 57.06)	-8.85 (-31.64 to 16.51)	25.89 (11.91 to 45.2)
Selig_05-2022	1 (Target)	-52.88 (-83.8 to -21.31)	55.81 (30.13 to 85.67)	-4.75 (-23.52 to 21.67)	23.23 (10.74 to 42.63)	5.95 (-14.07 to 29.1)	22.55 (10.52 to 39.19)
Selig_05-2022	2 (Heidenheim)	-47.07 (-72.63 to -19.53)	48.32 (24.61 to 74.82)	-7.32 (-29.27 to 15.81)	22.94 (10.44 to 40.83)	-3.59 (-21.83 to 21.95)	22.41 (10.88 to 36.67)
Selig_05-2022	3 (Regensburg)	-47.59 (-104.64 to 16.78)	83.81 (37.24 to 108.34)	2.78 (-24.42 to 18.33)	25.41 (10.03 to 39.58)	NA	NA
Selig_05-2022	4 (Munich) (II)	-58.28 (-91.35 to -18.51)	66.58 (39.61 to 98.18)	-37.06 (-60.59 to -9.15)	43.27 (23.97 to 66.37)	-16.26 (-36.9 to 8.05)	28.1 (13.06 to 47.54)
Selig_05-2022	1-3 (CI)	-49.25 (-79.48 to -20.49)	52.71 (27.6 to 80.42)	-6.24 (-25.36 to 19.12)	23.21 (10.69 to 42.08)	4.54 (-16.5 to 28.27)	22.53 (10.61 to 38.93)
Sukarnjanaset_2019	Total (Multicenter)	35.24 (1.94 to 68.03)	44.82 (22.54 to 75.21)	21.57 (-6.13 to 46.16)	32.89 (16.21 to 53.87)	14.86 (-9.16 to 37.4)	27.47 (12.78 to 48.58)
Sukarnjanaset_2019	1 (Target)	31.3 (-6.01 to 59.91)	41.02 (21.8 to 68.18)	19.38 (-4.16 to 45.86)	30.26 (13.22 to 53.27)	20.22 (-2.09 to 42.32)	27.27 (12.33 to 49.06)
Sukarnjanaset_2019	2 (Heidenheim)	45.42 (15.01 to 70.62)	46.4 (22.5 to 71.53)	23.21 (3.44 to 44.83)	31.02 (15.83 to 48)	13.9 (-3.56 to 38.76)	24.62 (11.08 to 42.58)
Sukarnjanaset_2019	3 (Regensburg)	43.57 (-26.3 to 98.53)	61.67 (33.09 to 98.53)	22.86 (-6.16 to 88.7)	25.78 (16.98 to 88.7)	NA	NA
Sukarnjanaset_2019	4 (Munich) (II)	34.23 (-0.91 to 70.57)	45.62 (23.04 to 82.45)	22.09 (-10.58 to 46.52)	34.91 (17.7 to 55.54)	12.63 (-14.8 to 35.08)	27.96 (13.4 to 48.67)
Sukarnjanaset_2019	1-3 (CI)	37 (4.29 to 66.67)	43.61 (22.2 to 70.11)	21.02 (-2.47 to 45.84)	30.59 (14.37 to 50.92)	18.81 (-2.74 to 41.71)	26.63 (12.04 to 47.68)
Tamme_2016	Total (Multicenter)	-3.44 (-58.41 to 40.62)	48.99 (23.31 to 87.29)	9.05 (-17.53 to 36.25)	27.61 (12.91 to 49.65)	7.44 (-17 to 34.1)	25.19 (11.77 to 48.63)
Tamme_2016	1 (Target)	1.75 (-49.62 to 49.43)	49.48 (22.37 to 84.59)	16.09 (-9.39 to 43.5)	30.37 (14.06 to 51.53)	24.04 (-0.86 to 49.74)	30.66 (14.91 to 53.52)
Tamme_2016	2 (Heidenheim)	-12.95 (-53.95 to 36.69)	46.76 (23.89 to 69.72)	9 (-16.02 to 32.39)	25.55 (11.99 to 45.89)	14.16 (-10.25 to 43.84)	25.25 (11.06 to 48.21)
Tamme_2016	3 (Regensburg)	7.82 (-77.75 to 44.32)	64.5 (29.72 to 89.19)	18.28 (-7.94 to 44.25)	24.17 (12.78 to 44.25)	NA	NA
Tamme_2016	4 (Munich) (II)	-3.07 (-69.5 to 37.62)	49.62 (23.51 to 99.51)	5.39 (-21.16 to 33.33)	27.35 (12.7 to 50.12)	0.31 (-22.2 to 25.24)	23.6 (10.85 to 46.85)
Tamme_2016	1-3 (CI)	-3.6 (-52.42 to 43.78)	48.88 (23.22 to 78.19)	13.98 (-12.1 to 39.73)	28.06 (13.22 to 48.77)	21.64 (-3.14 to 48.39)	29.82 (14.1 to 51.66)
Udy_2015	Total (Multicenter)	-10.69 (-44.85 to 24.85)	35.73 (17.32 to 64.33)	-3.61 (-27.82 to 23.99)	26.18 (12.07 to 49.84)	7.5 (-15.09 to 33.11)	23.8 (10.85 to 47.42)
Udy_2015	1 (Target)	-16.1 (-48.86 to 18.02)	35.14 (16.99 to 61.69)	9.77 (-12.44 to 35.05)	22.81 (10.92 to 49.68)	15.22 (-6.05 to 42.07)	26.48 (11.32 to 49.54)
Udy_2015	2 (Heidenheim)	-0.88 (-33.06 to 28.91)	31.16 (15.25 to 52.37)	4.92 (-22.21 to 31.89)	27.76 (11.64 to 49.95)	8.32 (-10.42 to 30.93)	20.32 (9.05 to 42.41)
Udy_2015	3 (Regensburg)	-17.13 (-53.32 to 67.26)	64.53 (47.88 to 88.21)	8.18 (-7.85 to 36.48)	21.38 (8.59 to 41.97)	NA	NA
Udy_2015	4 (Munich) (II)	-12.28 (-47.84 to 25.57)	37.58 (18.29 to 72.46)	-11.69 (-34.36 to 15.21)	27.42 (12.76 to 49.99)	3.47 (-18.77 to 28.02)	23.22 (10.8 to 47.82)
Udy_2015	1-3 (CI)	-9.41 (-42.39 to 23.9)	34.11 (16.45 to 58.66)	8.07 (-15.56 to 33.58)	24.12 (11.09 to 49.74)	14.18 (-7.55 to 40.54)	24.79 (10.96 to 46.96)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Wallenburg_2022	Total (Multicenter)	-28.96 (-65.22 to 9.83)	45.68 (21.48 to 76.31)	-19.61 (-50.17 to 11.62)	34.84 (16.24 to 59.86)	-11.34 (-41.04 to 15.89)	29.12 (13.1 to 53.33)
Wallenburg_2022	1 (Target)	-22.76 (-60.78 to 11.18)	39.13 (17.39 to 69.15)	-4.45 (-31.58 to 23.33)	27.2 (12.43 to 49.3)	4.17 (-21.19 to 27.51)	24.74 (10.8 to 41.62)
Wallenburg_2022	2 (Heidenheim)	-19.21 (-50.23 to 8.1)	31.99 (15.73 to 53.32)	-11.67 (-36.91 to 11.64)	25.88 (11.67 to 46.28)	-6.77 (-25.73 to 13.39)	20.22 (9.98 to 39.8)
Wallenburg_2022	3 (Regensburg)	-15.69 (-78.99 to 46.38)	75.22 (38.6 to 93.2)	-8.47 (-32.83 to 66.44)	46.9 (14.24 to 66.44)	NA	NA
Wallenburg_2022	4 (Munich) (II)	-38.21 (-74.09 to 8.87)	56.9 (29.63 to 89.28)	-31.45 (-60.21 to 3.31)	41.79 (21.17 to 68.02)	-18.61 (-50.07 to 9.29)	33.1 (14.89 to 59.19)
Wallenburg_2022	1-3 (CI)	-21.09 (-54.63 to 10.6)	36.25 (16.46 to 63.14)	-7.35 (-33.21 to 19.54)	26.79 (12.22 to 48.15)	1.62 (-21.73 to 25.59)	24.11 (10.49 to 41.26)

Table S4 Comparison of the predicted and observed plasma piperacillin concentrations of the 24 evaluated population pharmacokinetic models for piperacillin, based on the total dataset stratified by study.
A priori: Prediction of all concentrations based on dosing history and patient covariates only;
Bayesian 1: Prediction of concentrations considering the first therapeutic drug monitoring (TDM) sample for each patient;
Bayesian 2: Prediction of concentrations considering the first and second TDM sample for each patient;
MPE: Median relative prediction error (accuracy); MAPE: Median absolute relative prediction error (precision); II: intermittent infusion; CI: continuous infusion; NA: not available;
Values presented as median [%] (95%-confidence interval)

PopPK Model	**Study**	**MPE *A priori***	**MAPE *A priori***	**MPE Bayesian 1**	**MAPE Bayesian 1**	**MPE Bayesian 2**	**MAPE Bayesian 2**
Andersen_2018	Total (Multicenter)	24.92 (23.07 to 27.05)	43.51 (41.95 to 45.16)	18.6 (16.98 to 20.76)	34.61 (33.07 to 36.23)	22.83 (20.66 to 24.72)	36.92 (35.34 to 38.65)
Andersen_2018	1 (Target)	22.01 (19.09 to 25.52)	37.54 (34.81 to 39.69)	11.4 (8.51 to 14.22)	24 (22.07 to 25.93)	16.64 (11.94 to 19.04)	24.96 (22.42 to 27.02)
Andersen_2018	2 (Heidenheim)	28.11 (24.55 to 31.37)	35.23 (32.22 to 38.04)	6.84 (4.04 to 11.47)	23.51 (20.39 to 26.38)	6.73 (-1.68 to 14.37)	22.33 (16.25 to 25.41)
Andersen_2018	3 (Regensburg)	25.96 (-20.87 to 80.16)	68.93 (60.22 to 94.15)	20.99 (-11.66 to 48.8)	21.17 (-11.29 to 30.41)	NA	NA
Andersen_2018	4 (Munich) (II)	25.18 (22.67 to 29.09)	51.15 (49.29 to 53.59)	27.66 (25.12 to 29.89)	44.12 (42.72 to 46.33)	27.37 (24.42 to 30.28)	43.95 (42 to 46.57)
Andersen_2018	1-3 (CI)	24.82 (22.6 to 27.27)	37.26 (35.57 to 39.34)	9.46 (7.2 to 11.84)	23.79 (21.74 to 25.67)	15.12 (11.28 to 18.75)	24.5 (22.33 to 26.34)
Asin-Prieto_2013	Total (Multicenter)	-135.55 (-137.6 to -133.78)	135.59 (133.71 to 137.46)	-77.34 (-78.57 to -75.37)	77.68 (75.78 to 78.95)	-55.77 (-57.26 to -53.9)	57.41 (55.72 to 59.33)
Asin-Prieto_2013	1 (Target)	-137.5 (-139.76 to -135.6)	137.5 (135.66 to 139.9)	-79.5 (-82.87 to -77.18)	80.05 (77.85 to 83.41)	-50.23 (-52.57 to -46.72)	51.35 (47.83 to 53.49)
Asin-Prieto_2013	2 (Heidenheim)	-129.27 (-132.55 to -126.57)	129.27 (126.23 to 132.55)	-76.2 (-78.99 to -72.26)	76.2 (72.26 to 78.97)	-55.64 (-59.54 to -49.25)	55.64 (49.13 to 59.64)
Asin-Prieto_2013	3 (Regensburg)	-133.18 (-184.56 to -111.46)	133.18 (111.46 to 185.4)	-75.99 (-117.26 to -54.62)	75.99 (54.62 to 103.6)	NA	NA
Asin-Prieto_2013	4 (Munich) (II)	-138.47 (-142.85 to -133.37)	138.56 (133.68 to 143)	-77.11 (-79.15 to -74.91)	77.38 (75.24 to 79.31)	-58.15 (-60.39 to -55.19)	59.92 (57.89 to 62.19)
Asin-Prieto_2013	1-3 (CI)	-134.18 (-136.08 to -132.29)	134.18 (132.31 to 136.03)	-77.94 (-80.05 to -75.56)	78.49 (76.23 to 80.98)	-51.45 (-53.64 to -48.65)	52.43 (49.79 to 54.58)
Bue_2020	Total (Multicenter)	78.25 (76.44 to 80.82)	79.75 (77.79 to 81.82)	72.28 (70.2 to 75.02)	73.87 (71.89 to 76.17)	65.33 (63.31 to 67.77)	67.25 (64.76 to 69.11)
Bue_2020	1 (Target)	92.87 (88.53 to 98.06)	93.71 (89.73 to 98.26)	89.83 (85.56 to 95.08)	90.2 (86.08 to 94.44)	86.66 (79.25 to 90.31)	87.12 (80.4 to 91.07)
Bue_2020	2 (Heidenheim)	83.6 (76.61 to 89.01)	83.92 (75.9 to 90.1)	81.91 (73.13 to 88.66)	83 (74.76 to 90.58)	78.06 (67.17 to 91.93)	78.06 (67.17 to 90.69)
Bue_2020	3 (Regensburg)	93.32 (66.79 to 146.81)	93.32 (66.79 to 127.01)	82.18 (32.12 to 130.78)	82.18 (32.12 to 119.96)	NA	NA
Bue_2020	4 (Munich) (II)	69.41 (66.54 to 72.43)	71.9 (69.07 to 74.48)	62.87 (60.16 to 66.39)	65.58 (63.19 to 68.29)	56.63 (52.88 to 59.33)	60.64 (57.7 to 63.54)
Bue_2020	1-3 (CI)	89.56 (85.85 to 93.34)	90.69 (86.98 to 95.3)	87.73 (83.89 to 92.36)	88.33 (84.52 to 93.06)	85.24 (80.7 to 89.76)	85.31 (80.68 to 89.41)
Bulitta_2007	Total (Multicenter)	-62.97 (-67.21 to -58.9)	67.55 (64.47 to 69.58)	-28.06 (-30.75 to -26.19)	39.79 (38.25 to 41.39)	-17.65 (-19.69 to -15.56)	34.28 (32.56 to 36.18)
Bulitta_2007	1 (Target)	-53.43 (-57.64 to -49.37)	59.14 (54.62 to 61.79)	-10.56 (-15.8 to -7.12)	30.24 (27.05 to 32.38)	3.6 (-0.47 to 6.73)	25.58 (23.45 to 28.17)
Bulitta_2007	2 (Heidenheim)	-64.81 (-75.05 to -59.63)	65.9 (60.88 to 73.75)	-19.15 (-23.11 to -13.36)	32.51 (29.88 to 38.18)	-8.46 (-13.9 to -1.13)	27.69 (22.2 to 31.86)
Bulitta_2007	3 (Regensburg)	-56.14 (-90.75 to -11.76)	73.67 (38.17 to 106.47)	-16.59 (-70.02 to 12.18)	42.52 (26.59 to 60.52)	NA	NA
Bulitta_2007	4 (Munich) (II)	-79.8 (-91.57 to -68.76)	84.36 (74.6 to 93.02)	-41.36 (-44.42 to -37.93)	47.27 (44.8 to 49.56)	-29.37 (-32.15 to -25.69)	39.12 (36.23 to 41.48)
Bulitta_2007	1-3 (CI)	-55.96 (-59.35 to -51.54)	61.6 (57.79 to 65.02)	-13.98 (-16.75 to -10.9)	30.95 (28.71 to 33.15)	1.02 (-2.2 to 4.21)	26.2 (24.22 to 28.62)
Chen_2016	Total (Multicenter)	-90.23 (-92.35 to -88.1)	90.74 (88.71 to 92.7)	-17.16 (-19.09 to -15.05)	34.05 (31.87 to 35.46)	9.73 (7.34 to 11.75)	30.82 (29.17 to 32.45)
Chen_2016	1 (Target)	-88.15 (-90.16 to -84.88)	88.74 (85.37 to 90.91)	3.92 (-0.42 to 7.86)	26.01 (23.66 to 27.83)	16.67 (13.37 to 20.82)	27.91 (23.52 to 30.77)
Chen_2016	2 (Heidenheim)	-85.18 (-88.18 to -82.3)	85.18 (82.38 to 88.49)	-2.09 (-6.22 to 1.82)	24.8 (22.26 to 28.64)	3.06 (-3.29 to 10.7)	23.24 (16.32 to 28.4)
Chen_2016	3 (Regensburg)	-74.4 (-119.07 to -29.97)	104.7 (88.32 to 163.32)	-0.35 (-27.23 to 19.8)	22.58 (9.1 to 29.26)	NA	NA
Chen_2016	4 (Munich) (II)	-94.12 (-96.9 to -90.89)	94.82 (91.8 to 97.49)	-32.79 (-35.26 to -29.29)	42.19 (39.87 to 44.36)	7.68 (4.51 to 10.17)	32.47 (30.8 to 34.47)
Chen_2016	1-3 (CI)	-86.98 (-89.02 to -85.17)	87.19 (84.66 to 89.25)	1.37 (-2.6 to 3.86)	25.55 (24.07 to 27.07)	14.42 (11.72 to 18.79)	27.36 (24.48 to 30.09)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Chung_2015	Total (Multicenter)	-33.18 (-35.64 to -31.06)	46.43 (44.69 to 48.1)	-15.63 (-17.89 to -13.74)	33.42 (31.88 to 34.71)	-3.38 (-5.33 to -1.43)	28.01 (26.72 to 29.43)
Chung_2015	1 (Target)	-27.02 (-31.49 to -23.91)	41.84 (39.5 to 44.79)	-2.81 (-5.17 to 1.14)	27.54 (24.98 to 30.39)	8.25 (5.47 to 12.59)	24.79 (22.57 to 26.65)
Chung_2015	2 (Heidenheim)	-24.32 (-27.76 to -20.21)	34.99 (31.21 to 38.23)	-6.24 (-9.55 to -2.05)	22.09 (19.33 to 24.22)	-2.3 (-7.3 to 3.46)	21.35 (16.37 to 25.83)
Chung_2015	3 (Regensburg)	-14.85 (-59.16 to 50.09)	73.2 (56.2 to 108.84)	-1.63 (-64.51 to 30.44)	40.68 (19.91 to 70.92)	NA	NA
Chung_2015	4 (Munich) (II)	-40.27 (-42.6 to -36.86)	54.61 (51.98 to 57.29)	-27.16 (-30.41 to -24.75)	39.84 (38.08 to 41.58)	-7.87 (-9.97 to -5.89)	30.12 (28.14 to 31.64)
Chung_2015	1-3 (CI)	-25.53 (-28.36 to -22.3)	39.59 (37.7 to 41.44)	-4.34 (-6.64 to -1.85)	25.27 (23.18 to 27.18)	5.33 (2.05 to 8.46)	24.43 (22.19 to 26.38)
Delattre_2012	Total (Multicenter)	-19.26 (-21.18 to -16.77)	41.4 (39.51 to 43.03)	-13.98 (-15.42 to -12.27)	29.32 (27.97 to 30.78)	-6.16 (-7.98 to -4.61)	24.21 (23.09 to 25.23)
Delattre_2012	1 (Target)	-16.7 (-19.79 to -12.44)	36.48 (33.37 to 39.06)	0.49 (-3.11 to 2.67)	22.11 (19.23 to 23.75)	7.43 (4.09 to 10.65)	22.63 (20.62 to 25.32)
Delattre_2012	2 (Heidenheim)	-11.34 (-15.17 to -8.86)	30.35 (28.28 to 33.19)	-2.14 (-4.61 to 2.14)	21.49 (18.38 to 23.37)	-2.23 (-7.64 to 5.32)	22.32 (18.11 to 27.19)
Delattre_2012	3 (Regensburg)	-11.05 (-54.32 to 46.49)	72.49 (59.46 to 107.64)	6.82 (-41.16 to 35.8)	26.67 (-1.47 to 43.27)	NA	NA
Delattre_2012	4 (Munich) (II)	-26.97 (-30.78 to -23.48)	50.62 (48.27 to 52.93)	-25.3 (-27.13 to -23.05)	34.24 (33.11 to 35.47)	-11.81 (-13.65 to -9.71)	24.94 (23.44 to 26.04)
Delattre_2012	1-3 (CI)	-13.87 (-15.57 to -11.54)	33.45 (31.46 to 34.7)	-0.85 (-3.64 to 0.34)	21.93 (19.91 to 23.26)	5.83 (3.01 to 8.92)	22.53 (20.62 to 24.84)
Fillatre_2021	Total (Multicenter)	12.33 (8.69 to 15.29)	47.33 (45.19 to 48.88)	6.21 (4.85 to 8.16)	26.91 (25.64 to 27.99)	13.01 (11.34 to 15.16)	26.87 (25.42 to 28.02)
Fillatre_2021	1 (Target)	33.66 (29.25 to 37.95)	55.29 (52.28 to 58.81)	21.04 (18 to 25.04)	30.48 (28.39 to 33.44)	25.25 (21.52 to 28.54)	31.58 (27.98 to 34.17)
Fillatre_2021	2 (Heidenheim)	18.81 (11.33 to 25.61)	45.9 (42.4 to 50.5)	13.29 (10 to 17.14)	26.43 (24.13 to 29.12)	16.03 (5.78 to 23.44)	24.88 (17.61 to 29.92)
Fillatre_2021	3 (Regensburg)	36.66 (8.99 to 102.86)	60.09 (35.75 to 72.02)	8.74 (-23.7 to 23.85)	11.17 (-18.86 to 15.97)	NA	NA
Fillatre_2021	4 (Munich) (II)	-1.47 (-5.77 to 1.88)	44.11 (42.18 to 46.64)	-2.49 (-4.55 to -0.52)	25.59 (24.02 to 27.28)	7.31 (5.12 to 9.35)	25.55 (24.32 to 26.51)
Fillatre_2021	1-3 (CI)	28.25 (24.13 to 31.97)	51.5 (48.41 to 53.94)	17.35 (14.18 to 19)	28.58 (26.51 to 30.24)	23.86 (20.3 to 26.93)	30.46 (28.18 to 33.04)
Hahn_2021	Total (Multicenter)	-48.91 (-50.62 to -47.18)	53.03 (50.84 to 54.97)	-24.6 (-26.26 to -22.77)	33.16 (31.84 to 34.22)	-18.83 (-21.15 to -16.76)	29.33 (28.12 to 30.84)
Hahn_2021	1 (Target)	-51.51 (-54.29 to -48.06)	54.67 (51.42 to 57.76)	-22.65 (-26.63 to -20.4)	31.49 (29.29 to 33.44)	-10.06 (-14.52 to -7.72)	24.83 (22.57 to 26.32)
Hahn_2021	2 (Heidenheim)	-45.51 (-49.35 to -41.96)	47.96 (45.39 to 51.46)	-23.43 (-26 to -20.52)	28.63 (27.28 to 30.98)	-17.29 (-22.07 to -11.96)	25.38 (21.89 to 30.06)
Hahn_2021	3 (Regensburg)	-46.02 (-94.86 to -5.21)	79.23 (61.37 to 112.1)	-15.98 (-86.97 to 16.75)	48.71 (30.05 to 74.52)	NA	NA
Hahn_2021	4 (Munich) (II)	-48.32 (-51.57 to -44.38)	54.78 (51.41 to 57.94)	-27.08 (-29.83 to -24.68)	35.74 (34.01 to 37.51)	-23.62 (-26.09 to -21.31)	31.8 (30.17 to 33.64)
Hahn_2021	1-3 (CI)	-49.18 (-51.22 to -47.25)	51.67 (48.97 to 53.35)	-22.77 (-24.81 to -20.92)	29.97 (27.64 to 31.17)	-11.31 (-13.85 to -8.84)	25.1 (23.22 to 26.94)
Hamada_2013	Total (Multicenter)	-24.02 (-26.72 to -20.54)	41.84 (39.99 to 43.76)	6.25 (4.01 to 7.81)	29.32 (28.05 to 30.62)	12.45 (10.33 to 14.35)	30.89 (29.22 to 32.36)
Hamada_2013	1 (Target)	-20.78 (-25.6 to -17.53)	40.03 (37.09 to 42.6)	5.2 (2.7 to 8.48)	24.1 (21.34 to 26.11)	14.44 (10.63 to 17.64)	25.23 (22.66 to 27.43)
Hamada_2013	2 (Heidenheim)	-20.28 (-23.3 to -15.7)	35.55 (33.23 to 38.67)	0.24 (-4.13 to 4.19)	20.12 (15.99 to 22.13)	3.57 (-3.65 to 10.12)	22.48 (18.23 to 26.7)
Hamada_2013	3 (Regensburg)	-8.61 (-55.34 to 57.21)	74.44 (63.09 to 110.67)	11.44 (-27.92 to 40.55)	23.04 (-4.73 to 33.02)	NA	NA
Hamada_2013	4 (Munich) (II)	-31.05 (-36.79 to -27.17)	47.18 (43.76 to 50.41)	9.89 (6.85 to 13.28)	34.74 (32.94 to 36.76)	12.55 (9.27 to 15.16)	34.74 (33.1 to 36.5)
Hamada_2013	1-3 (CI)	-20.5 (-23.36 to -18.21)	38.22 (36.19 to 40.21)	3.24 (1 to 5.58)	23.33 (21.64 to 25.68)	12.26 (9.41 to 14.76)	24.69 (22.9 to 26.76)
Ishihara_2020	Total (Multicenter)	4.13 (2.32 to 6.1)	38.82 (37.04 to 40.41)	8.95 (7.52 to 11.1)	33.44 (32.31 to 34.93)	6.52 (4.65 to 8.54)	28.87 (27.5 to 30.23)
Ishihara_2020	1 (Target)	-0.21 (-4.07 to 3.85)	34.76 (32.52 to 37.28)	10.09 (6.59 to 12.34)	28.58 (26.44 to 31.46)	17.25 (13.65 to 20.71)	28.17 (25.73 to 30.78)
Ishihara_2020	2 (Heidenheim)	5.39 (3.01 to 8.59)	29.89 (28.01 to 32.55)	9.81 (5.45 to 13.54)	25.09 (21.91 to 28.18)	9.81 (4.88 to 14.52)	23.95 (18.68 to 28.14)
Ishihara_2020	3 (Regensburg)	6.21 (-40.87 to 63.12)	62.68 (38.81 to 76.7)	16.22 (-34.99 to 48.29)	23.22 (-30.56 to 30.57)	NA	NA
Ishihara_2020	4 (Munich) (II)	5.93 (1.97 to 9.23)	46.58 (44.05 to 48.42)	6.92 (3.48 to 9.63)	38.6 (36.51 to 40.29)	1.52 (-0.86 to 4.03)	29.83 (28.01 to 31.86)
Ishihara_2020	1-3 (CI)	2.57 (-0.13 to 4.96)	32.52 (30.81 to 33.95)	10.02 (6.96 to 11.7)	27.35 (25.68 to 29.61)	15.48 (12.46 to 18.3)	27.3 (25.26 to 29.47)

PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Jeon_2014	Total (Multicenter)	-26.08 (-28.61 to -24.01)	44.97 (43.38 to 46.84)	-7.66 (-9.24 to -5.72)	30.18 (28.84 to 31.49)	7.9 (5.74 to 9.73)	26.53 (25.3 to 28.03)
Jeon_2014	1 (Target)	-26.56 (-30.67 to -23.51)	39.58 (37.28 to 42.69)	2.56 (-0.49 to 4.79)	22.52 (20.47 to 23.93)	9.73 (6.87 to 13.03)	23.72 (20.31 to 26.87)
Jeon_2014	2 (Heidenheim)	-13.78 (-17.06 to -8.63)	33.23 (30.54 to 35.95)	4.33 (-1.87 to 7.44)	25.35 (22.83 to 27.82)	4.99 (-1.7 to 11.05)	22.63 (18.34 to 27.6)
Jeon_2014	3 (Regensburg)	-31.06 (-115.65 to -1.29)	61.37 (34.14 to 66.04)	9.26 (-25.99 to 37.54)	22.18 (-0.14 to 30.38)	NA	NA
Jeon_2014	4 (Munich) (II)	-33.28 (-38.62 to -29.26)	56.63 (54.32 to 58.91)	-20.49 (-23.68 to -17.95)	35.86 (33.82 to 37.83)	7.88 (5.64 to 9.92)	27.61 (26.05 to 29.09)
Jeon_2014	1-3 (CI)	-21.18 (-23.89 to -17.77)	36.67 (34.61 to 38.65)	3.55 (1.35 to 5.67)	23.6 (22.25 to 25.35)	8.06 (4.44 to 10.36)	23.47 (20.8 to 25.97)
Kim_2016	Total (Multicenter)	-32.03 (-34.39 to -29.81)	44.11 (42.8 to 45.82)	-19.16 (-20.34 to -17.89)	30.26 (29.01 to 31.34)	-9.94 (-11.37 to -8.24)	25.39 (24.2 to 26.64)
Kim_2016	1 (Target)	-29.96 (-33.27 to -26.15)	40.89 (37.65 to 43.54)	-7.49 (-10.74 to -5.02)	24.48 (22.39 to 26.16)	0.32 (-3.76 to 3.18)	21.56 (19.56 to 22.88)
Kim_2016	2 (Heidenheim)	-24.47 (-27.34 to -21.32)	32.88 (29.71 to 36.26)	-11.49 (-15.87 to -7.69)	23.57 (20.13 to 26.61)	-10.13 (-16.97 to -4.31)	23.99 (21.66 to 26.76)
Kim_2016	3 (Regensburg)	-24.72 (-72.81 to 31)	76.61 (62.33 to 102.19)	-0.04 (-48.15 to 30.02)	34.11 (19.57 to 61.8)	NA	NA
Kim_2016	4 (Munich) (II)	-37.14 (-40.01 to -34.13)	50.42 (47.71 to 52.58)	-28.73 (-31.1 to -26.71)	36.01 (33.56 to 37.37)	-14.39 (-16.74 to -11.79)	27.19 (24.72 to 28.64)
Kim_2016	1-3 (CI)	-27.43 (-30.01 to -25.16)	38.11 (35.8 to 39.78)	-8.57 (-10.73 to -6.12)	24.32 (22.61 to 25.85)	-1.21 (-4.1 to 2.7)	22.22 (20.69 to 23.44)
Kim_2022	Total (Multicenter)	-9.83 (-12.19 to -8.01)	37 (35.33 to 38.43)	-5.85 (-7.82 to -3.93)	27.28 (26.23 to 28.67)	-0.92 (-2.83 to 0.81)	23.71 (22.68 to 24.72)
Kim_2022	1 (Target)	-8.43 (-12.64 to -4.3)	39.32 (36.72 to 42.41)	3.29 (-0.52 to 7.15)	26.62 (24.94 to 29.53)	11.23 (7.79 to 14.59)	24.68 (21.95 to 26.96)
Kim_2022	2 (Heidenheim)	-1.05 (-5.25 to 3.55)	29.85 (27.83 to 31.68)	1.92 (-0.95 to 6.1)	23.03 (20.77 to 25)	1.39 (-5.44 to 4.43)	22.81 (19.56 to 27.26)
Kim_2022	3 (Regensburg)	5.63 (-37.48 to 68.84)	59.87 (45.08 to 70.85)	11.26 (-34.88 to 47.46)	32.27 (6.73 to 45.94)	NA	NA
Kim_2022	4 (Munich) (II)	-14.41 (-17.48 to -11.48)	39.44 (37.3 to 41.59)	-12.83 (-15.36 to -10.52)	28.68 (27.5 to 30.03)	-5.37 (-7.39 to -3.49)	23.5 (22.08 to 25.01)
Kim_2022	1-3 (CI)	-5.35 (-8.58 to -2.33)	35.33 (33.82 to 37.72)	2.84 (0.6 to 5.56)	25.19 (23.54 to 26.85)	9.57 (6.74 to 12.95)	24.39 (22.72 to 26.28)
Klastrup_2020	Total (Multicenter)	-12.98 (-15.09 to -11.21)	37.51 (36.03 to 39)	-7.71 (-9.43 to -6.05)	28.04 (26.96 to 29.31)	6.65 (4.7 to 8.67)	28.37 (26.74 to 29.64)
Klastrup_2020	1 (Target)	-11.84 (-14.91 to -8.51)	33.98 (31.35 to 35.85)	3.32 (0.56 to 6.22)	20.87 (18.22 to 22.3)	10.64 (7.11 to 14.09)	22.05 (19.53 to 24)
Klastrup_2020	2 (Heidenheim)	-2.8 (-6.44 to 1.67)	28.14 (25.04 to 30.17)	-0.88 (-4.41 to 2.11)	22.21 (19.01 to 23.99)	0.29 (-10.47 to 7.1)	22.14 (17.32 to 25.38)
Klastrup_2020	3 (Regensburg)	-10.52 (-69.88 to 36.98)	69 (56.41 to 89.16)	2.2 (-26.42 to 23.66)	21.65 (8.33 to 28.97)	NA	NA
Klastrup_2020	4 (Munich) (II)	-18.64 (-22.41 to -15.16)	43.34 (40.46 to 45.66)	-17.9 (-19.94 to -15.48)	32.04 (29.86 to 33.57)	5.47 (2.62 to 7.87)	31.51 (29.48 to 33.34)
Klastrup_2020	1-3 (CI)	-8.3 (-10.87 to -5.43)	32.18 (30.75 to 33.74)	1.52 (-0.94 to 3.87)	21.71 (20.09 to 23.2)	8.96 (5.8 to 12.59)	22.08 (20.04 to 23.79)
Li_2005	Total (Multicenter)	-64.85 (-67.04 to -62.31)	66.66 (64.87 to 69.4)	-19.27 (-20.88 to -17.88)	30.47 (29.26 to 31.97)	-5.97 (-8.65 to -4.42)	26.84 (25.62 to 28.05)
Li_2005	1 (Target)	-62.95 (-67.12 to -58.94)	65.45 (62.38 to 69.82)	-8.67 (-11.67 to -6.03)	25.41 (23.65 to 27.8)	4.33 (0.74 to 7.65)	22.22 (20.08 to 23.91)
Li_2005	2 (Heidenheim)	-59.18 (-62.29 to -55.83)	59.88 (56.67 to 62.89)	-13.47 (-17.52 to -10.35)	24.49 (21.81 to 26.89)	-6.76 (-12.83 to 0.23)	23.94 (20.97 to 28.01)
Li_2005	3 (Regensburg)	-55.19 (-101.44 to -2.31)	78.93 (47.32 to 116.58)	-0.72 (-28.72 to 28.38)	28.59 (17.58 to 47.08)	NA	NA
Li_2005	4 (Munich) (II)	-70.1 (-73.4 to -65.59)	71.6 (67.79 to 75.64)	-27.84 (-30.26 to -25.56)	36.03 (34.08 to 37.95)	-9.5 (-11.84 to -7.26)	30.09 (28.51 to 32.17)
Li_2005	1-3 (CI)	-60.96 (-63.35 to -58.55)	62.58 (59.11 to 65.34)	-10.23 (-12.46 to -7.84)	25.08 (23.34 to 26.94)	2.55 (-0.38 to 5.84)	22.61 (20.47 to 24.36)
Roberts-DM_2015	Total (Multicenter)	42.26 (39.32 to 45.92)	56.74 (54.68 to 59.3)	23.49 (20.93 to 26.35)	39.9 (38.7 to 41.62)	19.52 (17.73 to 21.58)	34.59 (33.14 to 36.2)
Roberts-DM_2015	1 (Target)	66.18 (62.33 to 70.98)	70.82 (67.29 to 73.75)	48.57 (43.18 to 51.24)	53.95 (49.46 to 57.97)	42.83 (37 to 47.03)	47.52 (44.48 to 50.98)
Roberts-DM_2015	2 (Heidenheim)	52.51 (45.55 to 58.98)	57.69 (52.92 to 62.05)	38.57 (33.56 to 45.27)	44.79 (39.83 to 49)	28.13 (20.83 to 35.07)	34.91 (29.35 to 41.15)
Roberts-DM_2015	3 (Regensburg)	71.23 (48.92 to 136.82)	81.26 (51.87 to 116.02)	43.69 (-9.08 to 84.81)	57.82 (19.7 to 89.88)	NA	NA
Roberts-DM_2015	4 (Munich) (II)	23.11 (19.14 to 26.68)	44.12 (40.39 to 46.89)	7.82 (4.83 to 9.98)	32.1 (30.12 to 33.83)	10.35 (8.34 to 13.62)	30.44 (28.87 to 31.86)
Roberts-DM_2015	1-3 (CI)	61.15 (57.35 to 64.6)	66.67 (63.76 to 70.13)	45.92 (42.88 to 48.91)	50.63 (47.26 to 53.63)	39.12 (34.04 to 42.51)	45.19 (42.45 to 48.85)

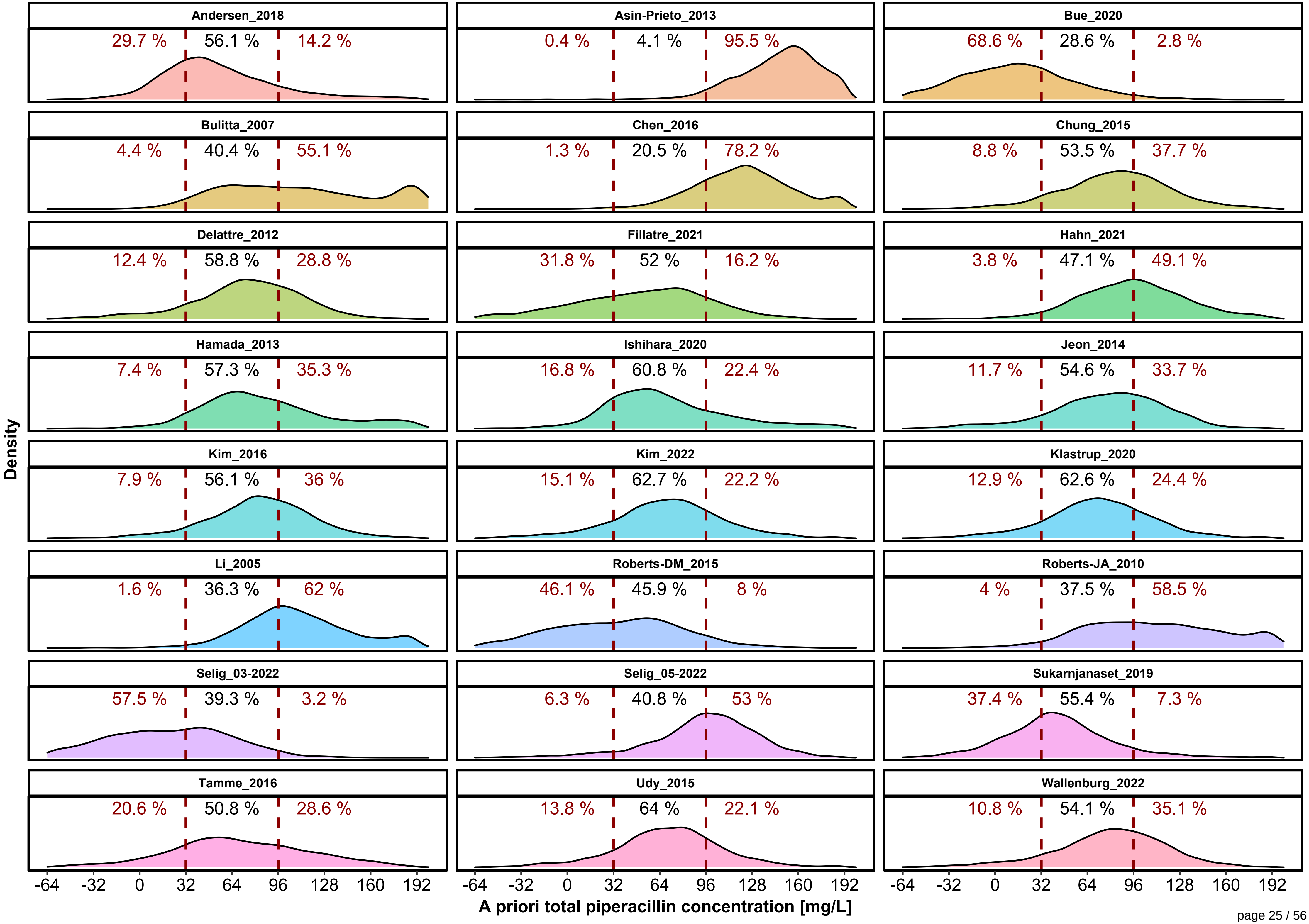
PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Roberts-JA_2010	Total (Multicenter)	-66.97 (-70.4 to -62.92)	71.73 (68.7 to 75.29)	-22.34 (-24.61 to -20.26)	37.67 (35.59 to 39.4)	-18.48 (-20.98 to -16.51)	35.61 (33.36 to 37.64)
Roberts-JA_2010	1 (Target)	-63.75 (-67.59 to -57)	69.8 (65.33 to 75.74)	1.45 (-2.78 to 4.32)	26.92 (24.25 to 29.29)	12.83 (8.82 to 17.11)	26.48 (24.35 to 28.43)
Roberts-JA_2010	2 (Heidenheim)	-73.53 (-79.91 to -67.86)	73.65 (68.02 to 79.61)	-48.96 (-55.83 to -41.67)	52.33 (46.5 to 58.93)	-5.36 (-12.79 to 0.24)	22.61 (19.82 to 25.66)
Roberts-JA_2010	3 (Regensburg)	-66.81 (-109.45 to -21.18)	75.33 (36.04 to 98.75)	-4.04 (-21.12 to 15.85)	21.78 (6.23 to 37.49)	NA	NA
Roberts-JA_2010	4 (Munich) (II)	-65.37 (-72.15 to -59.27)	72.78 (67.04 to 78.34)	-32.12 (-35.8 to -27.43)	43.11 (39.85 to 46.65)	-36.73 (-40.53 to -33.02)	44.54 (42.44 to 48.07)
Roberts-JA_2010	1-3 (CI)	-68.05 (-72.32 to -63.73)	71.14 (67.5 to 75.14)	-12.27 (-15.34 to -9.36)	33.15 (30.92 to 35.09)	8.89 (5.23 to 12.29)	25.4 (23.34 to 27.13)
Selig_03-2022	Total (Multicenter)	64.71 (61.41 to 67.76)	67.72 (64.53 to 70.7)	12.89 (10.41 to 15.03)	36.23 (34.47 to 37.93)	15.07 (13.04 to 17.82)	30.31 (28.71 to 31.62)
Selig_03-2022	1 (Target)	88.5 (84.81 to 92.74)	89.38 (85.97 to 92.32)	43.97 (39.72 to 48.28)	47.48 (43.72 to 51)	28.76 (25.7 to 31.94)	34.01 (31.75 to 36.46)
Selig_03-2022	2 (Heidenheim)	75.39 (69.28 to 81.16)	75.97 (70.29 to 81.28)	33.18 (27.99 to 37.68)	38.44 (33.79 to 42.11)	17.64 (12.82 to 24.76)	26.66 (23.01 to 32.51)
Selig_03-2022	3 (Regensburg)	85.5 (57.68 to 148.45)	93.25 (73.17 to 125.2)	32.39 (-27.12 to 59.17)	37.49 (-23.97 to 52.22)	NA	NA
Selig_03-2022	4 (Munich) (II)	41.89 (38.14 to 45.52)	50.24 (47.52 to 53.9)	-6.35 (-9.27 to -3.64)	30.93 (29.27 to 32.94)	6.87 (4.15 to 9.31)	29.28 (27.63 to 31.17)
Selig_03-2022	1-3 (CI)	83.65 (80.31 to 86.25)	84.7 (81.32 to 87.65)	39.65 (36.16 to 41.82)	43.75 (41.6 to 46.23)	26.74 (24.2 to 29.77)	32.36 (29.89 to 34.82)
Selig_05-2022	Total (Multicenter)	-53.36 (-55.17 to -51.73)	60.01 (58.2 to 61.94)	-22.03 (-24.2 to -20.36)	34.54 (33.32 to 35.92)	-8.85 (-10.88 to -7.04)	25.89 (24.67 to 27.02)
Selig_05-2022	1 (Target)	-52.88 (-57.31 to -49.65)	55.81 (52.99 to 58.75)	-4.75 (-7.43 to -2.42)	23.23 (21.77 to 24.83)	5.95 (2.23 to 8.82)	22.55 (20.49 to 25.59)
Selig_05-2022	2 (Heidenheim)	-47.07 (-50.41 to -44.9)	48.32 (45.14 to 50.77)	-7.32 (-10.04 to -3.44)	22.94 (20.12 to 25.16)	-3.59 (-10.72 to 3.63)	22.41 (19.64 to 25.4)
Selig_05-2022	3 (Regensburg)	-47.59 (-94.05 to 3.57)	83.81 (63.85 to 122.78)	2.78 (-20.83 to 30.96)	25.41 (10.27 to 41.02)	NA	NA
Selig_05-2022	4 (Munich) (II)	-58.28 (-62.17 to -56.08)	66.58 (64 to 68.39)	-37.06 (-38.7 to -35.64)	43.27 (41.21 to 44.87)	-16.26 (-19.01 to -14.8)	28.1 (26.44 to 29.38)
Selig_05-2022	1-3 (CI)	-49.25 (-51.25 to -46.26)	52.71 (50.89 to 54.96)	-6.24 (-8.72 to -4.63)	23.21 (21.88 to 24.43)	4.54 (2.29 to 8.3)	22.53 (21.07 to 25.32)
Sukarnjanaset_2019	Total (Multicenter)	35.24 (33.31 to 36.97)	44.82 (43.3 to 46.52)	21.57 (19.7 to 23.27)	32.89 (31.47 to 34.12)	14.86 (12.79 to 16.82)	27.47 (26.17 to 28.97)
Sukarnjanaset_2019	1 (Target)	31.3 (28.45 to 35.59)	41.02 (38.34 to 43.09)	19.38 (16.64 to 22.51)	30.26 (27.99 to 33.45)	20.22 (17.21 to 23.48)	27.27 (24.81 to 29.66)
Sukarnjanaset_2019	2 (Heidenheim)	45.42 (42.71 to 48.59)	46.4 (43.23 to 48.99)	23.21 (18.62 to 27.17)	31.02 (27.62 to 34.01)	13.9 (8.98 to 17.57)	24.62 (19.94 to 31.9)
Sukarnjanaset_2019	3 (Regensburg)	43.57 (2.24 to 106.01)	61.67 (42.53 to 75.83)	22.86 (-48.51 to 54.82)	25.78 (-42.66 to 34.6)	NA	NA
Sukarnjanaset_2019	4 (Munich) (II)	34.23 (32.08 to 36.86)	45.62 (42.73 to 47.71)	22.09 (19.78 to 24.58)	34.91 (33.5 to 36.26)	12.63 (10.12 to 14.37)	27.96 (26.64 to 29.86)
Sukarnjanaset_2019	1-3 (CI)	37 (34.4 to 39.79)	43.61 (41.19 to 45.61)	21.02 (18.84 to 23.57)	30.59 (28.93 to 33.28)	18.81 (16.07 to 22.09)	26.63 (24.49 to 28.64)
Tamme_2016	Total (Multicenter)	-3.44 (-6.75 to -0.58)	48.99 (46.97 to 51.04)	9.05 (7.2 to 10.76)	27.61 (26.18 to 28.94)	7.44 (5.97 to 9.63)	25.19 (23.58 to 26.18)
Tamme_2016	1 (Target)	1.75 (-2.59 to 6.42)	49.48 (46.17 to 51.81)	16.09 (12.46 to 18.28)	30.37 (27.71 to 33.52)	24.04 (20.12 to 27.75)	30.66 (28.24 to 32.69)
Tamme_2016	2 (Heidenheim)	-12.95 (-20.49 to -5.98)	46.76 (42.48 to 50.29)	9 (5.57 to 12.57)	25.55 (23.41 to 28.79)	14.16 (8.56 to 21.96)	25.25 (20.24 to 30.74)
Tamme_2016	3 (Regensburg)	7.82 (-18.64 to 75.47)	64.5 (46.7 to 91.7)	18.28 (-12.18 to 44.55)	24.17 (-0.41 to 37.12)	NA	NA
Tamme_2016	4 (Munich) (II)	-3.07 (-7.8 to 2.17)	49.62 (46.13 to 53.07)	5.39 (2.82 to 7.72)	27.35 (25.42 to 28.93)	0.31 (-1.74 to 1.82)	23.6 (22.26 to 24.74)
Tamme_2016	1-3 (CI)	-3.6 (-7.39 to 0.16)	48.88 (46.72 to 51)	13.98 (12.32 to 16.55)	28.06 (26.06 to 30.29)	21.64 (17.74 to 24.5)	29.82 (27.86 to 31.88)
Udy_2015	Total (Multicenter)	-10.69 (-12.69 to -8.45)	35.73 (34.52 to 37.09)	-3.61 (-5.43 to -2.32)	26.18 (24.87 to 27.39)	7.5 (5.9 to 8.97)	23.8 (22.44 to 25.19)
Udy_2015	1 (Target)	-16.1 (-20.37 to -12.46)	35.14 (32.75 to 37.29)	9.77 (5.96 to 12.44)	22.81 (20.49 to 25.15)	15.22 (11.82 to 18.7)	26.48 (23.22 to 29.4)
Udy_2015	2 (Heidenheim)	-0.88 (-4.68 to 3.15)	31.16 (28.11 to 33.36)	4.92 (1.57 to 10.03)	27.76 (23.52 to 31.74)	8.32 (0.39 to 13.42)	20.32 (14.72 to 23.45)
Udy_2015	3 (Regensburg)	-17.13 (-96.8 to 16.82)	64.53 (48.27 to 78.75)	8.18 (-21.4 to 24.59)	21.38 (-0.86 to 34.54)	NA	NA
Udy_2015	4 (Munich) (II)	-12.28 (-15.5 to -9.69)	37.58 (34.8 to 39.46)	-11.69 (-13.27 to -9.56)	27.42 (25.59 to 28.7)	3.47 (0.92 to 5.61)	23.22 (21.58 to 24.59)
Udy_2015	1-3 (CI)	-9.41 (-13.03 to -6.02)	34.11 (32.53 to 35.91)	8.07 (6.24 to 10.34)	24.12 (22.47 to 25.89)	14.18 (11.72 to 17.28)	24.79 (21.55 to 27.16)

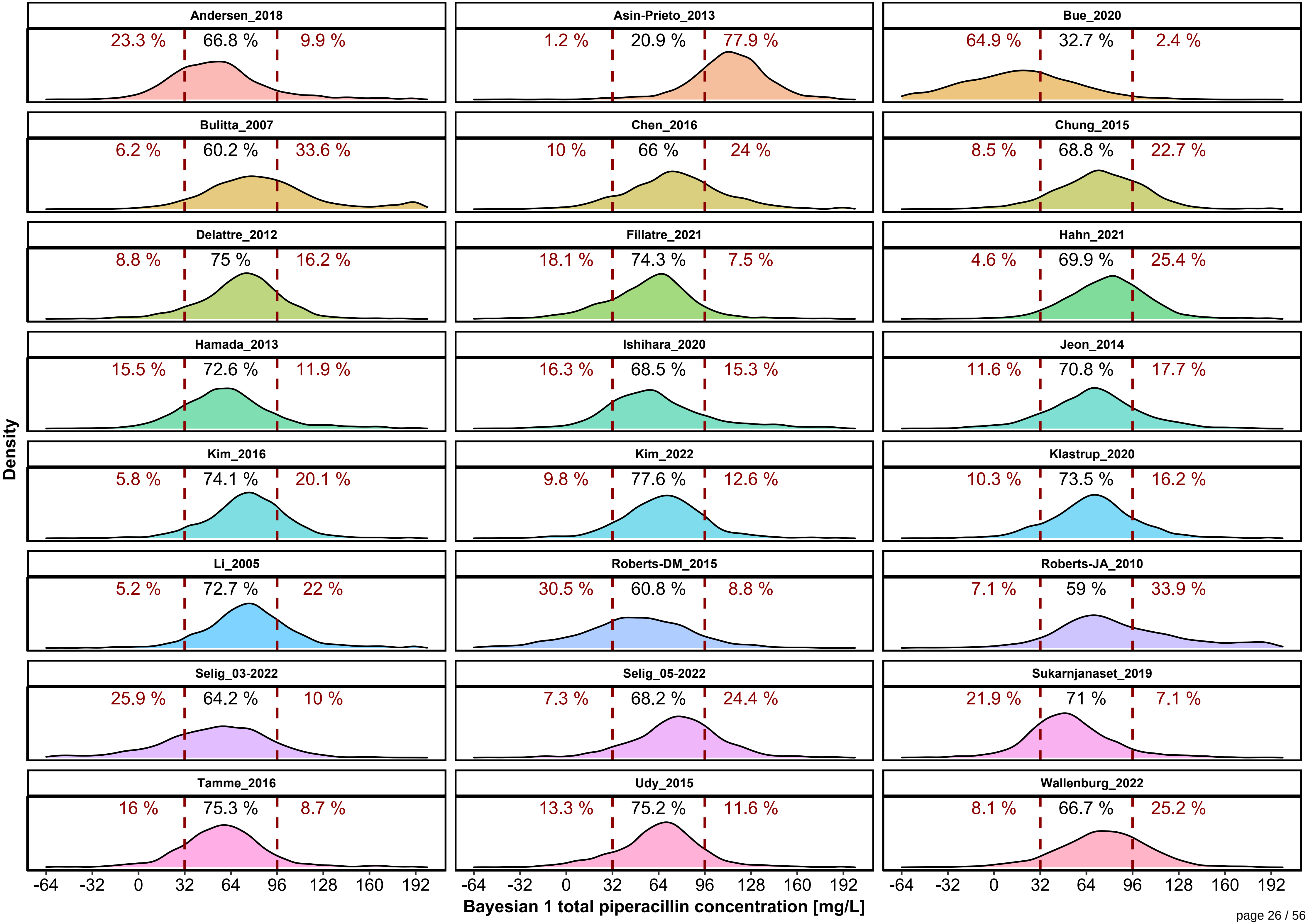
PopPK Model	Study	MPE *A priori*	MAPE *A priori*	MPE Bayesian 1	MAPE Bayesian 1	MPE Bayesian 2	MAPE Bayesian 2
Wallenburg_2022	Total (Multicenter)	-28.96 (-30.91 to -26.74)	45.68 (44.09 to 47.43)	-19.61 (-22.01 to -17.28)	34.84 (33.01 to 36.06)	-11.34 (-13.11 to -9.59)	29.12 (27.39 to 30.69)
Wallenburg_2022	1 (Target)	-22.76 (-26.87 to -19.51)	39.13 (36.49 to 41.99)	-4.45 (-6.82 to -0.37)	27.2 (23.65 to 29.6)	4.17 (1.44 to 7.78)	24.74 (22.94 to 26.8)
Wallenburg_2022	2 (Heidenheim)	-19.21 (-22.49 to -14.81)	31.99 (28.84 to 34.19)	-11.67 (-15.97 to -8.25)	25.88 (22.41 to 28.33)	-6.77 (-12.14 to -1.75)	20.22 (14.9 to 23.29)
Wallenburg_2022	3 (Regensburg)	-15.69 (-59.98 to 46.01)	75.22 (64.35 to 107.65)	-8.47 (-87.01 to 18.29)	46.9 (23.73 to 80.14)	NA	NA
Wallenburg_2022	4 (Munich) (II)	-38.21 (-41.11 to -34.86)	56.9 (54.12 to 60.16)	-31.45 (-33.68 to -28.85)	41.79 (40.03 to 43.94)	-18.61 (-21.12 to -15.53)	33.1 (30.63 to 35.15)
Wallenburg_2022	1-3 (CI)	-21.09 (-23.75 to -18.15)	36.25 (33.87 to 38.54)	-7.35 (-10.27 to -4.75)	26.79 (24.33 to 28.59)	1.62 (-1.35 to 4.94)	24.11 (22.45 to 26.56)

Fig. S2 Attainment of piperacillin target concentrations based on *A priori*, Bayesian 1 and Bayesian 2 predictions of the 24 evaluated population pharmacokinetic models for piperacillin, using the total evaluation dataset.

Relative prediction errors were translated to total piperacillin concentrations to enable a more clinical interpretation of model performance;
A priori (A): Prediction of all concentrations based on dosing history and patient covariates only;
Bayesian 1 (B1): Prediction of concentrations considering the first TDM sample for each patient;
Bayesian 2 (B2): Prediction of concentrations considering the first and second TDM sample for each patient; red dashed lines represent the lower and upper limits of the target concentration range; percentages indicate the proportion of predicted concentrations within (black color) and outside (red color) the target range

(see figure on page 25-27)


Andersen_2018
29.7 %
56.1 %
14.2 %
Asin-Prieto_2013
0.4 %
4.1 %
95.5 %
Bue_2020
68.6 %
28.6 %
2.8 %
Bulitta_2007
4.4 %
40.4 %
55.1 %
Chen_2016
1.3 %
20.5 %
78.2 %
Chung_2015
8.8 %
53.5 %
37.7 %
Delattre_2012
12.4 %
58.8 %
28.8 %
Fillatre_2021
31.8 %
52 %
16.2 %
Hahn_2021
3.8 %
47.1 %
49.1 %
Hamada_2013
7.4 %
57.3 %
35.3 %
Ishihara_2020
16.8 %
60.8 %
22.4 %
Jeon_2014
11.7 %
54.6 %
33.7 %
Kim_2016
7.9 %
56.1 %
36 %
Kim_2022
15.1 %
62.7 %
22.2 %
Klastrup_2020
12.9 %
62.6 %
24.4 %
Li_2005
1.6 %
36.3 %
62 %
Roberts-DM_2015
46.1 %
45.9 %
8 %
Roberts-JA_2010
4 %
37.5 %
58.5 %
Selig_03-2022
57.5 %
39.3 %
3.2 %
Selig_05-2022
6.3 %
40.8 %
53 %
Sukarnjanaset_2019
37.4 %
55.4 %
7.3 %
Tamme_2016
20.6 %
50.8 %
28.6 %
Udy_2015
13.8 %
64 %
22.1 %
Wallenburg_2022
10.8 %
54.1 %
35.1 %
Density
-64 -32 0 32 64 96 128 160 192
A priori total piperacillin concentration [mg/L]


Andersen_2018
23.3 %
66.8 %
9.9 %
Asin-Prieto_2013
1.2 %
20.9 %
77.9 %
Bue_2020
64.9 %
32.7 %
2.4 %
Bulitta_2007
6.2 %
60.2 %
33.6 %
Chen_2016
10 %
66 %
24 %
Chung_2015
8.5 %
68.8 %
22.7 %
Delattre_2012
8.8 %
75 %
16.2 %
Fillatre_2021
18.1 %
74.3 %
7.5 %
Hahn_2021
4.6 %
69.9 %
25.4 %
Hamada_2013
15.5 %
72.6 %
11.9 %
Ishihara_2020
16.3 %
68.5 %
15.3 %
Jeon_2014
11.6 %
70.8 %
17.7 %
Kim_2016
5.8 %
74.1 %
20.1 %
Kim_2022
9.8 %
77.6 %
12.6 %
Klastrup_2020
10.3 %
73.5 %
16.2 %
Li_2005
5.2 %
72.7 %
22 %
Roberts-DM_2015
30.5 %
60.8 %
8.8 %
Roberts-JA_2010
7.1 %
59 %
33.9 %
Selig_03-2022
25.9 %
64.2 %
10 %
Selig_05-2022
7.3 %
68.2 %
24.4 %
Sukarnjanaset_2019
21.9 %
71 %
7.1 %
Tamme_2016
16 %
75.3 %
8.7 %
Udy_2015
13.3 %
75.2 %
11.6 %
Wallenburg_2022
8.1 %
66.7 %
25.2 %
Density
-64
-32
0
32
64
96
128
160
192
Bayesian 1 total piperacillin concentration [mg/L]

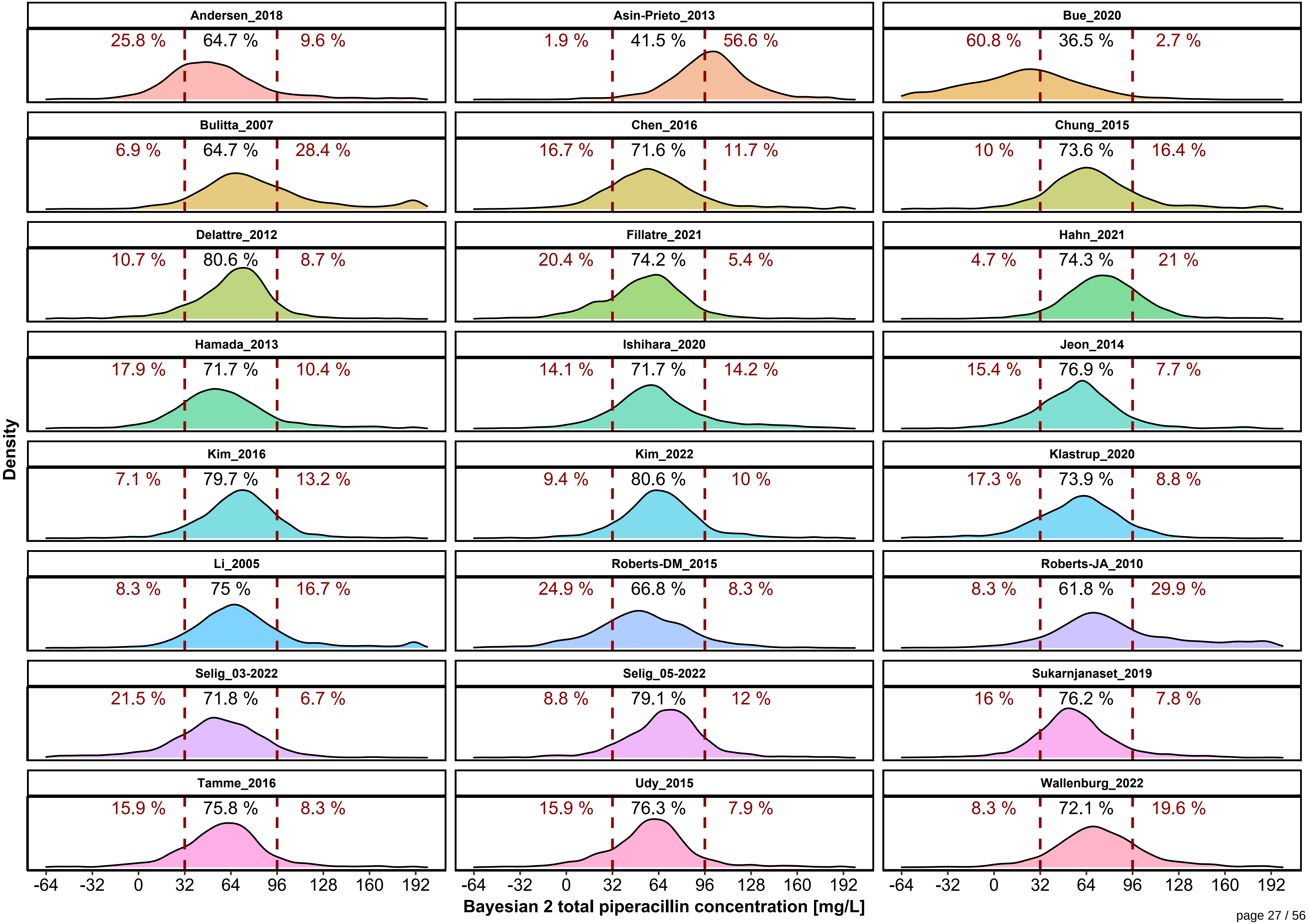
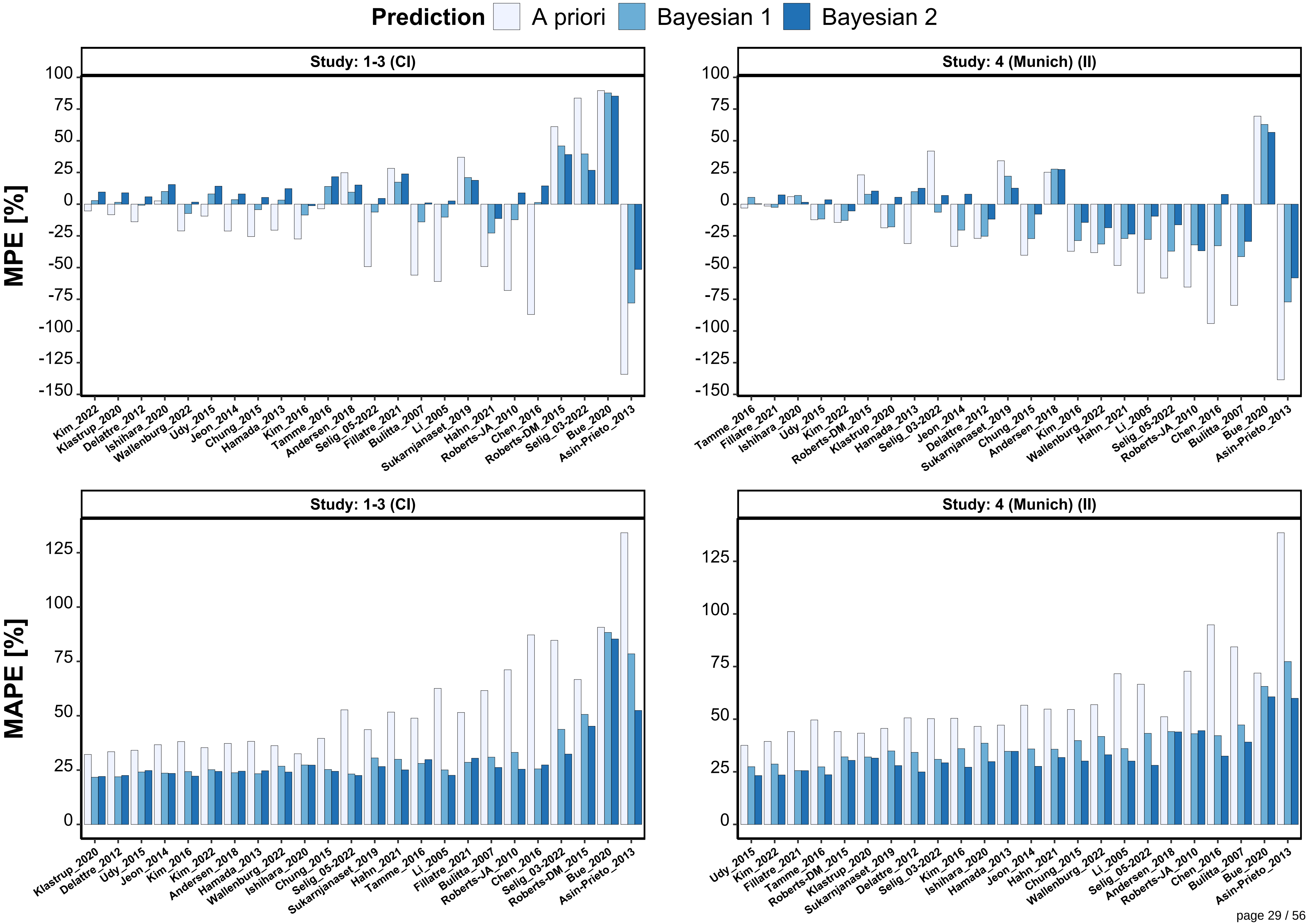

Andersen_2018
25.8 %
64.7 %
9.6 %
Asin-Prieto_2013
1.9 %
41.5 %
56.6 %
Bue_2020
60.8 %
36.5 %
2.7 %
Bulitta_2007
6.9 %
64.7 %
28.4 %
Chen_2016
16.7 %
71.6 %
11.7 %
Chung_2015
10 %
73.6 %
16.4 %
Delattre_2012
10.7 %
80.6 %
8.7 %
Fillatre_2021
20.4 %
74.2 %
5.4 %
Hahn_2021
4.7 %
74.3 %
21 %
Hamada_2013
17.9 %
71.7 %
10.4 %
Ishihara_2020
14.1 %
71.7 %
14.2 %
Jeon_2014
15.4 %
76.9 %
7.7 %
Kim_2016
7.1 %
79.7 %
13.2 %
Kim_2022
9.4 %
80.6 %
10 %
Klastrup_2020
17.3 %
73.9 %
8.8 %
Li_2005
8.3 %
75 %
16.7 %
Roberts-DM_2015
24.9 %
66.8 %
8.3 %
Roberts-JA_2010
8.3 %
61.8 %
29.9 %
Selig_03-2022
21.5 %
71.8 %
6.7 %
Selig_05-2022
8.8 %
79.1 %
12 %
Sukarnjanaset_2019
16 %
76.2 %
7.8 %
Tamme_2016
15.9 %
75.8 %
8.3 %
Udy_2015
15.9 %
76.3 %
7.9 %
Wallenburg_2022
8.3 %
72.1 %
19.6 %
Density
-64 -32 0 32 64 96 128 160 192
Bayesian 2 total piperacillin concentration [mg/L]

Fig. S3 Predictive performance of the 24 evaluated population pharmacokinetic (PopPK) models for piperacillin, based on the total evaluation dataset stratified by different piperacillin/tazobactam administration methods.

Median relative prediction error (MPE) [%] as a measure for accuracy and median absolute relative prediction error (MAPE) [%] reflecting precision of the predictions, comparing the observed plasma piperacillin (PIP) concentrations for each PopPK model;
A priori (A): Prediction of all concentrations based on dosing history and patient covariates only;
Bayesian 1 (B1): Prediction of concentrations considering the first TDM sample for each patient;
Bayesian 2 (B2): Prediction of concentrations considering the first and second TDM sample for each patient; CI: continuous infusion; II: intermittent infusion; Models were ordered (from left to right) according to the model-specific sum (A+B1+B2) of (absolute) MPE and MAPE values, respectively

(see figure on page 29)



Prediction
A priori
Bayesian 1
Bayesian 2
Study: 1-3 (CI)
Study: 4 (Munich) (II)
MPE [%]
MAPE [%]
Kim_2022
Klastrup_2020
Delattre_2012
Ishihara_2020
Wallenburg_2022
Udy_2015
Jeon_2014
Chung_2015
Hamada_2013
Kim_2016
Tamme_2016
Andersen_2018
Selig_05-2022
Fillatre_2021
Bulitta_2007
Li_2005
Sukarnjanaset_2019
Hahn_2021
Roberts-JA_2010
Chen_2016
Roberts-DM_2015
Selig_03-2022
Bue_2020
Asin-Prieto_2013

Table S5 Median MPE and median MAPE of all 24 population pharmacokinetic models for piperacillin, based on the total evaluation dataset stratified by study and piperacillin/tazobactam administration method.

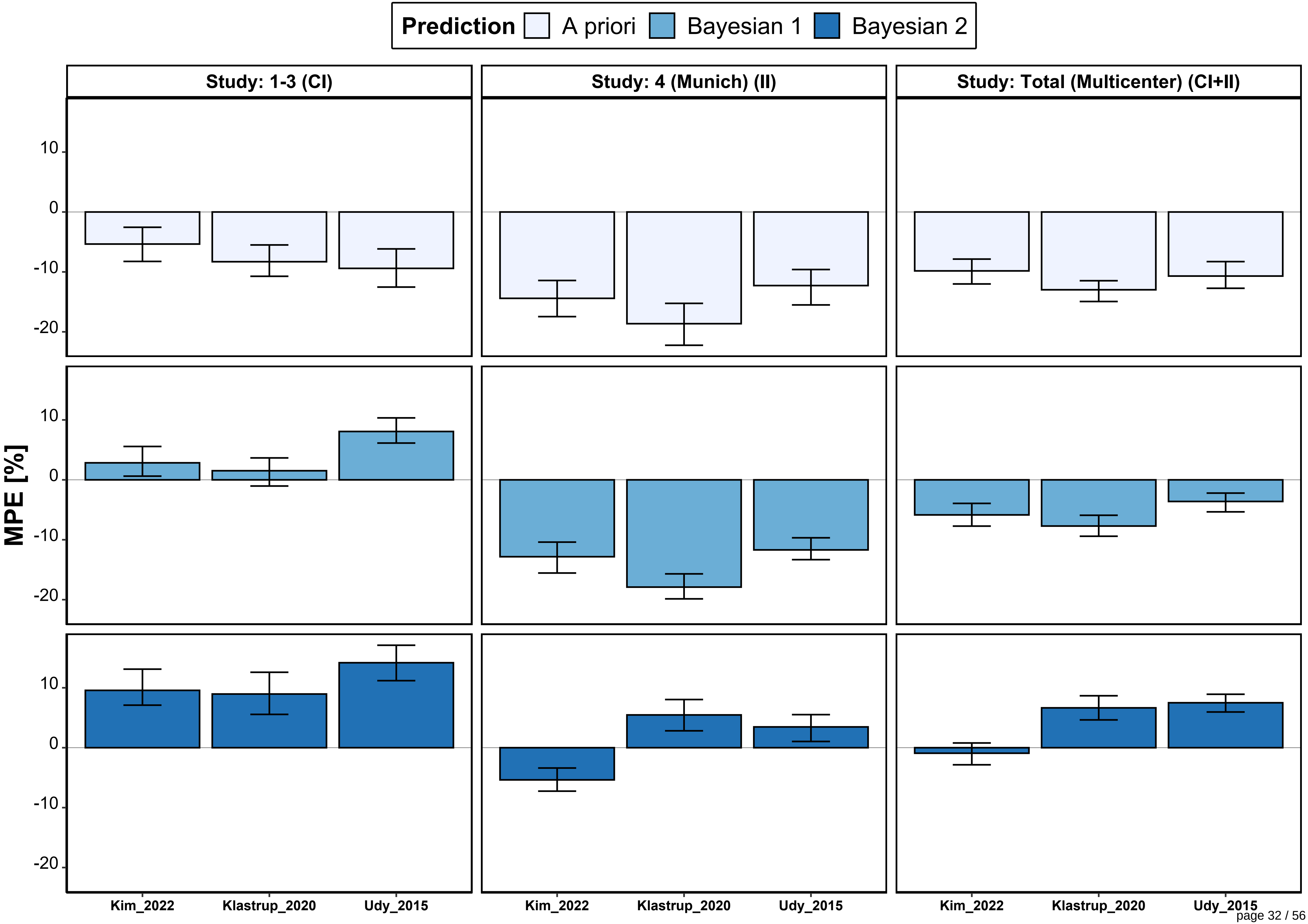
Median relative prediction error (MPE) [%] as a measure for accuracy and median absolute relative prediction error (MAPE) [%] reflecting precision of the predictions, comparing the observed plasma piperacillin (PIP) concentrations for each PopPK model;
A priori (A): Prediction of all concentrations based on patient covariates and dosing history only;
Bayesian 1 (B1): Prediction of concentrations considering the first TDM sample per patient;
Bayesian 2 (B2): Prediction of concentrations considering the first and second TDM sample per patient; CI: continuous infusion; II: intermittent infusion; NA: not available

Study	**A**	**B1**	**B2**	**Relative Improvement [%]**	
				B1 vs. A	B2 vs. A
$\|MPE\|_{median\_all\_models}$					
Total (Multicenter)	32.6	17.9	10.6	45.1	67.5
1 (Target)	30.6	9.2	13.6	69.9	55.6
2 Heidenheim	26.3	10.7	7.5	59.3	71.5
3 Regensburg	33.9	10.3	NA	69.6	NA
4 Munich (II)	35.7	26.2	11.1	26.6	68.9
1-3 (CI)	27.8	9.7	11.8	65.1	57.6
$MAPE_{median\_all\ models}$					
Total (Multicenter)	46.9	33.3	28.6	29.0	39.0
1 (Target)	45.7	27.1	25.4	40.7	44.4
2 Heidenheim	46.1	25.5	23.6	44.7	48.8
3 Regensburg	74.1	26.2	NA	64.6	NA
4 Munich (II)	50.9	35.9	30.1	29.5	40.9
1-3 (CI)	46.2	26.2	24.9	43.3	46.1

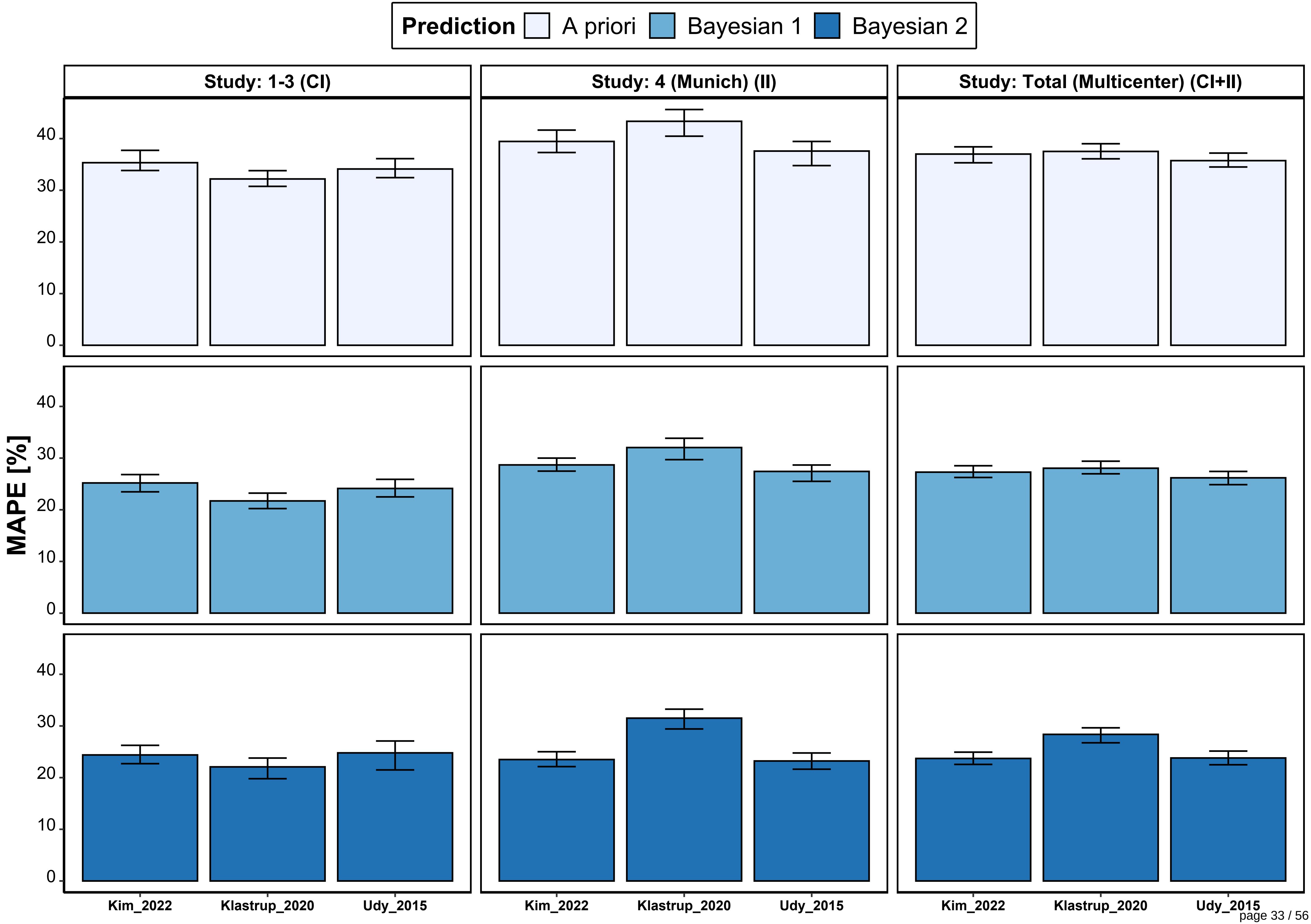
Fig. S4 Predictive performance of the three best-predicting population pharmacokinetic models for piperacillin, stratified by piperacillin/tazobactam administration method

Median relative prediction error (MPE) [%] as a measure for accuracy and median absolute relative prediction error (MAPE) [%] reflecting precision of the predictions, comparing the observed plasma piperacillin (PIP) concentrations for each PopPK model.
Additional presentation of the 95%-confidence intervals of MPE and MAPE, respectively;
A priori (A): Prediction of all concentrations based on dosing history and patient covariates only;
Bayesian 1 (B1): Prediction of concentrations considering the first TDM sample for each patient;
Bayesian 2 (B2): Prediction of concentrations considering the first and second TDM sample for each patient; CI: continuous infusion; II: intermittent infusion; black line: reference line (PE=0%).

(see figure on page 32-33)



Prediction
A priori
Bayesian 1
Bayesian 2
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Study: Total (Multicenter) (CI+II)
MPE [%]
10
0
-10
-20
Kim_2022
Klastrup_2020
Udy_2015

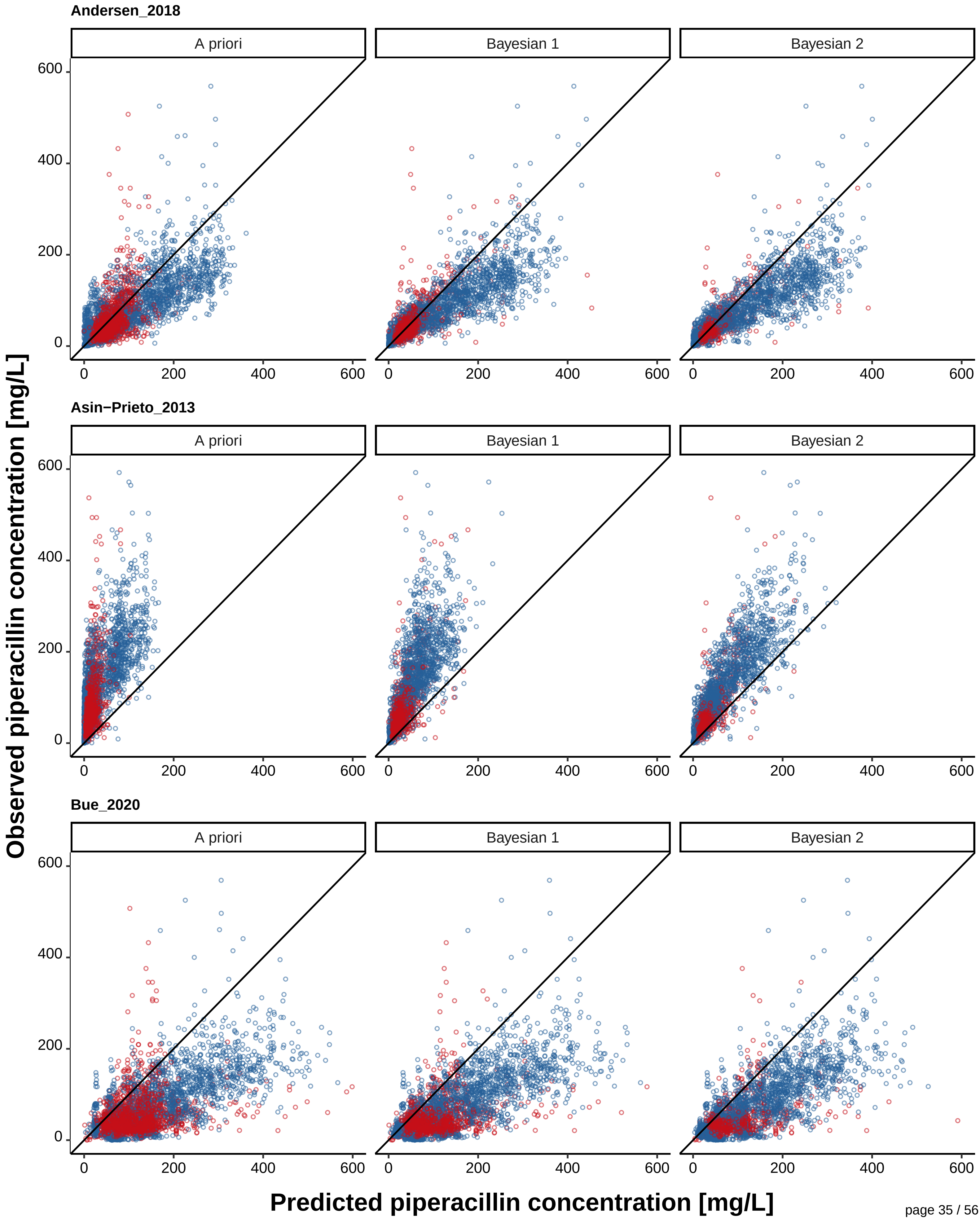


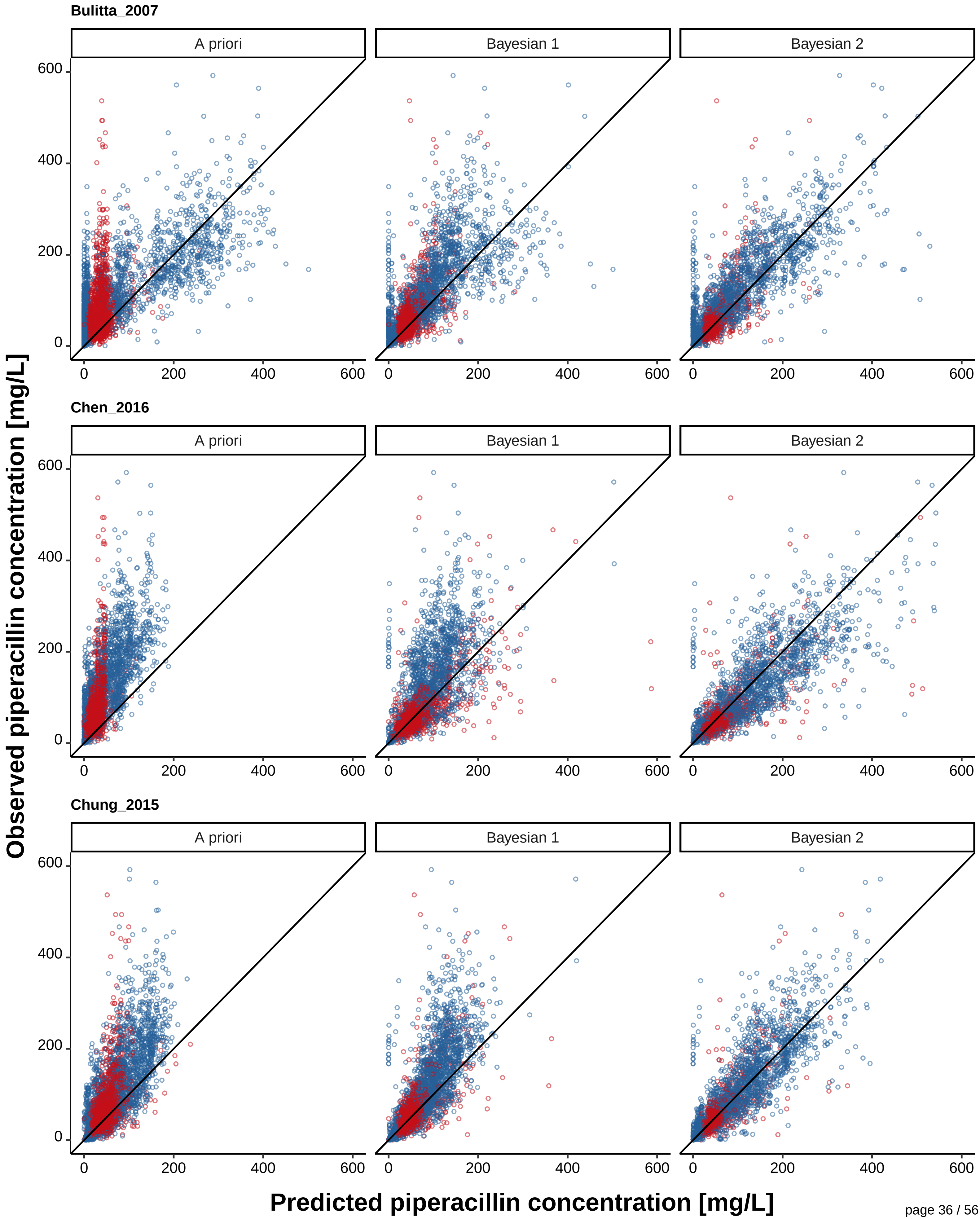
Prediction
A priori
Bayesian 1
Bayesian 2
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Study: Total (Multicenter) (CI+II)
MAPE [%]
0
10
20
30
40
Kim_2022
Klastrup_2020
Udy_2015

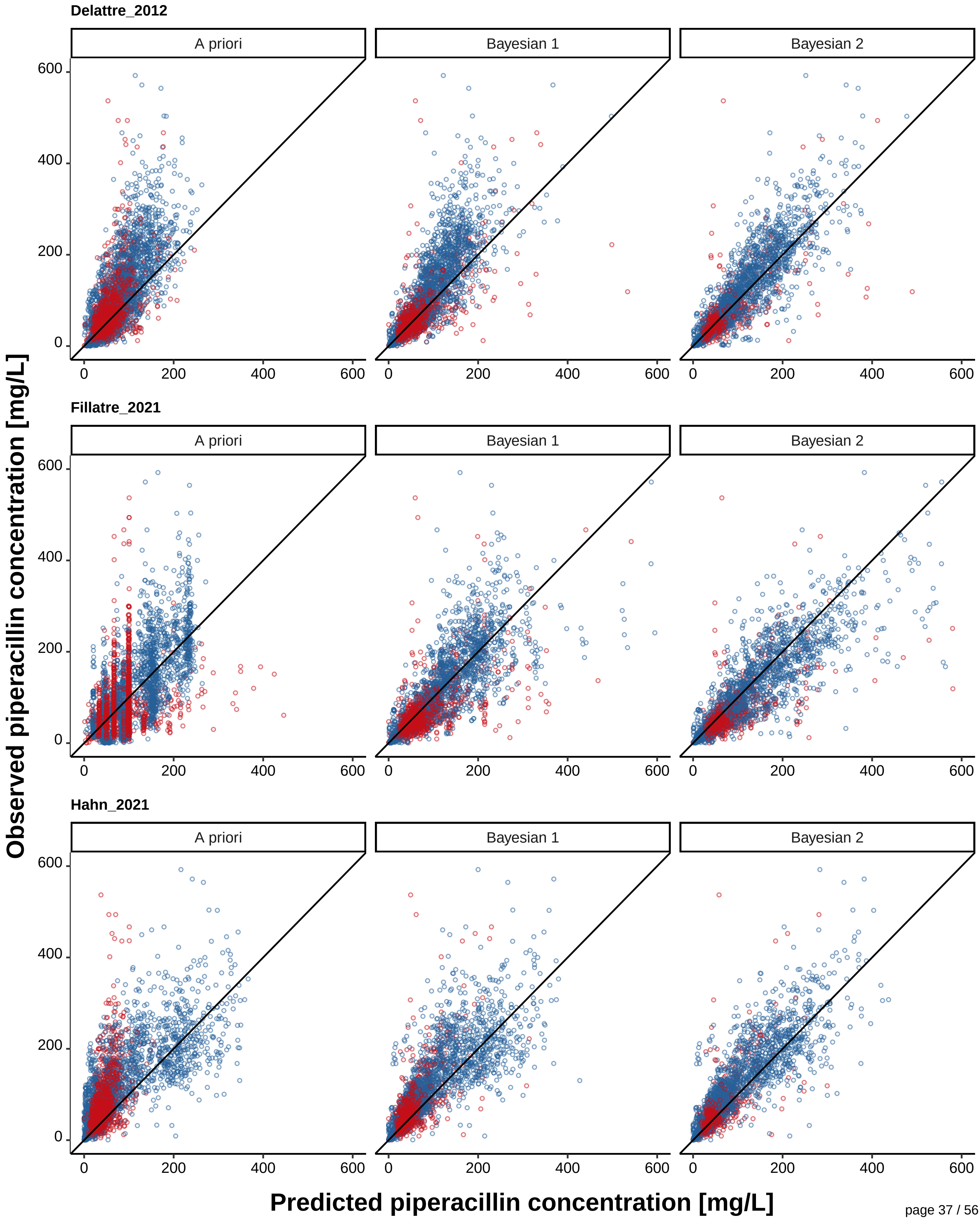
Fig. S5 Goodness-of-fit plots of the 24 population pharmacokinetic models for piperacillin, based on the total evaluation dataset stratified by type of prediction.

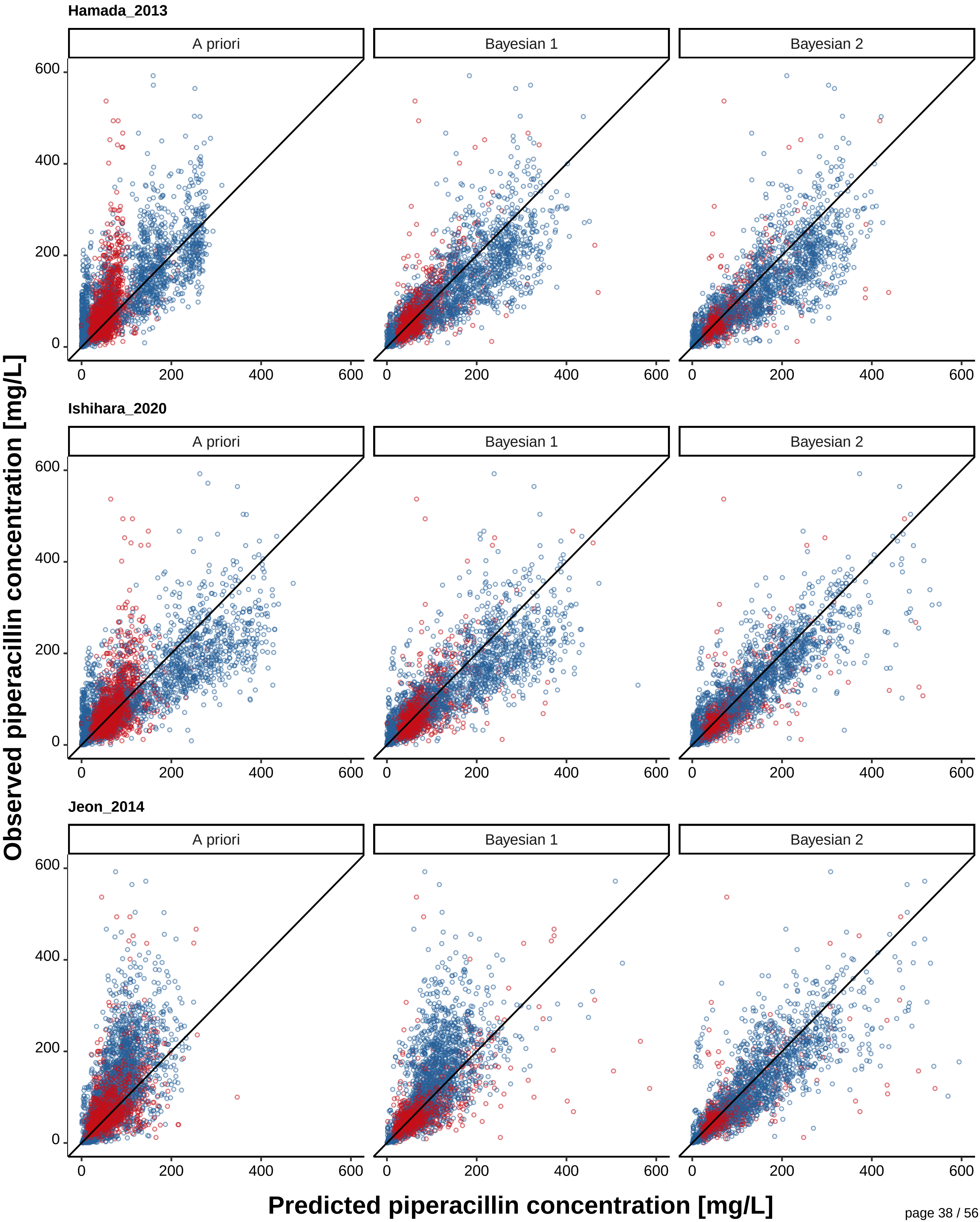
Piperacillin observations versus population (*A priori*) and individual predicted (Bayesian 1, Bayesian 2) concentrations;
A priori: Prediction of all concentrations based on dosing history and patient covariates only;
Bayesian 1: Prediction of concentrations considering the first TDM sample for each patient;
Bayesian 2: Prediction of concentrations considering the first and second TDM sample for each patient; CI: continuous infusion; II: intermittent infusion; Color indicates concentrations in patients receiving CI (red) and II (blue); black line: Line of identity

(see figure on page 35-42)

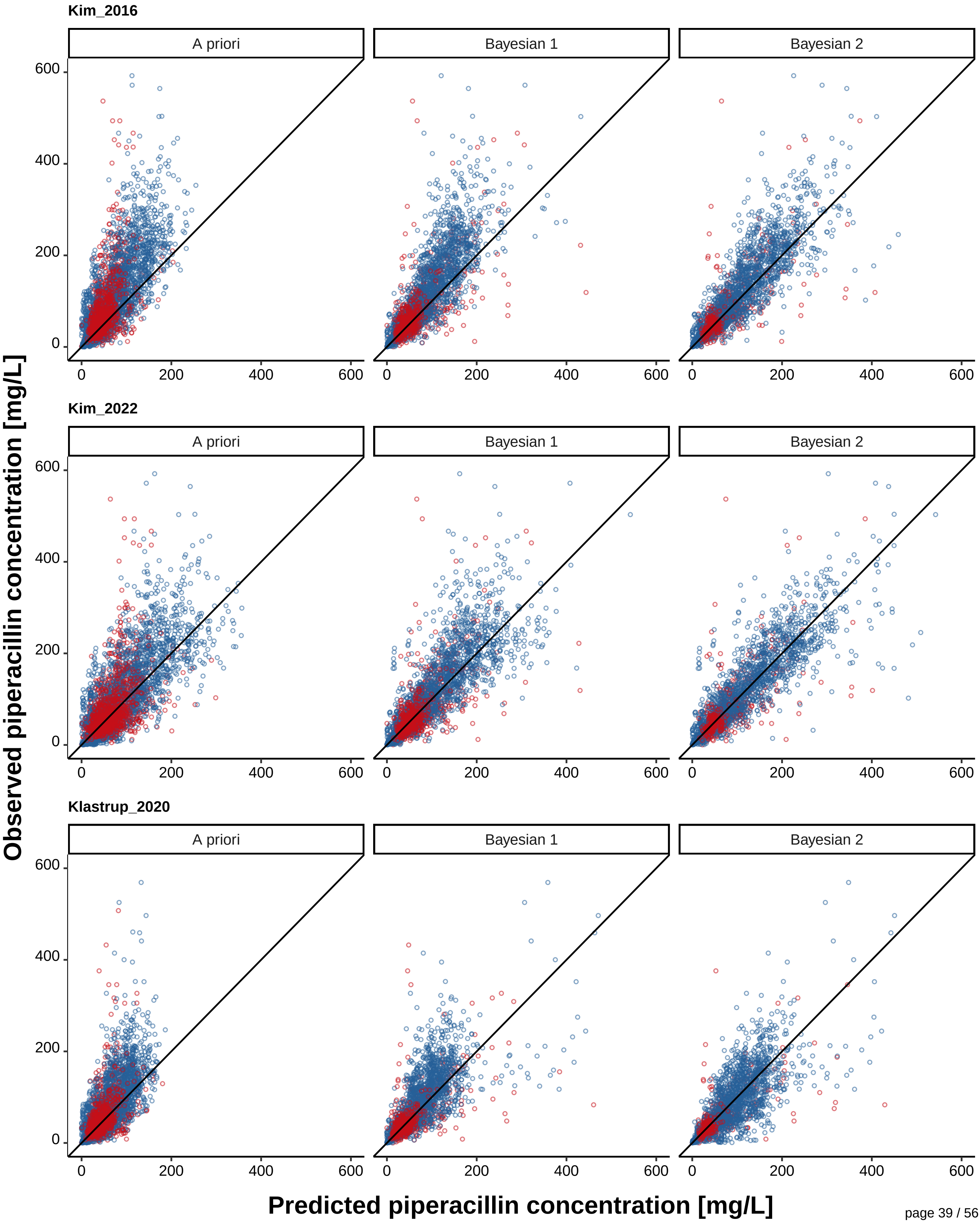

Andersen_2018
A priori
Bayesian 1
Bayesian 2
Asin−Prieto_2013
A priori
Bayesian 1
Bayesian 2
Bue_2020
A priori
Bayesian 1
Bayesian 2
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]


Bulitta_2007
Chen_2016
Chung_2015
A priori
Bayesian 1
Bayesian 2
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]

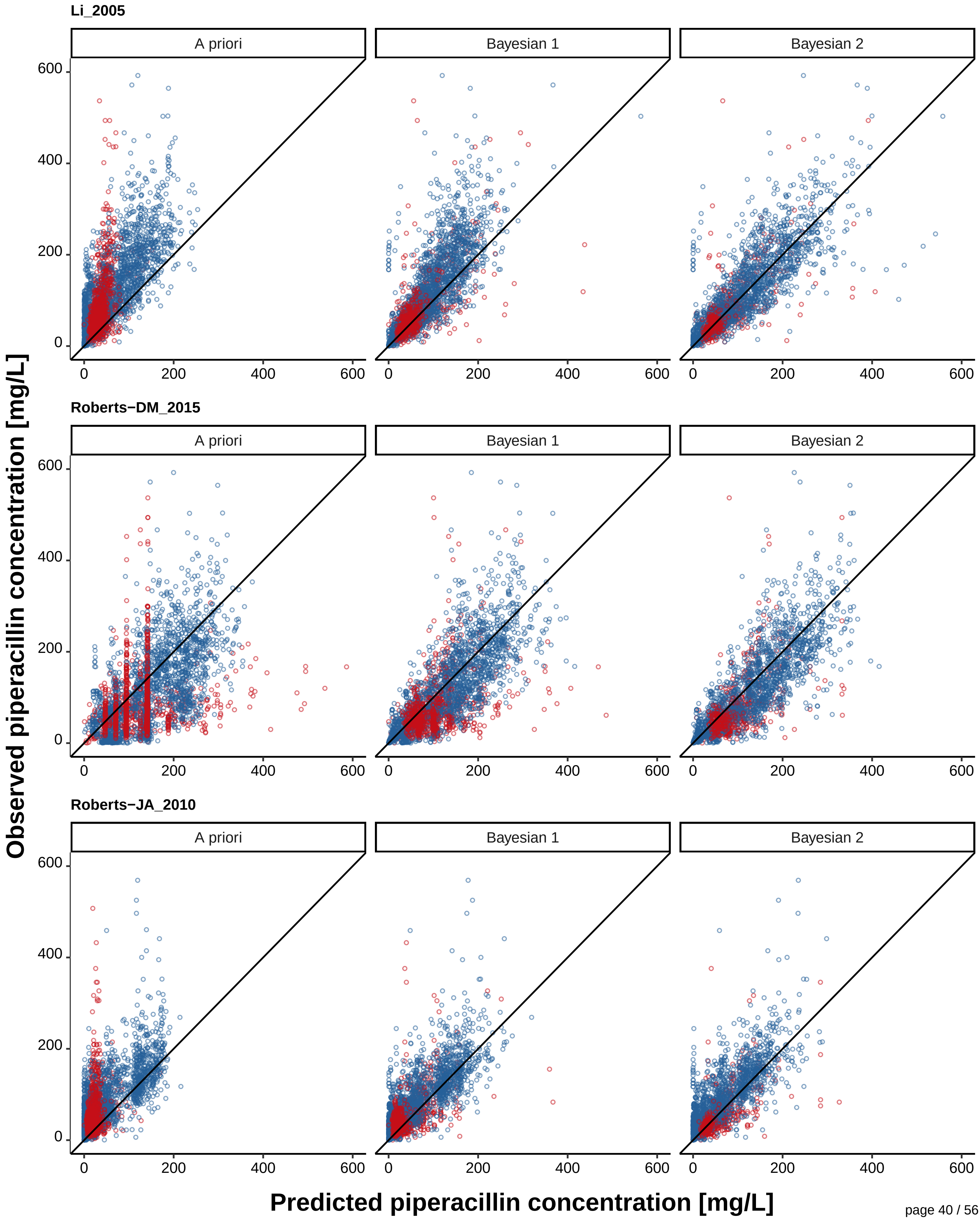

Delattre_2012
A priori
Bayesian 1
Bayesian 2
Fillatre_2021
Hahn_2021
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]

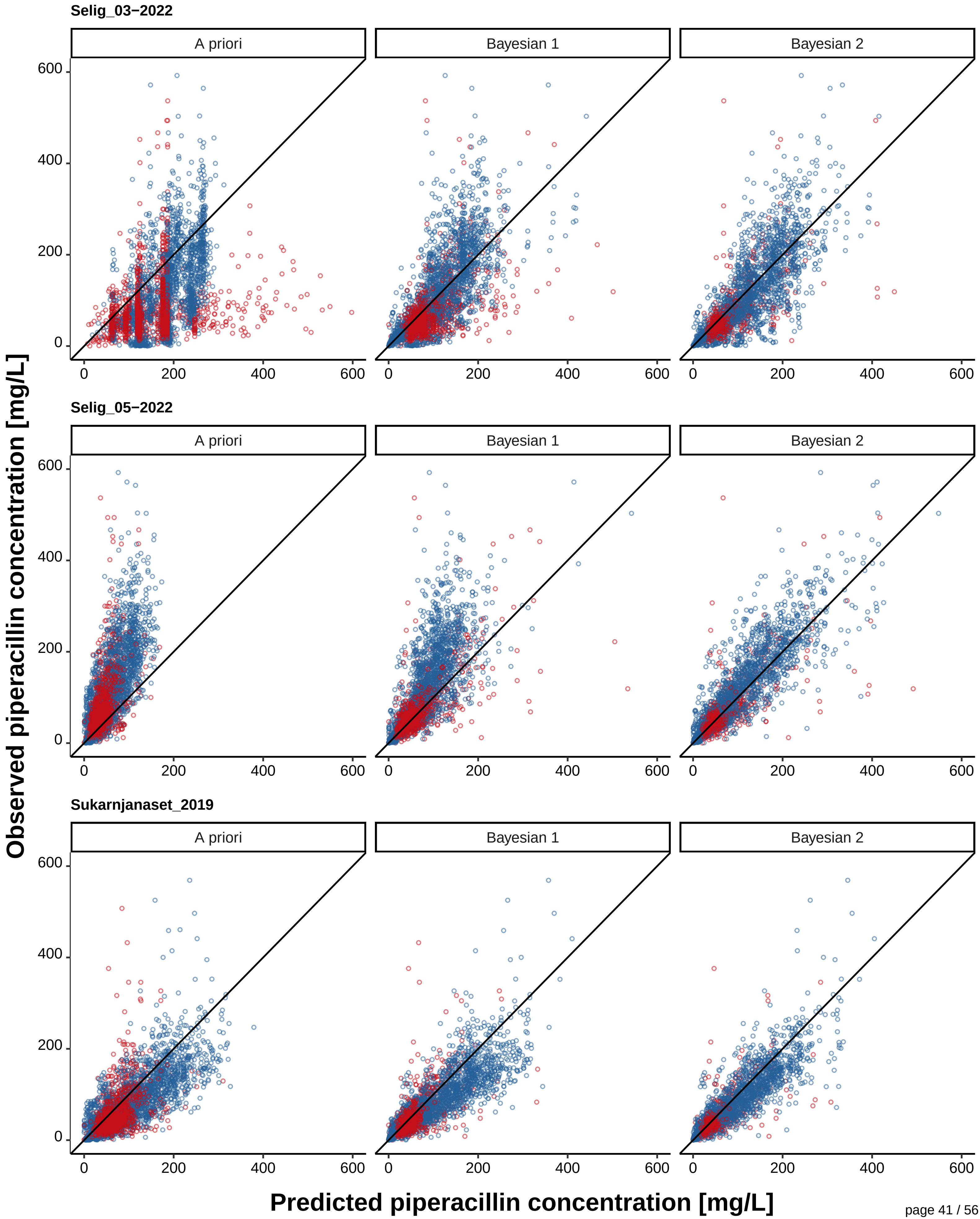


Hamada_2013
A priori
Bayesian 1
Bayesian 2
Ishihara_2020
Jeon_2014
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]



Kim_2016
A priori
Bayesian 1
Bayesian 2
Kim_2022
A priori
Bayesian 1
Bayesian 2
Klastrup_2020
A priori
Bayesian 1
Bayesian 2
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]


Li_2005
A priori
Bayesian 1
Bayesian 2
Roberts−DM_2015
Roberts−JA_2010
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]



Selig_03−2022
A priori
Bayesian 1
Bayesian 2
Selig_05−2022
A priori
Bayesian 1
Bayesian 2
Sukarnjanaset_2019
A priori
Bayesian 1
Bayesian 2
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]

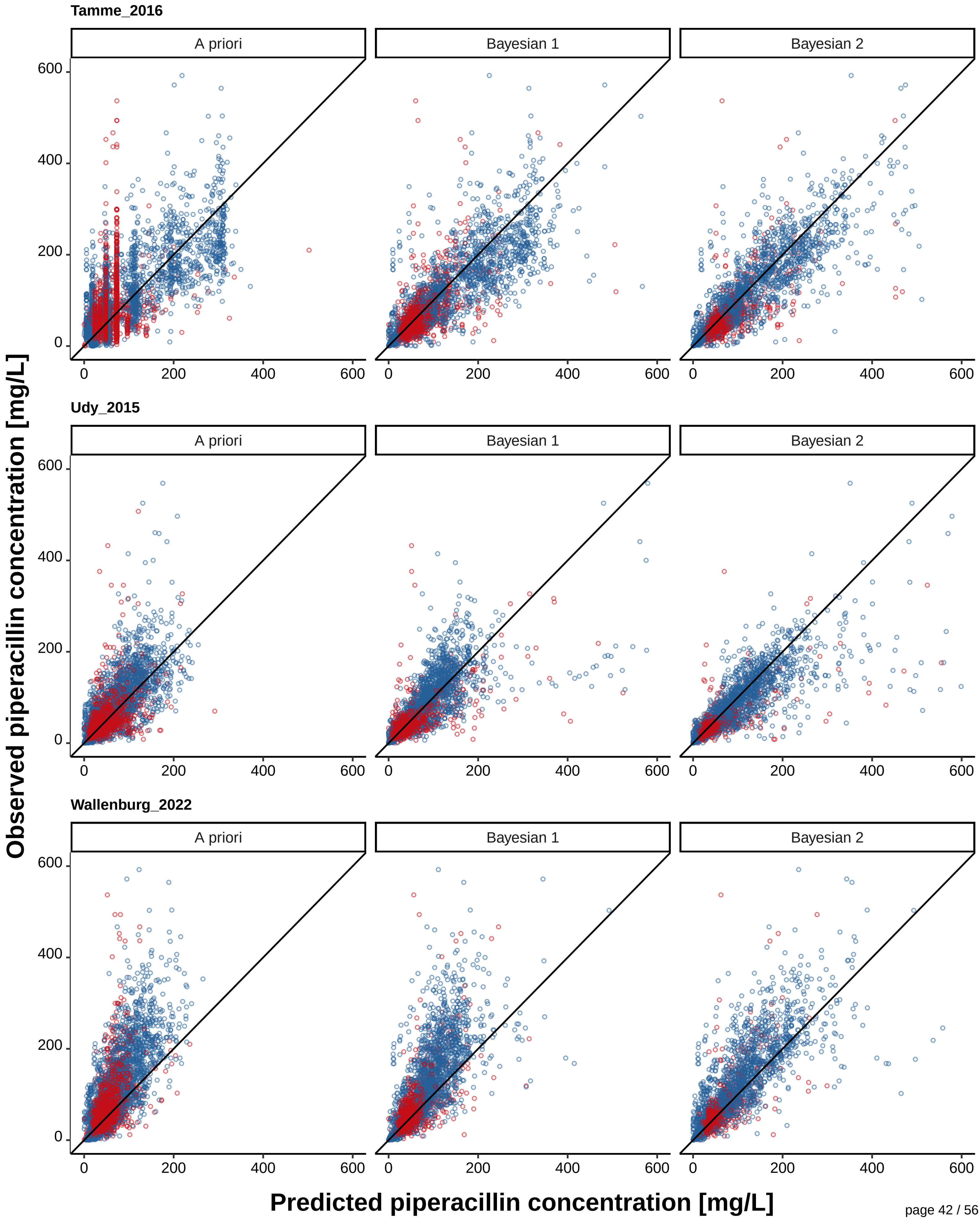
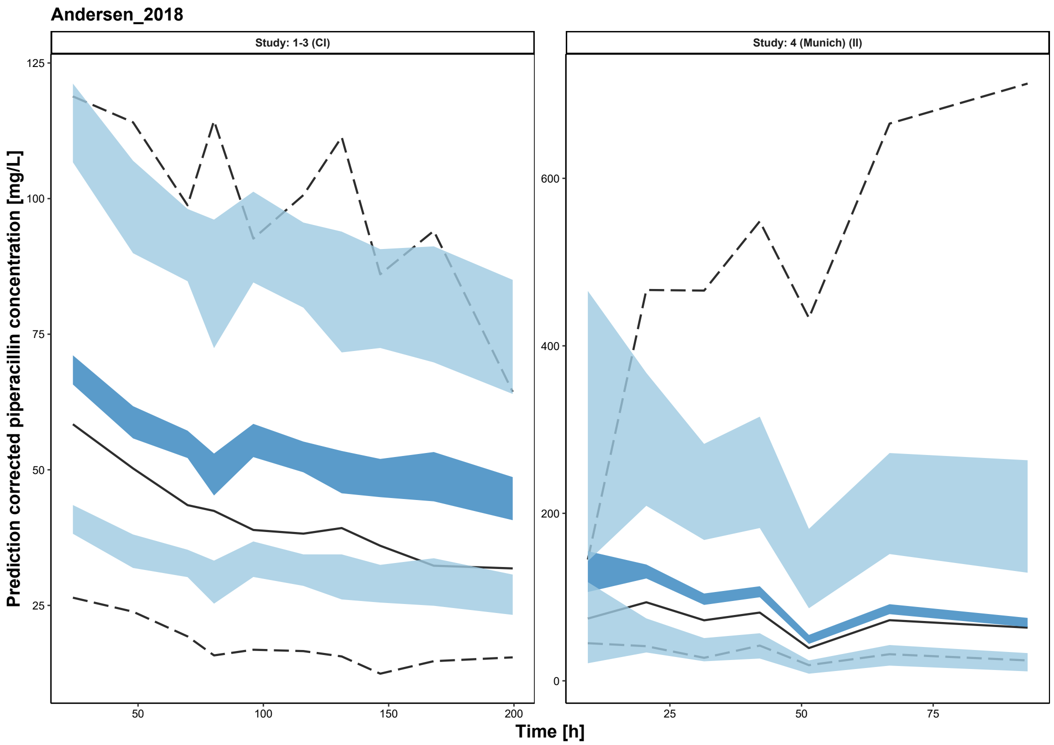
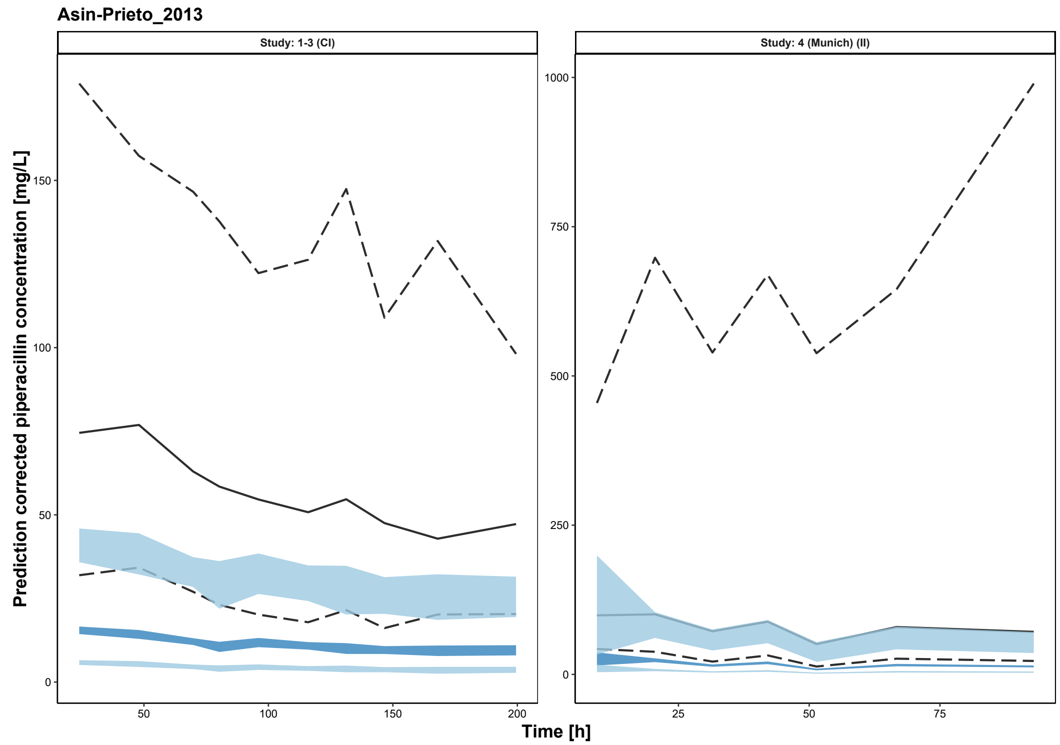

Tamme_2016
A priori
Bayesian 1
Bayesian 2
Udy_2015
A priori
Bayesian 1
Bayesian 2
Wallenburg_2022
A priori
Bayesian 1
Bayesian 2
0
200
400
600
Observed piperacillin concentration [mg/L]
Predicted piperacillin concentration [mg/L]

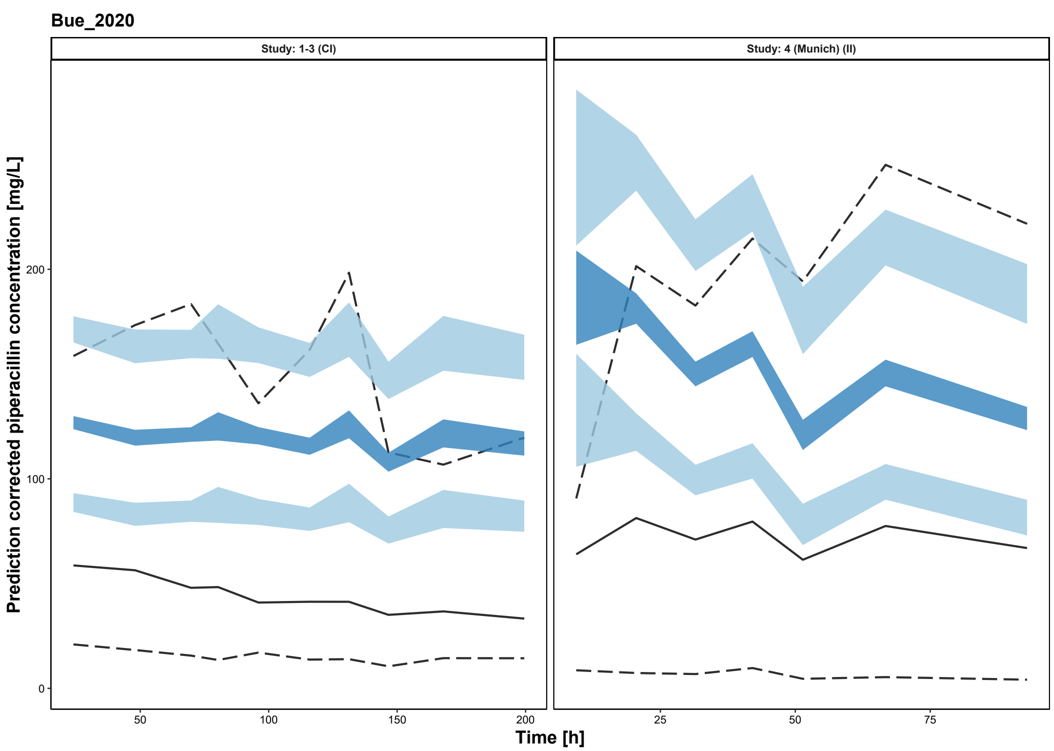
Fig. S6 Prediction-corrected visual-predictive-checks (pcVPC, 1000 simulations) of the 24 population pharmacokinetic models for piperacillin using the total evaluation dataset, stratified by different piperacillin/tazobactam administration methods.

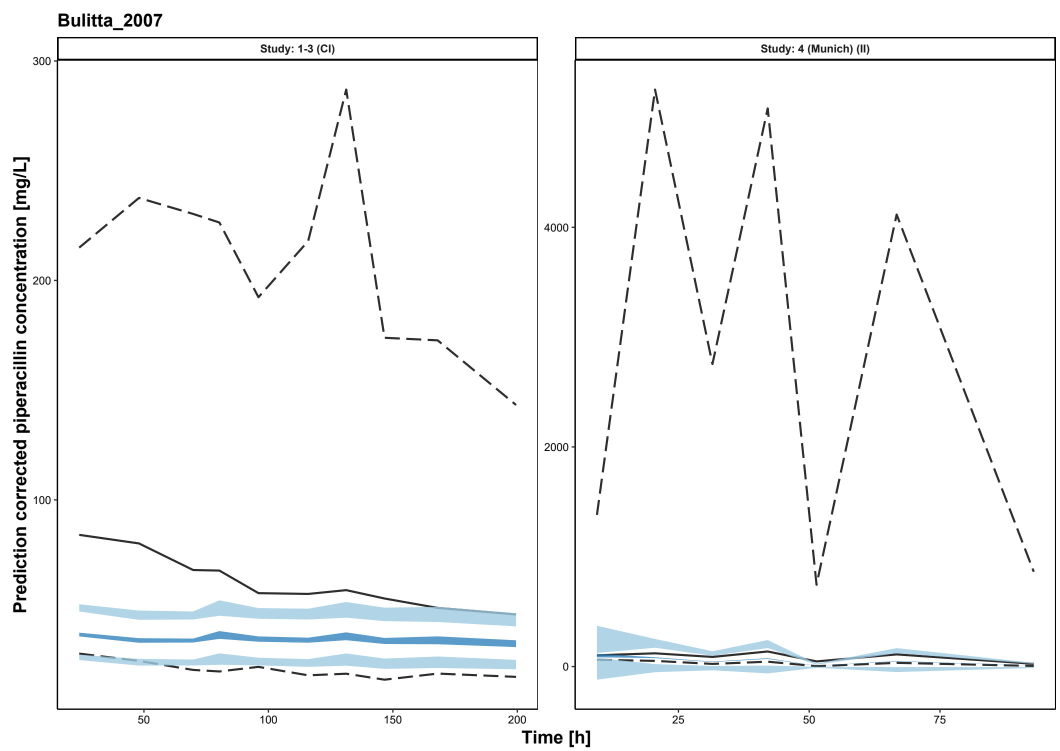
CI: continuous infusion; II: intermittent infusion; black dashed lines: 90%-interval of the observed concentrations (95^{th} and 5^{th} percentile, respectively); black solid line: median of the observed concentrations (50^{th} percentile); blue areas: 95%-confidence interval of the simulated concentration percentiles (confidence interval of 95^{th} and 5^{th} percentile (light blue) and 50^{th} percentile (dark blue), respectively)

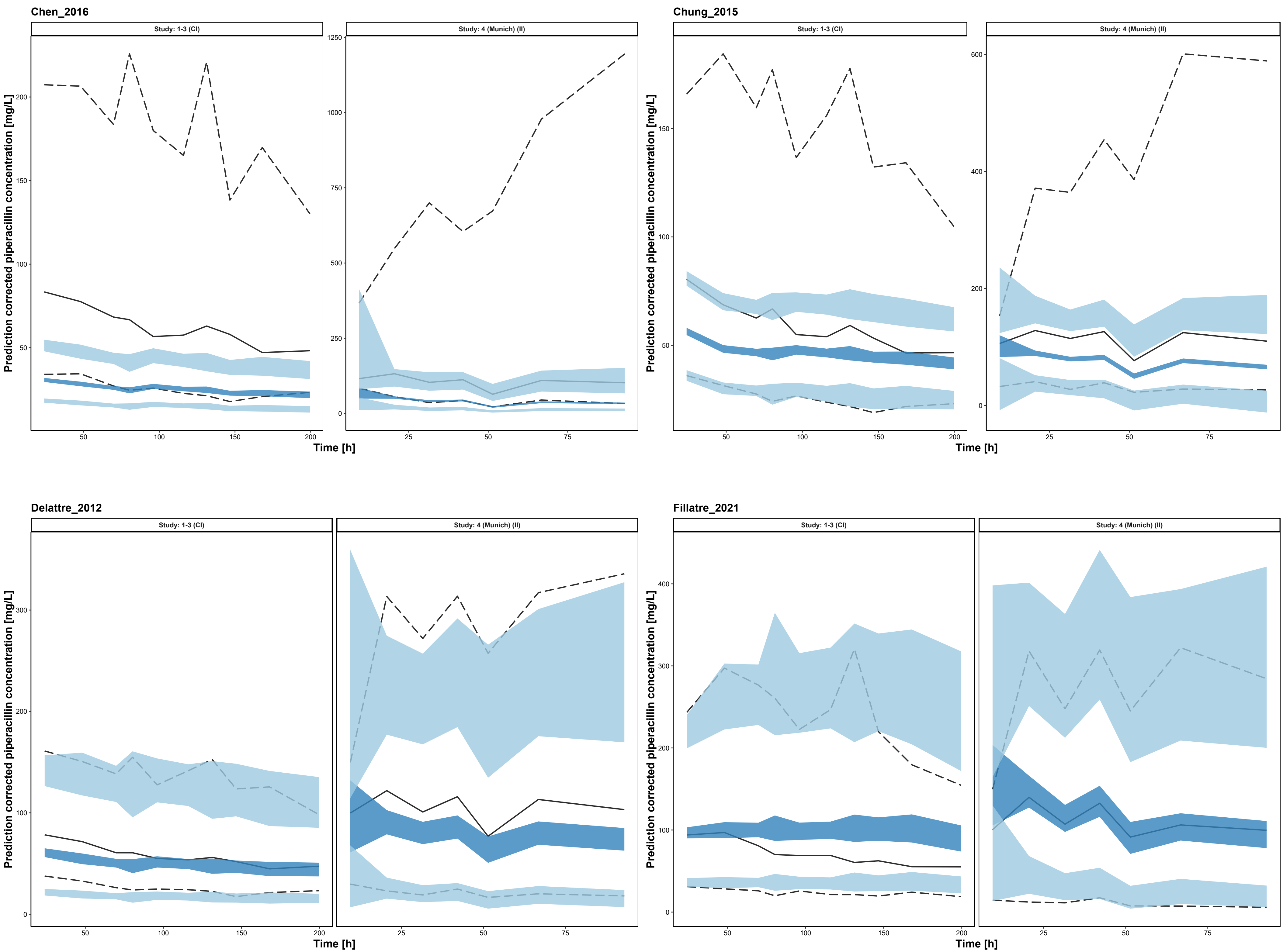
(see figure on page 44-49)


Andersen_2018
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
Time [h]

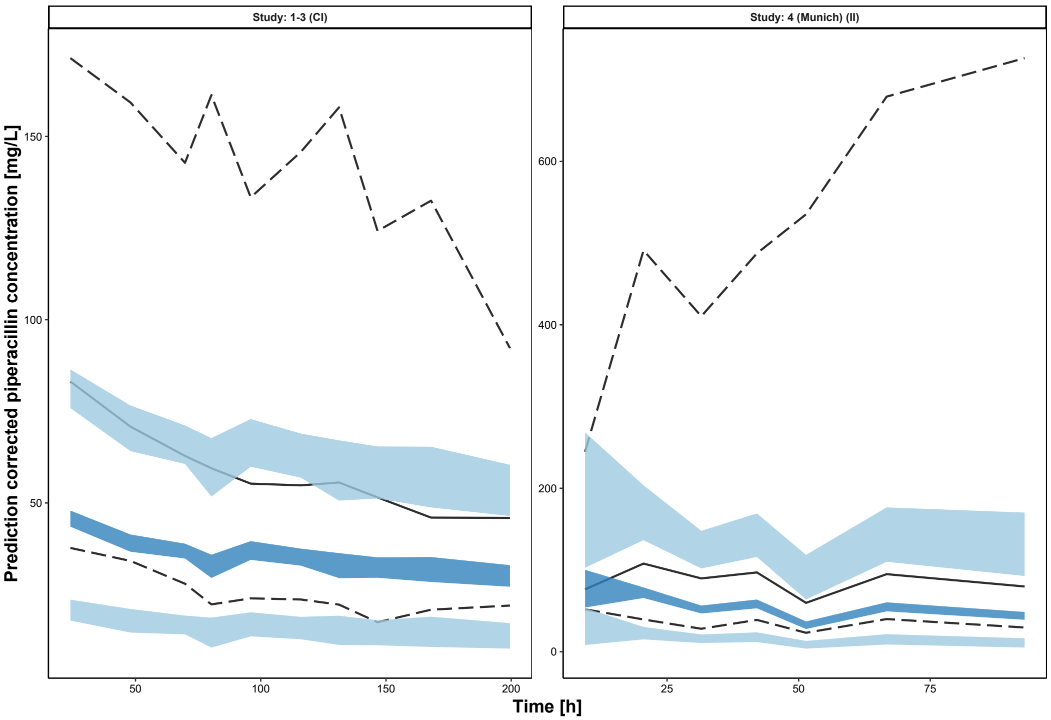

Asin-Prieto_2013
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
Time [h]


Bue_2020
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
Time [h]

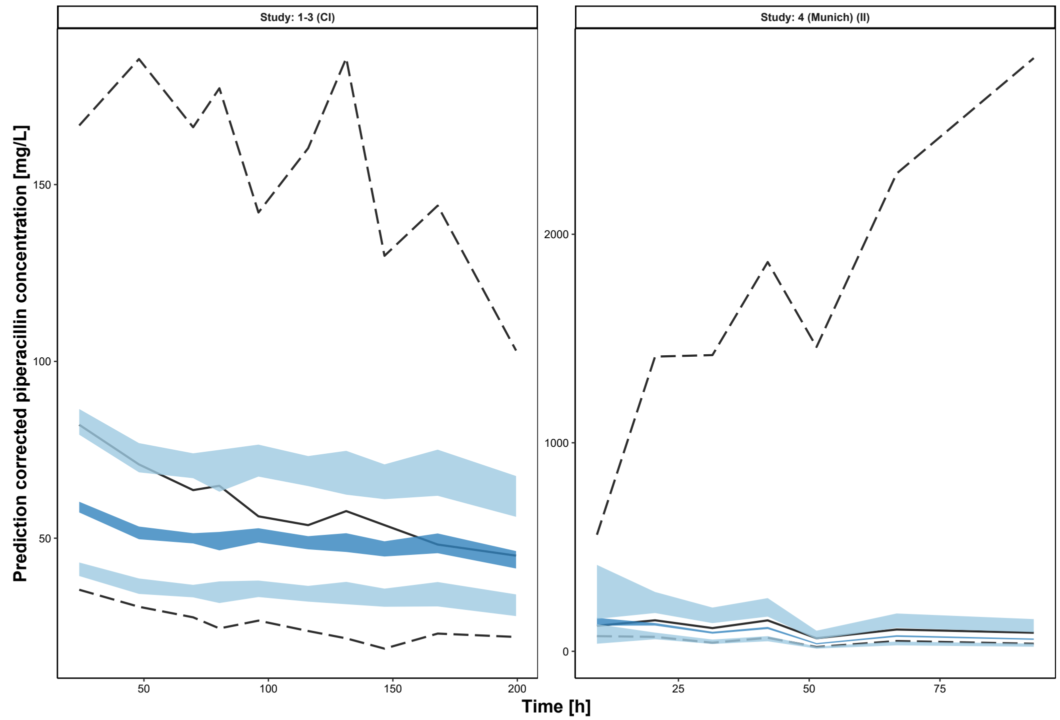

Bulitta_2007
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
Time [h]


Chen_2016
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Chung_2015
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Delattre_2012
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Fillatre_2021
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
Time [h]

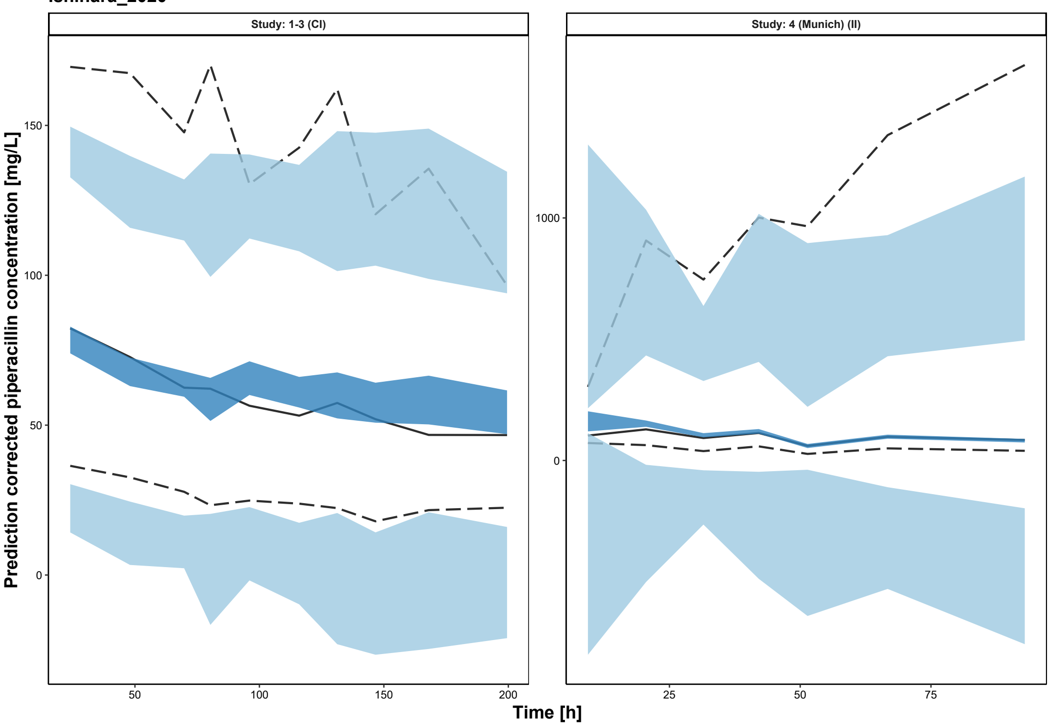
Hahn_2021



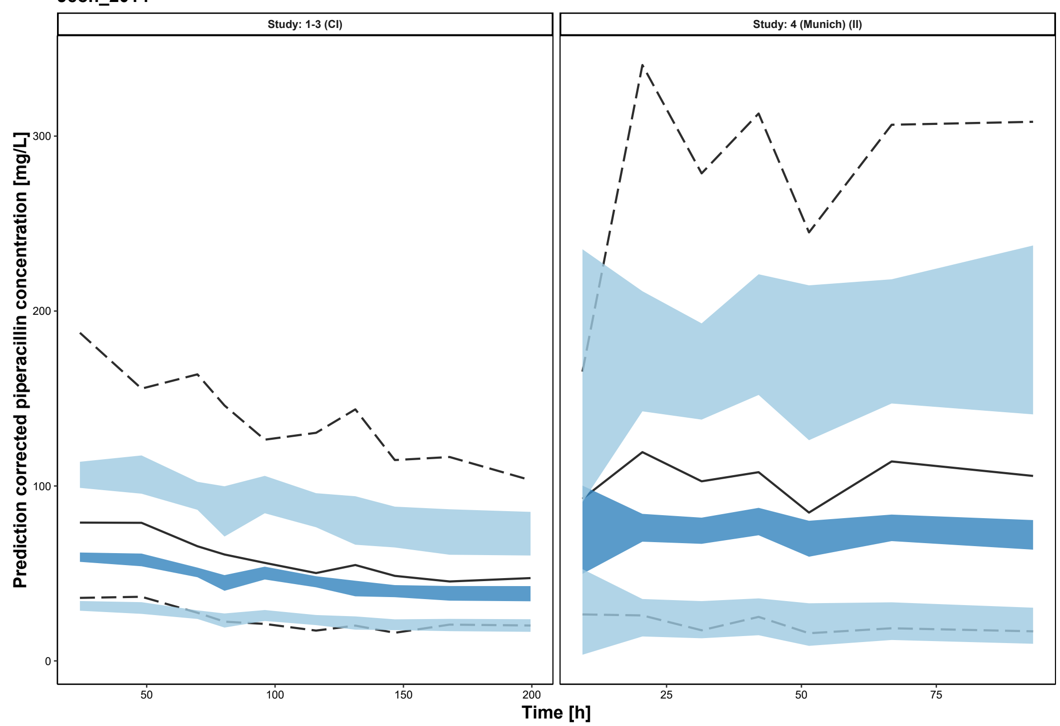
Hamada_2013

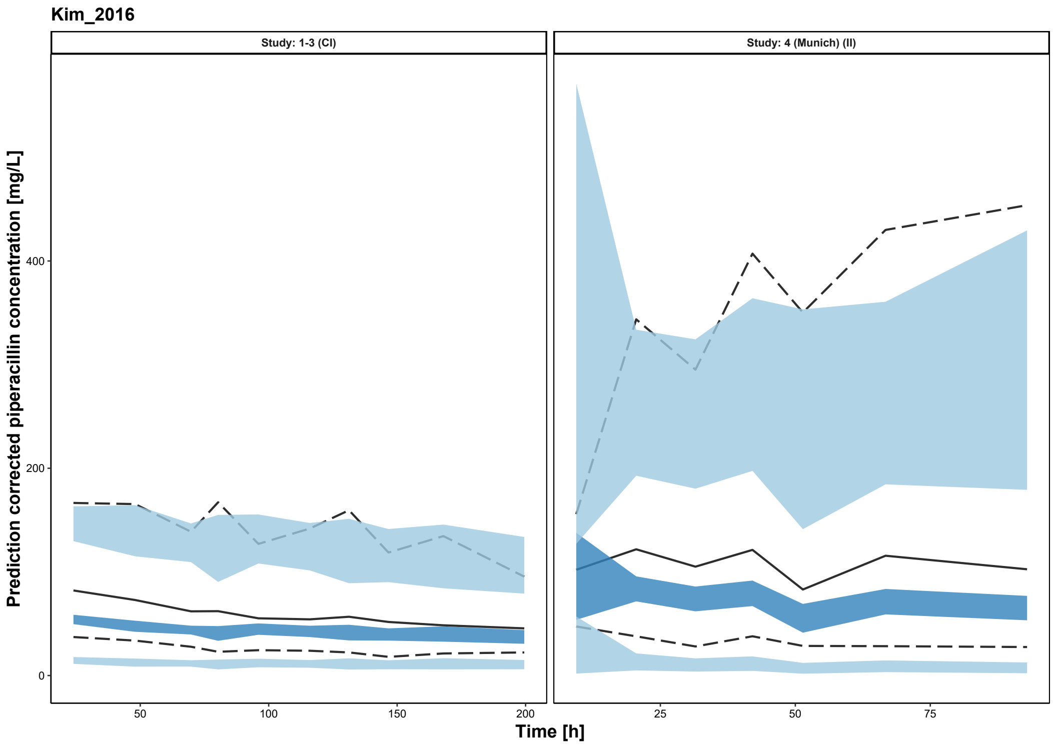


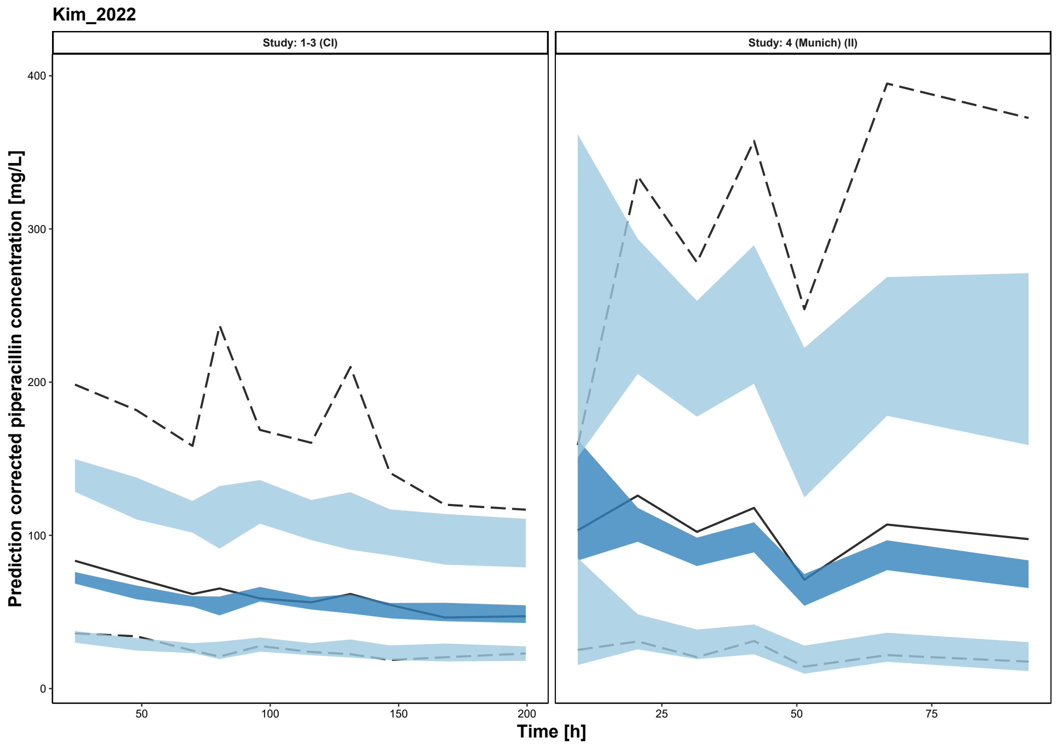
Ishihara_2020

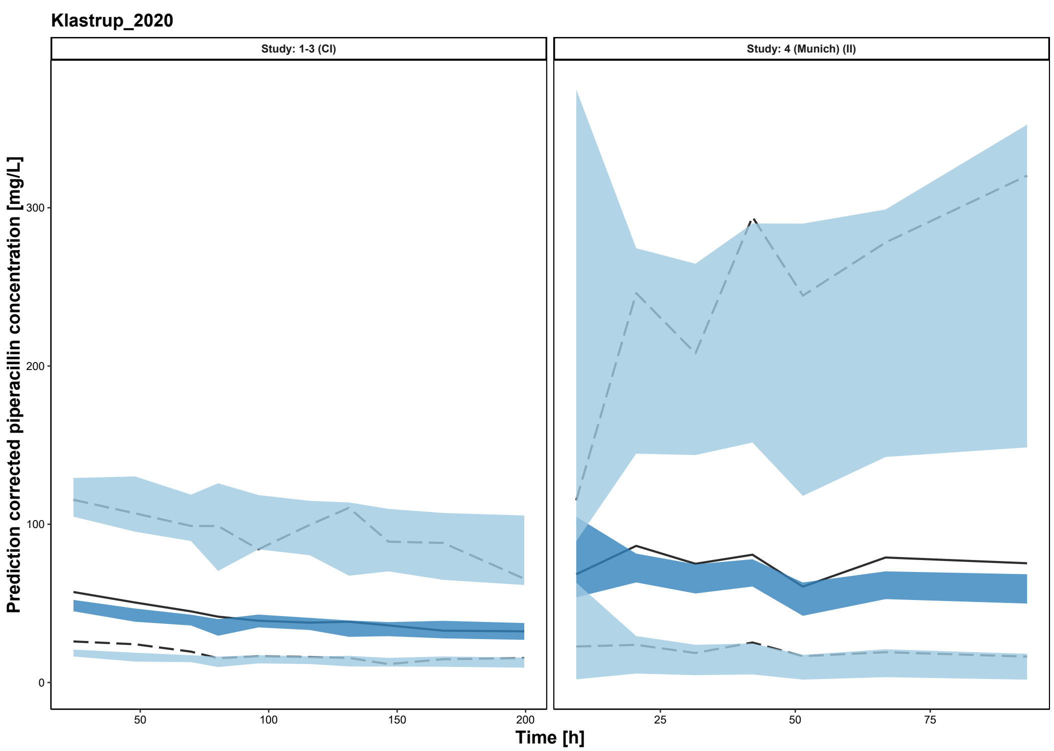


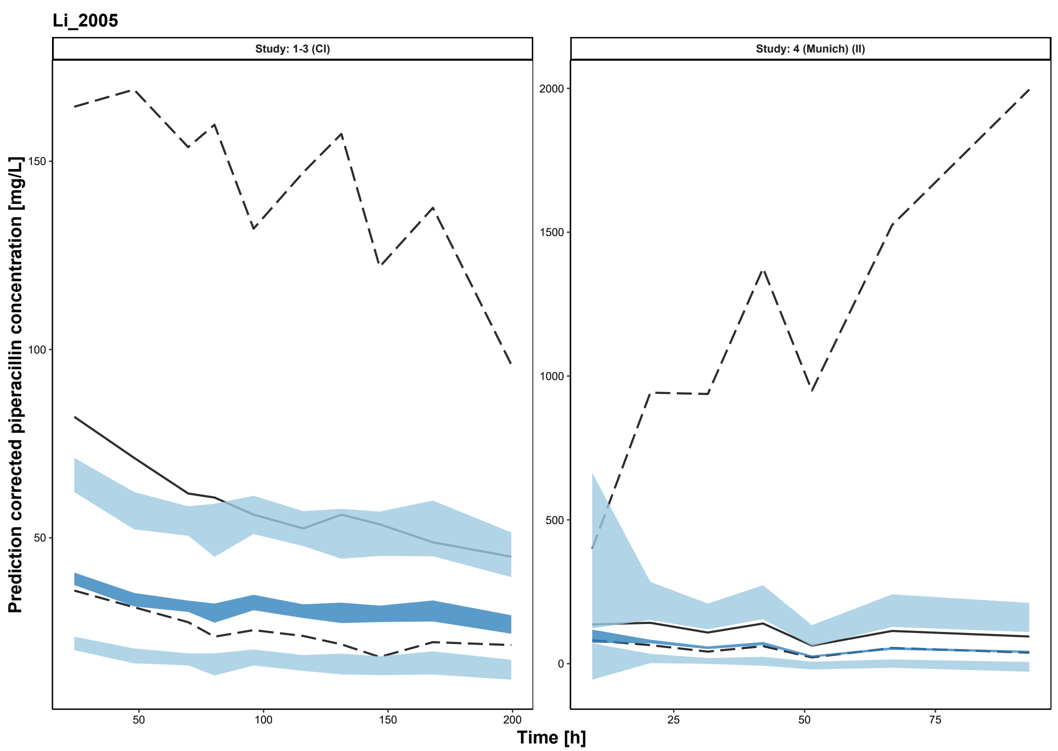
Jeon_2014



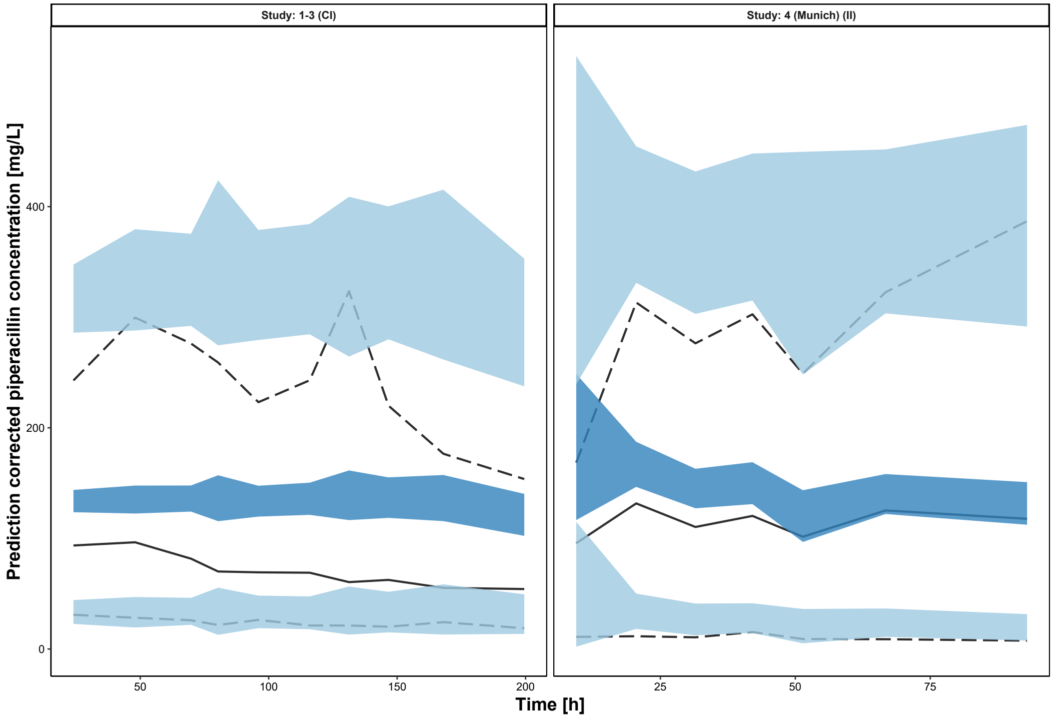

Kim_2016
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
0
200
400
50
100
150
200
25
50
75
Time [h]


Kim_2022
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
0
100
200
300
400
50
100
150
200
25
50
75
Time [h]

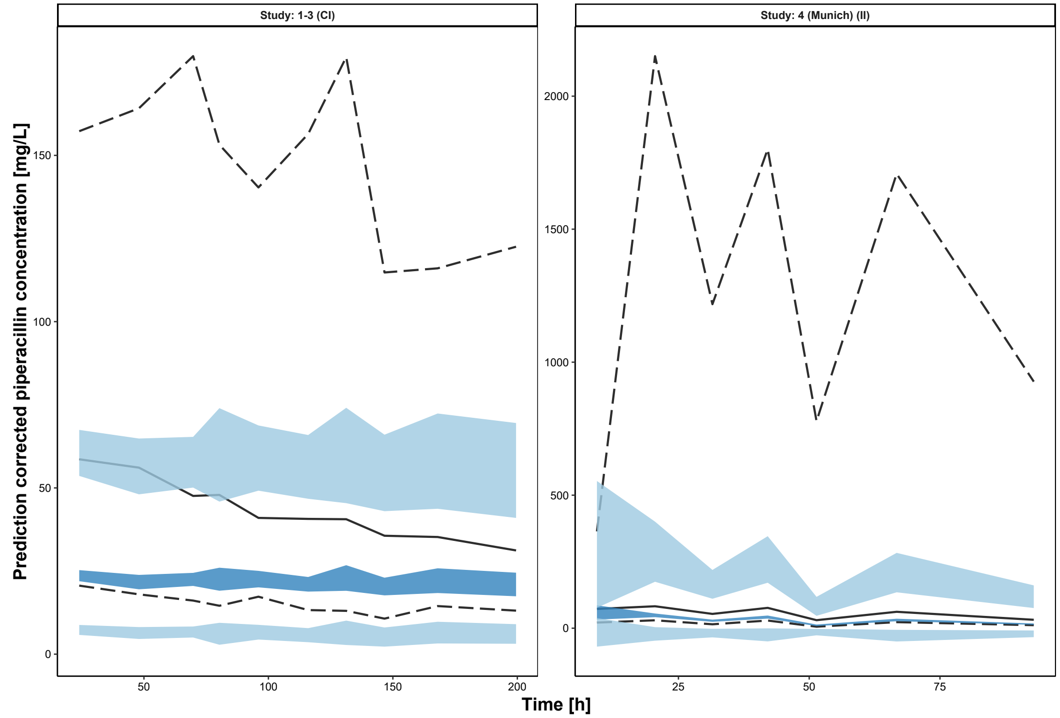

Klastrup_2020
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
0
100
200
300
50
100
150
200
25
50
75
Time [h]


Li_2005
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
50
100
150
0
500
1000
1500
2000
50
100
150
200
25
50
75
Time [h]

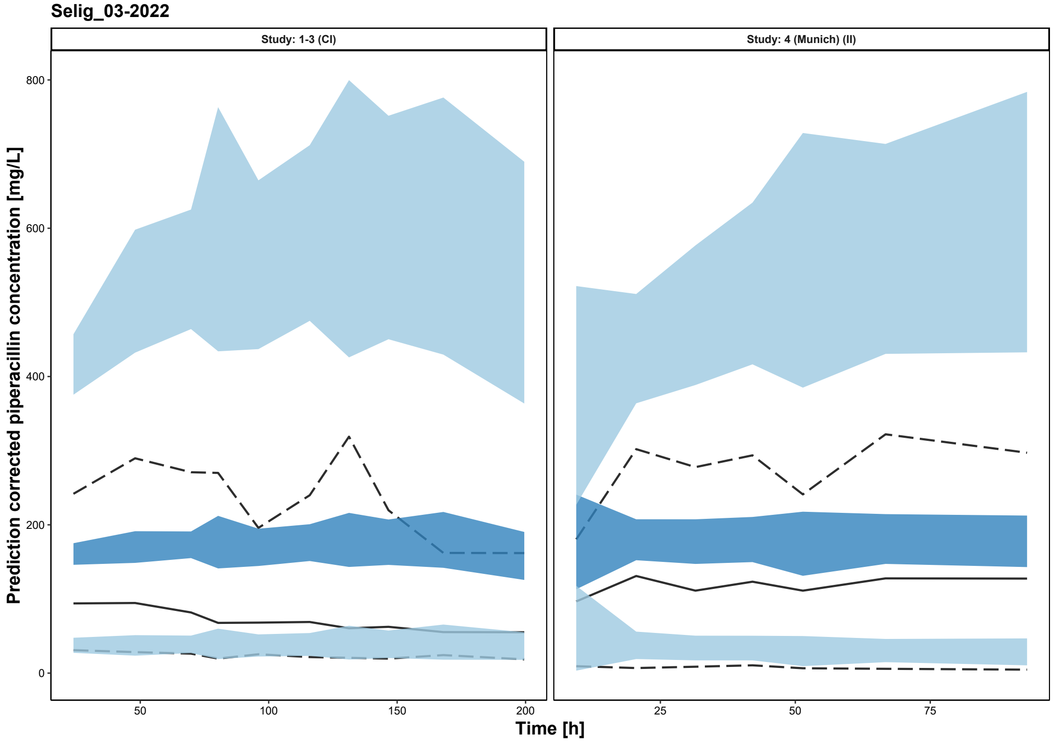
Roberts-DM_2015



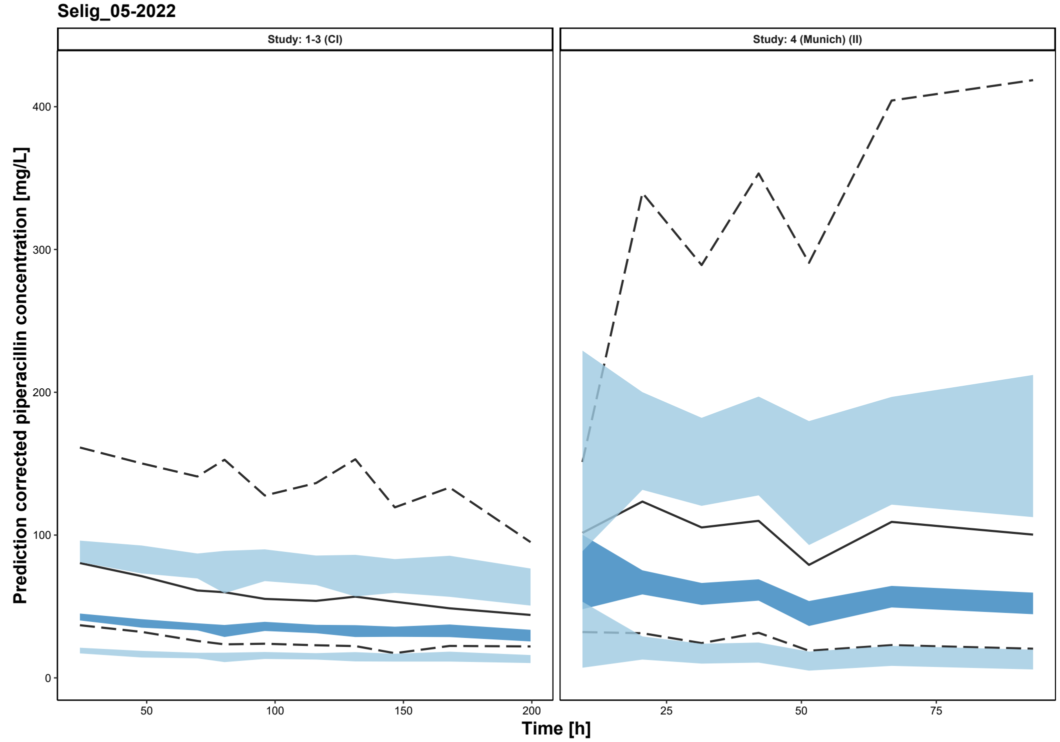
Roberts-JA_2010



Selig_03-2022



Selig_05-2022



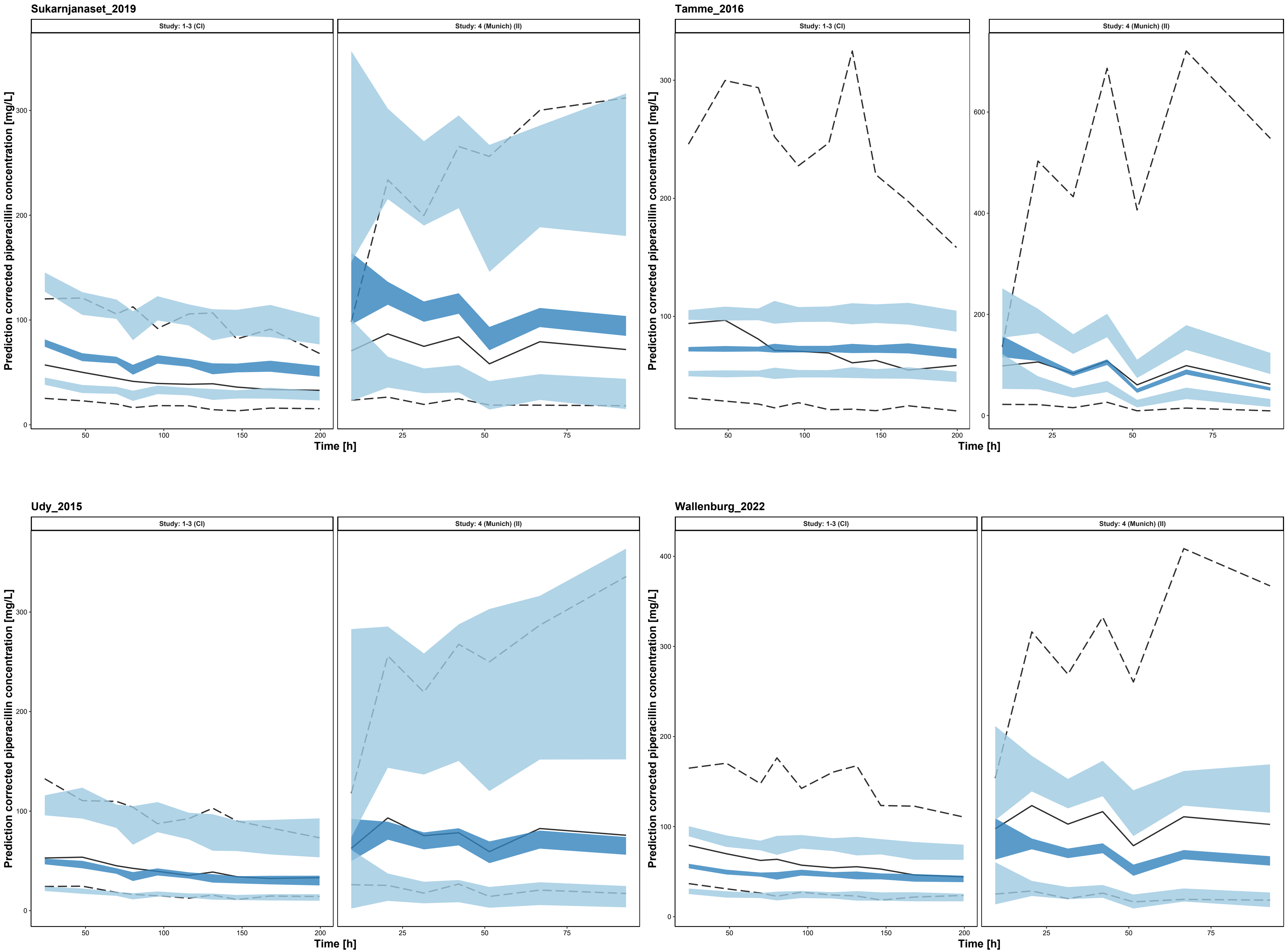
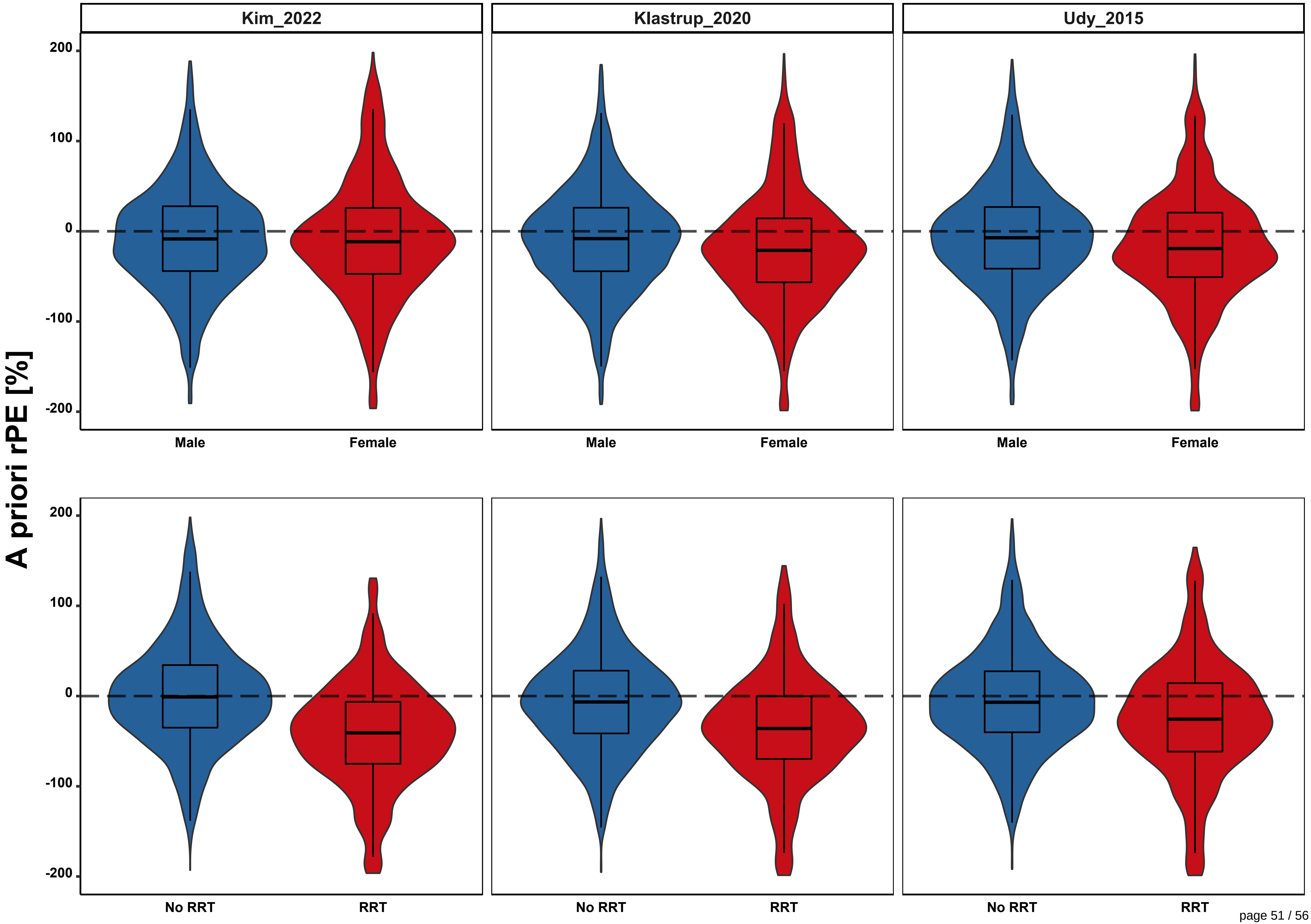
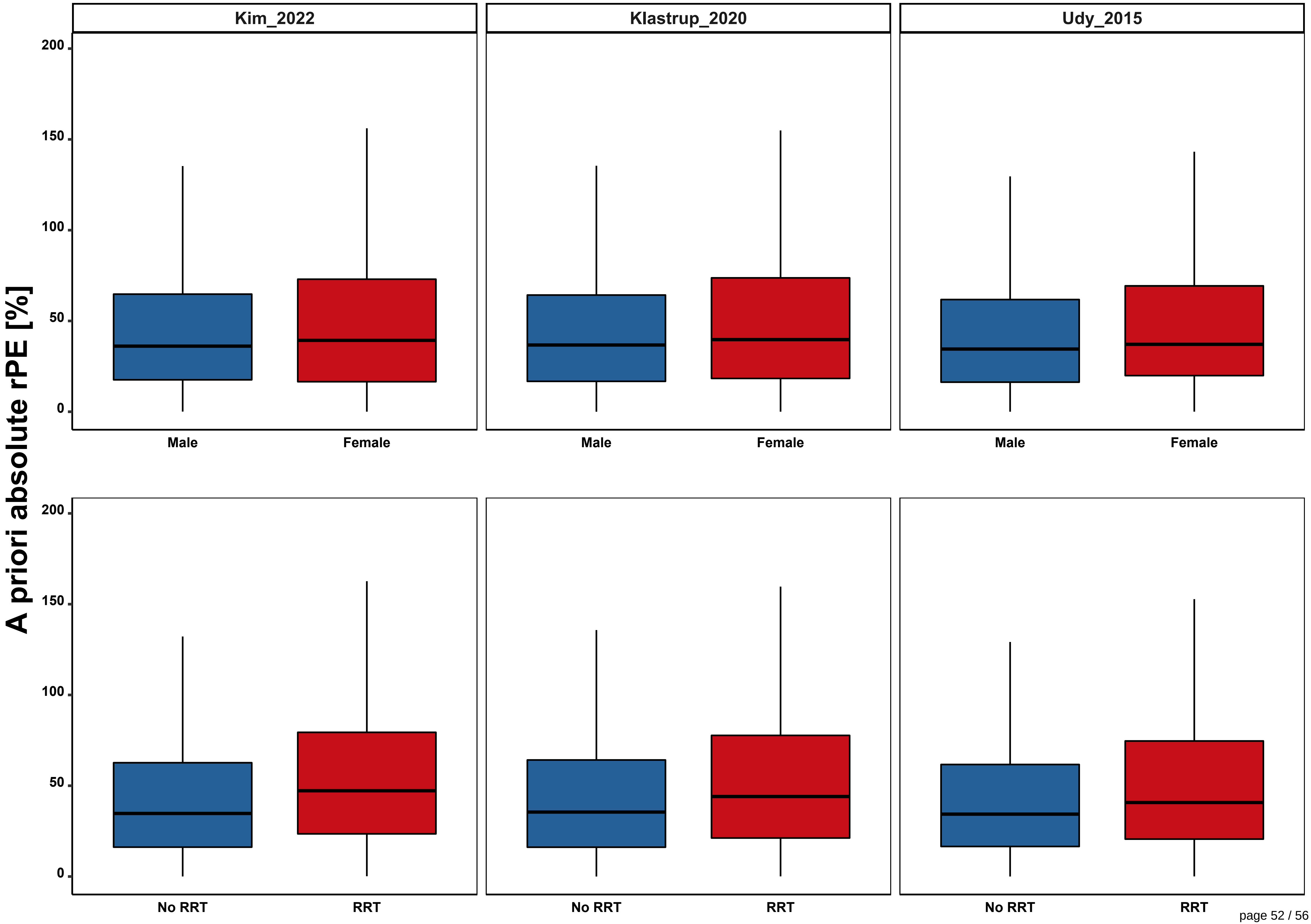

Sukarnjanaset_2019
Tamme_2016
Udy_2015
Wallenburg_2022
Study: 1-3 (CI)
Study: 4 (Munich) (II)
Prediction corrected piperacillin concentration [mg/L]
Time [h]
0
100
200
300
400
600
25
50
75
150

Fig. S7 Associations between the *a priori* relative prediction error (rPE) and absolute rPE and dichotomous patient covariates given the population pharmacokinetic models by Kim, Klastrup and Udy et al.

Colored violin plots show the relative distribution of prediction errors for the respective covariate manifestation (rPE approximately normally distributed); boxplots represent the 25th, 50th and 75th percentiles, whiskers the 1.5-fold interquartile range; long-dashed lines: reference lines (rPE = 0%); RRT: renal replacement therapy

(see figure on page 51-52)


A priori rPE [%]
Kim_2022
Klastrup_2020
Udy_2015
200
100
0
-100
-200
Male
Female
No RRT
RRT



Kim_2022
Klastrup_2020
Udy_2015
A priori absolute rPE [%]
200
150
100
50
0
Male
Female
No RRT
RRT

Table S6 Associations between the *a priori* relative prediction error (rPE) and dichotomous patient covariates given the population pharmacokinetic models by Kim, Klastrup and Udy et al.

A priori (A): prediction of all concentrations based on dosing history and patient covariates only;
MPE: median relative prediction error; MAPE: median absolute relative prediction error; CI: confidence interval; RRT: renal replacement therapy;
r_{pb}: point biserial correlation coefficient; $r_{pb\_corrected}$: corrected point biserial correlation coefficient; red fields highlight significant $r_{pb\_corrected}$

PopPK model	Covariate		No. of patients	No. of observed concentrations	(A) MPE (95%-CI)	(A) MAPE (95%-CI)	r_{pb}	p-value (α = 0.05)	$r_{pb\_corrected}$[a]
Kim_2022	**Sex**	male	370	2498 (68.4 %)	-8.49 (-11.09 to -5.53)	36.13 (34.38 to 37.97)	-0.005	0.787	-0.01
		female	191	1156 (31.6 %)	-11.51 (-15.03 to -8.79)	39.31 (36.1 to 41.72)			
	RRT	no	465	2840 (77.7 %)	-0.97 (-3.65 to 1.58)	34.74 (33.26 to 36.61)	-0.297	< 0.001	-0.35
		yes	96	814 (22.3 %)	-40.79 (-45.48 to -36.75)	47.23 (43.7 to 50.9)			
Klastrup_2020	**Sex**	male	370	2498 (68.4 %)	-8.15 (-10.69 to -5.35)	36.76 (35.13 to 38.59)	-0.084	< 0.001	-0.09
		female	191	1156 (31.6 %)	-21.15 (-24.73 to -17.43)	39.73 (37.0 to 42.89)			
	RRT	no	465	2840 (77.7 %)	-6.54 (-8.66 to -4.19)	35.51 (33.99 to 37.29)	-0.218	< 0.001	-0.26
		yes	96	814 (22.3 %)	-35.96 (-40.26 to 31.89)	44.07 (40.95 to 47.02)			
Udy_2015	**Sex**	male	370	2498 (68.4 %)	-7.17 (-10.03 to -4.54)	34.53 (32.63 to 36.04)	-0.075	0.005	-0.08
		female	191	1156 (31.6 %)	-19.15 (-22.92 to -15.38)	37.14 (33.69 to 39.01)			
	RRT	no	465	2840 (77.7 %)	-6.82 (-9.61 to -4.16)	34.36 (32.82 to 35.67)	-0.145	< 0.001	-0.17
		yes	191	814 (22.3 %)	-25.65 (-30.53 to -22.15)	40.76 (36.88 to 44.02)			

[a] To account for possible attenuation of the r_{pb} due to unequal group sizes of dichotomous variables (e.g. TDM samples of male 68.4 vs. female 31.6%), a correction formula according to Hunter et al. was applied [29]:

$$r_{pb\_corrected} = ar/\sqrt{[(a^2 - 1)r^2 + 1]}$$

where

$r = r_{pb}$
$a = \sqrt{[.25/pq]}$
p, q are the relative proportions of the dichotomous covariate (No. of observed concentrations)

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